



DEPARTMENT OF ELECTRICAL
AND COMPUTER ENGINEERING

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Bachelor of Science in Electrical and Computer Engineering

INTELLIGENT PRODUCTION PLANNING IN FOOD MANUFACTURING: ENHANCING MRP SYSTEM WITH AI

MASTER IN ELECTRICAL AND COMPUTER ENGINEERING
SPECIALIZATION IN ROBOTICS & AUTONOMOUS SYSTEMS

NOVA University Lisbon
September, 2026



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Examination Committee

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To my family, whose continuous support, patience, and sacrifices from the very beginning of my academic path gave me the freedom to explore, learn, and pursue my passions, serving as the driving force to complete this chapter.

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Finally, I thank my family and friends for their continuous encouragement and support throughout my academic studies.

”

*“To me, defeat in anything is merely temporary.
Defeat simply tells me that something is wrong in
my doing; it is a path leading to success and truth.”*

— **Bruce Lee**, *Striking Thoughts*
(Martial Artist and Philosopher)

”

*“Knowing is not enough; we must apply. Willing is
not enough; we must do.”*

— **Johann Wolfgang von Goethe**, *Wilhelm Meister's
Apprenticeship*
(Writer and Statesman)

ABSTRACT

Food manufacturers have to plan production around seasonal demand and perishable materials, and must react when market disruptions occur. Traditional Material Requirements Planning System (MRP system) follow static, rule-based logic and adapt poorly to these conditions, which exposes companies to stock shortages, and food waste.

This thesis proposes a planning platform that integrates Artificial Intelligence (AI) and Machine Learning (ML) into an existing MRP system, providing data-driven demand forecasts and production planning recommendations. Six forecasting models (Random Forest, XGBoost, LightGBM, Prophet, Chronos, and Granite) are combined through Inverse Error Weighting into a single forecast with a 90% prediction interval. The platform converts this forecast into raw-material requirements through recursive Bill of Materials (BOM) decomposition and schedules stock consumption with First-Expired-First-Out (FEFO) logic to reduce waste. It can also simulate how external disruptions would affect the production plan. Following Industry 5.0 principles, managers can manually adjust the weight given to each model.

The prototype was tested on 2014–2026 data from a Portuguese food manufacturer in three scenarios. In the controlled 2025 baseline, the combined forecast for the analyzed items stayed within or close to its prediction interval for most of the year. When a demand shock based on the 2022 war in Ukraine was applied, the platform updated the material requirements and raised alerts for the affected items without manual recalculation. On live 2026 data, three of the six models failed, and the platform continued planning with the remaining three. The industrial partner considered the outputs credible for operational use. These results describe how the platform behaves rather than how much it improves planning. Forecast error was not measured across the product catalog, and the effect on waste and material accuracy was not compared with the company's actual plan. The live test also showed that changes in product identifiers break the demand history the models depend on. Comparing planned against actual purchases, as requested by the industrial partner, is the next step toward measuring that benefit.

Keywords: Industry 4.0 · Industry 5.0 · Artificial Intelligence (AI) · Machine Learning (ML) · Material Requirements Planning (MRP)

RESUMO

No setor alimentar, a produção tem de ser planeada em função da procura sazonal, de matérias perecíveis e de perturbações de mercado. Os *Material Requirements Planning Systems* (MRP systems) tradicionais, baseados em regras estáticas, adaptam-se mal a estas condições, o que gera ruturas de stock e desperdício.

Esta tese propõe uma plataforma de planeamento que integra *Artificial Intelligence* (AI) e *Machine Learning* (ML) num MRP system existente e que fornece previsões de procura baseadas em dados e recomendações de planeamento da produção. Seis modelos de previsão são combinados através de *Inverse Error Weighting* numa única previsão com um intervalo de predição de 90%. A plataforma converte esta previsão em necessidades de matérias-primas através da decomposição recursiva da *Bill of Materials* (BOM), planeia o consumo de stock com lógica *First-Expired-First-Out* (FEFO) para reduzir o desperdício e permite simular o efeito de perturbações externas no plano de produção. Numa logica de *Industry 5.0*, os gestores podem ajustar manualmente o peso de cada modelo.

O protótipo foi testado com dados de 2014 a 2026 de uma empresa alimentar portuguesa, em três cenários. No cenário de referência de 2025, a previsão combinada para os artigos analisados manteve-se dentro ou próxima do intervalo de predição na maior parte do ano. Ao aplicar uma perturbação na procura baseada na guerra da Ucrânia de 2022, a plataforma atualizou automaticamente as necessidades de materiais e emitiu alertas. Com dados de 2026 não controlados, três dos seis modelos falharam e a plataforma continuou com os três restantes. O parceiro industrial considerou os resultados operacionalmente credíveis. Estes resultados descrevem o comportamento da plataforma, não o seu impacto no planeamento: o erro de previsão não foi medido em todo o catálogo, e o efeito no desperdício e na exatidão das necessidades de materiais não foi comparado com o plano real da empresa. O teste de 2026 mostrou também que as alterações nos identificadores de produto comprometem o histórico de procura de que os modelos dependem. A comparação entre compras planeadas e efetivas, pedida pelo parceiro, é o próximo passo.

Palavras-chave: Indústria 4.0 · Indústria 5.0 · *Artificial Intelligence* (AI) · *Machine Learning* (ML) · *Material Requirements Planning* (MRP)

DISCLOSURE OF USAGE OF GENERATIVE ARTIFICIAL INTELLIGENCE

The author, João António Simão Alves, acknowledges the use of the following generative artificial intelligence tools in the development of this thesis. These technologies were utilized under strict supervision, and all AI-generated outputs were critically reviewed and verified for accuracy. The author accepts full responsibility for the validity and originality of the content, confirming that the use of these tools aligns with all institutional policies and standards of academic integrity.¹

Generative AI tools used: ChatGPT-5.2, Gemini 3 Pro, Claude 4.5 Haiku, Claude 4.5 Opus, Claude 5 Sonnet, DeepL, and Paperpal.

Activities supported by Generative Artificial Intelligence Tools:²

Conceptualization		
☒ Idea generation	☒ Formulating research questions and hypotheses	☐ Hypothesis viability assessment
☒ Defining the research objective	☐ Feasibility assessment and risk evaluation	☐ Simulated debate and argument testing
Literature Review		
☒ Search and Discovery	☐ Concept Mapping and Systematization	☐ Gap Identification and Novelty Evaluation
☒ Literature summarization and synthesis	☐ Market/patent landscape analysis	☒ Cross-lingual literature comprehension
Methodology		
☐ Experimental Design Optimization	☐ Development of experimental or research protocols	☒ Methodological Instrument Selection

¹This declaration was generated using pdfL^AT_EX The L^AT_EX Project team, 1985 and the L^AT_EX package aidisclose Lourenço, 2025.

²This list extends the GAIDeT taxonomy Suchikova et al., 2025.

Software Development and Automation

- | | | |
|--|---|--|
| ☒ Code Generation | ☒ Debugging and Repair | ☐ Algorithm design |
| ☐ Refactoring and Optimization | ☐ Process automation | ☐ Code documentation and comment generation |

Data Management

- | | | |
|---|--|--|
| ☐ Data collection | ☐ Data analysis | ☐ Synthetic data generation |
| ☐ Validation | ☐ Quantitative Plotting and Charting | ☐ De-identification and anonymization support |
| ☐ Data cleaning | ☐ Reproducibility and rerun checks | ☐ Audio-to-text transcription and diarization |
| ☐ Data curation and organization | ☐ Data labeling and annotation assistance | |

Visuals and Multimedia

- | | | |
|---|--|--|
| ☒ Synthetic Asset Generation | ☐ Image Enhancement and Editing | ☐ Diagrammatic and Schematic Design |
|---|--|--|

Writing and Editing

- | | | |
|--|--|--|
| ☒ Drafting Text | ☐ Formulation of conclusions | ☐ Citation formatting and bibliography management |
| ☒ Polishing and Editing | ☒ Tone Adjustment | ☒ Press releases and outreach materials |
| ☐ Abstract and Executive Summary Generation | ☒ Translation | ☐ Title Generation |

Ethics Review

- | | | |
|--|---|--|
| ☐ Bias analysis and discrimination assessment | ☐ Ethical risk analysis | ☐ Data confidentiality monitoring |
| | ☐ Monitoring compliance with ethical standards | |

Quality Assurance

- | | | |
|--|--|--|
| ☐ Simulated Peer Review | ☒ Identification of limitations | ☐ Publication support |
| ☐ Consistency checking against field trends | ☐ Publication strategy and journal selection | |

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GLOSSARY

ARIMA	AutoRegressive Integrated Moving Average; a statistical time-series forecasting method that combines autoregression, differencing, and moving averages to model and predict future values based on past observations Box and Jenkins, 1970 (pp. 32, 36)
Chronos	A family of pretrained time series foundation models developed by Amazon, based on transformer language-model architectures adapted to tokenize and forecast numerical time series data in a zero-shot manner Ansari et al., 2024 (pp. vi, xii, xiii, 57, 61, 62, 66–68, 70, 83–85, 92)
Cobot	A collaborative robot designed to work safely alongside human operators in shared workspaces, assisting with repetitive, physically demanding, or hazardous tasks rather than replacing the human worker Demir et al., 2019 (pp. 18, 20)
Croston’s Method	A forecasting technique designed for intermittent demand, in which sales occur sporadically with many zero-demand periods. It separately estimates the average size of non-zero demand and the average interval between demand occurrences, combining both to produce a smoothed forecast Croston, 1972 (p. 94)
Digital Twin	A virtual, data-driven representation of a physical asset, process, or system, continuously updated with real-world data to enable simulation, monitoring, and predictive analysis, distinct from augmented or virtual reality technologies RAMI 4.0 - ISA n.d. (pp. 18–20, 24, 41)
E2E.pt	A commercial middleware platform (E2E.pt) that provides a unified, OpenAPI-based interface for integrating enterprise Business Software systems, such as ERP platforms, enabling standardized data exchange between shop-floor and business-planning systems (pp. 17, 27)

Granite	A family of foundation models developed by IBM, including compact time series variants (e.g. Granite TTM) designed for efficient, low-resource forecasting deployments Ekambaram et al., 2024 (pp. vi, xiii, 57, 61, 62, 66–68, 70, 83–85, 92)
IEC 61512	An international standard for batch control, extending the Hierarchy Levels axis of the RAMI 4.0 model alongside ISO/IEC 62264 RAMI 4.0 - ISA n.d. (p. 11)
IEC 62890	An international standard for life-cycle management of systems and products used in industrial-process measurement, control, and automation. It defines the Life Cycle & Value Stream axis of the RAMI 4.0 model RAMI 4.0 - ISA n.d. (p. 13)
Industry 3.0	The Third Industrial Revolution, characterized by the introduction of automation, electronic components, computers, and the first digital networks into industrial production processes, beginning in the second half of the 20th century Michelsen, 2020 (pp. xii, 12)
Industry 4.0	A paradigm of industrial digitalization characterized by the integration of cyber-physical systems, the Internet of Things (IoT), and real-time data exchange into manufacturing processes, enabling automated and interconnected production environments Barton et al., 2022 (pp. xii, xiv, 3, 5, 7–14, 17–21, 24, 31)
Industry 5.0	An evolution of Industry 4.0 that reintroduces human-centricity, sustainability, and resilience as core industrial objectives, positioning workers as collaborators with, rather than being replaced by, intelligent systems Bazel et al., 2024 (pp. vi, vii, xiv, 3–5, 11, 18–22, 24, 32, 37, 77)
ISO/IEC 62264	An international standard series for the integration of enterprise and control systems. Together with IEC 61512, it defines the Hierarchy Levels axis of the RAMI 4.0 model RAMI 4.0 - ISA n.d. (pp. 11, 12, 14)
LightGBM	A gradient boosting framework developed by Microsoft that uses histogram-based, leaf-wise tree growth to achieve faster training speed and lower memory usage than traditional gradient boosting implementations Ke et al., 2017 (pp. vi, xiii, 57, 62, 66, 67, 71)
Prophet	An open-source forecasting procedure, originally developed by Facebook, that decomposes a time series into trend, seasonality, and holiday effects using an additive regression model Taylor and Letham, 2018 (pp. vi, xiii, 32, 57, 61, 62, 66–68, 71)

Random Forest	An ensemble learning method that constructs multiple decision trees during training and outputs the average (regression) or majority vote (classification) of their individual predictions, improving accuracy and reducing overfitting Breiman, 2001 (pp. vi, xiii, 32, 57, 62, 66, 67, 72)
REST	Representational State Transfer; an architectural style for designing networked applications, in which systems communicate through stateless HTTP requests that operate on resources identified by URLs, commonly used to build web APIs (p. 49)
RFID	Radio-Frequency Identification; a wireless technology that uses radio waves to automatically identify and track tags attached to objects, commonly used for inventory tracking and traceability in supply chains Rodriguez et al., 2026 (p. 36)
Shop Floor	The physical area of a manufacturing facility where production activities, machinery operation, and material handling take place, typically corresponding to Levels 0-2 of the ISA-95 hierarchy and contrasted with the higher-level business planning and decision-making layers <i>ISA-95 Standard: Enterprise-Control System Integration</i> n.d. (pp. 16, 17, 24, 27, 33, 37, 38, 41, 44, 45, 54)
SKU	Stock Keeping Unit; a unique alphanumeric code assigned to a distinct product or item variant, used to track inventory levels, sales, and stock movements at the individual product level (pp. xiii, 59, 69, 73, 74, 78, 93)
XGBoost	eXtreme Gradient Boosting; a scalable and regularized implementation of gradient boosted decision trees, widely used for structured/tabular data prediction tasks Chen and Guestrin, 2016 (pp. vi, xiii, 32, 57, 62, 66, 67, 72, 83, 85, 92)

ACRONYMS

AI	Artificial Intelligence (<i>pp. vi, vii, 3–6, 8, 17–20, 22, 24, 27, 29–34, 37, 41, 92</i>)
AI-enhanced MRP system	Artificial Intelligence-enhanced Materials Requirements Planning (<i>pp. 4, 12, 14, 17, 35</i>)
API	Application Programming Interface (<i>pp. 46, 49</i>)
BOM	Bill of Materials (<i>pp. vi, vii, 28, 37, 38, 40, 41, 47, 54, 64, 76, 91–93</i>)
CPS	Cyber Physical Systems (<i>p. 7</i>)
EOQ	Economic Order Quantity (<i>p. 26</i>)
ERP system	Enterprise Resource Planning (<i>pp. 14–16, 23, 27, 33</i>)
FEFO	First-Expired-First-Out (<i>pp. vi, vii, xiii, 28, 37, 38, 40, 47, 64, 77, 78, 91–93</i>)
HMC	Human-Machine Collaboration (<i>p. 20</i>)
IIoT	Industrial Internet of Things (IIoT) (<i>p. 8</i>)
IoT	Internet of Things (<i>pp. 3, 6–8, 12, 20, 21, 27</i>)
ISA-95	International Society of Automation Standard 95 (<i>pp. xii, xiv, xvii, 5, 12, 14–17, 22, 27</i>)
IT	Information Technology (<i>p. 15</i>)
MES system	Manufacturing Execution System (<i>pp. 3, 5, 6, 14–17, 24, 27, 28, 33, 34</i>)
ML	Machine Learning (<i>pp. vi, vii, 4, 6, 17–19, 26, 29, 31–35, 92</i>)
MRP 4.0 systems	Material Requirements Planning 4.0 (<i>pp. 24, 25</i>)
MRP II systems	Material Requirements Planning system II (<i>p. 23</i>)

MRP system	Material Requirements Planning System (pp. vi , vii , 3–6 , 14 , 15 , 17 , 23–25 , 27 , 32–34 , 36 , 37 , 91 , 92)
OT	Operational Technology (p. 15)
PLC	Programmable Logic Controller (pp. 17 , 27)
POQ	Period Order Quantity (pp. 26 , 34)
RAMI 4.0	Reference Architecture Model Industry 4.0 (pp. xiv , xvi , 5 , 12 , 14 , 17 , 22 , 27)
RBAC	Role-Based Access Control (p. 58)
SCADA	Supervisory Control and Data Acquisition (pp. 17 , 27)
SOA	Service-Oriented Architecture (p. 11)

INTRODUCTION

1.1 Context and Motivation

Nowadays, it is globally recognized that the food industry holds one of the most essential roles in the global economy. This sector holds significant responsibilities, such as providing the basic needs of the population, ensuring food security, environmental sustainability and driving economic growth (*How has the food industry changed over the last decade?* - *Bio.Tech.Foods*. N.d.). However, given the markets expansion and environmental concerns, the question remains: How resilient is the current industry model, and can it continue to meet these global demands in the long term?

Ensuring a safe, sufficient, and sustainable food supply in the present moment is harder to manage. Beyond the satisfaction of consumer needs, the industry also deals with how that food is produced, transported and consumed, while keeping the marked competitiveness. But a thorough examination reveals that the industry currently faces serious obstacles, from geopolitical and logistical instabilities to critical resource and safety deficits, creating limitations. These limitations are making it harder to keep operations efficient and to respond quickly to what consumers need today, and the demand keeps constantly changing.

Some of the key issues that require immediate attention are:

- **Fragile supply chains:**

The present moment is characterized by series of events that have exposed the inherent fragility of contemporary supply chains. Events like the COVID-19 pandemic and international conflicts, have exposed how the actual structure is vulnerable, resulting in a series of adverse outcomes, like stock-outs, transportation bottlenecks and surging costs in logistics and raw materials. The 2022 Russian-Ukraine war had significant impact on the cost of essential ingredients ([Sharma, 2025](#)). More recently, the disruption to shipping through the Strait of Hormuz, following the escalation of conflict in the Middle East, has been identified by the Food and Agriculture Organization as a systemic shock to global agrifood systems, driving up energy, fertilizer, and transport costs across the food supply chain ([Food and Agriculture Organization of the United Nations, 2026](#)).

- **Food waste:**

It is estimated that around 1.3 billion tons of food are lost or wasted each year. Besides the huge economic problem it creates, it also brings environment burdens in terms of resources and greenhouse gas emissions ([Roy et al., 2023](#)).

- **Labor shortages:**

Global challenge has emerged for the food manufacturing and processing sectors. These sectors are struggling to recruit and retain enough skilled workers. Working conditions and shifts in labor markets are affecting the efficiency, productivity and quality of services. According to the FPSC Labor Market Information Survey, a survey revealed approximately 70% of firms have experienced labor shortages ([Sharma, 2025](#)).

- **Concerns regarding food quality & safety:**

Another serious issue, seen in the food sector, is the lack of proper investment on resources for inspection and quality control. It is imperative to acknowledge that these challenges amplify the potential for foodborne disease outbreaks. In the United States, as example, was concomitant with an increase in contamination incidents in 2024 ([Semuels, 2025](#)).

These persistent challenges have given rise to an urgent need for innovation. This innovation must find new ways to strengthen its supply chains, reduce waste, adapt to workforce deficits, and guarantee product integrity. This study emerged from the recognition of the need to rethink the way industry deals with these problems, especially by integrating smart technologies to help deal with challenges and limitations, such as to respond effectively to the market demand.

1.2 Problem and Objectives

The food industry currently faces some challenges such as supply chain disruptions, labor shortages, unexpected changes in demand and stringent quality standards. The traditional production planning and control systems are facing challenges in their ability to respond effectively to the demand and challenges. These systems, including MRP system for inventory or stock management and supply management are unable to adapt in real-time or effectively extract information from current data. All these factors result in a range of problems, like waste, delays, stock problems, and overproduction.

The goal of this thesis is to develop an intelligent decision-support tool that works alongside existing MRP system to extract and analyze historical sales, inventory, and production data, improving real-time decision-making in production planning. This tool aims to improve the effectiveness of production planning and execution processes, thus facilitating a more agile and informed response in the face of the challenges above described.

1.3 Approach and Contributions

As previously mentioned, the present thesis is driven by the pressing need for enhanced efficiency and resilience in the food industry. To answer this pressing need, is proposed a strategic approach to integrate artificial intelligence into traditional production planning systems combining the concepts of Industry 4.0 and Industry 5.0. This work recognizes the transformative impact of digital technologies, such as the Internet of Things (IoT), AI, and cloud computing are fundamental characteristics of Industry 4.0. It also aligns with the human-centered, sustainable and resilient, Industry 5.0's characteristics.

Rather than replacing the existing systems, the objective is to enhance them with predictive and adaptive capabilities to address the principal food industry challenges. The strategic integration of AI capabilities into traditional MRP system and Manufacturing Execution System (MES system) systems aims to enhance the decision-making and operational agility within the food supply chain. This approach not only take advantage of advanced technological capabilities, but also expands the human knowledge, particularly in the context of addressing critical societal challenges, including reduction of food waste, and improving supply chain resilience.

The integration of ML, and AI models into the existing MRP system, aims to meet the following objectives:

- **Improve the prediction of production and raw materials needs:**

Utilization of historical data, consumption trends, seasons and external data collected in real time by all components, and devices within the system, leads to a more accurate demand forecasting. Anticipating potential market fluctuations is a strategy to avoid disruptions, reduce stress and optimize resource allocations across the supply chain.

- **Optimize inventory and stock levels:**

According to the data it keeps collecting and the continuous alterations from Artificial Intelligence-enhanced Materials Requirements Planning (AI-enhanced MRP system) it recalculates and adjusts the stock levels. This results in reduction of the stock-outs or overstocking and preventing losses from possible expired products.

- **Minimize Food waste and Production Losses:**

AI-enhanced MRP system prioritizes the products with imminent expiration, recalibrating the production plans, contributing to reducing the food waste. It is expected intellectual scheduling will ensure the products are used in time, addressing one of the most significant challenges in food manufacturing.

- **Improved responsiveness and resilience to disruptions:**

The utilization of artificial intelligence in decision-making will contribute to recommending adjustments to lead with sudden demands or failures in real time.

With all these points, the system's design explicitly considers sustainability by aiming to reduce food waste and inventory management, resulting in lower emissions through optimized logistics.

The contributions of the present thesis are both theoretical and practical:

- **Innovation:**

The study presents a framework that incorporates AI capabilities into established MRP system.

- **Qualifiable Operational Impact:**

The expected advancements in operational performance are to be empirically validated via rigorous case studies or simulations, with a focus on forecast accuracy, reduction of lead times and responsiveness to unexpected fluctuations in demand or supply.

- **Human-Centered with Artificial Intelligence decision-making support:**

The thesis proposes a collaboration between human operators and AI systems within an industrial context, contributing to the conceptualization of Industry 5.0.

- **Sustainability and Resilience Assessment:**

Will be provided, including qualitative and quantitative evaluations of the system's capacity to contribute to the reduction of food waste and the enhancement of the speed and effectiveness of recovery from disruptions. These evaluations will contribute to research on sustainable and resilient manufacturing systems.

1.4 Document Organization

Chapter 1 presents the context and motivation behind this research, highlighting the main challenges faced by the food production industry. The specific problems that the thesis aims to solve are defined, and the objectives and contributions of the proposed solution, based on the integration of AI into MRP system are described.

Chapter 2 presents the state of the art, including a literature review and context on the evolution of the industrial landscape. The transition from Industry 4.0 to Industry 5.0 is addressed, along with a detailed exploration of frameworks such as RAMI 4.0 and ISA-95. The functionalities, architecture, advantages and limitations of MRP system, and MES system systems are also addressed, and the role and impact of Artificial Intelligence and Machine Learning in modern industrial environments are examined.

Chapter 3 translates the identified problem into a concrete conceptual design. The functional requirements driving the proposed solution are defined, followed by a use case analysis that models how these requirements translate into user interactions across the system's three operational roles — the Office Room, DockStation, and WorkStation. The chapter concludes with the conceptual architecture that satisfies these requirements, organized into four progressive layers, from foundational infrastructure to user interaction.

Chapter 4 describes the technical implementation of the proposed platform. It details the three-tire system architecture, the sequence of system communication flows that drive the forecasting, refinement, and planning lifecycle, and the technologies selected to support the system. The chapter also defines the current implementation scope of the functional prototype, distinguishing between the components that were fully coded and validated and those that remain high-fidelity conceptual blueprints for future development phases.

Chapter 5 presents the validation and results obtained from testing the prototype across three distinct operational scenarios: a controlled baseline performance analysis using historical data, an external shock stress test simulating geopolitical crises, and a live, real-time verification using uncontrolled production data.

Chapter 6 presents the conclusions of this thesis, summarizing the study's contributions and empirical findings, discussing the industrial challenges and structural limitations identified during validation, and outlining the future implementation phases and enhancements identified to guide the system's evolution into a production-ready framework.

STATE OF THE ART

2.1 Contextualization of the Food Industry

In recent years, the food industry has suffered from the following problems: the need to produce, respond to consumer preferences and there are also labor and social issues. Finally, one of the most important factors in this sector is the effectiveness of response to the globalization of supply chains ([Kittipanya-ngam and Tan, 2020](#)).

To answer these problems, the food industry is trying to keep up with the evolution of technology and other industries, for example the integration of Internet of Things (IoT) in agriculture. The adaptation of digital and smart technologies, such as Material Requirements Planning System (MRP system), Manufacturing Execution System (MES system), Artificial Intelligence (AI), and Machine Learning (ML) will allow the food industry to improve the operations, routines and processes, and innovate products ([Rejeb et al., 2022](#)). In this way, the companies can give a better response and adapt to unpredictable changes in the market, allowing them to evolve in this sector.

2.2 Industry 4.0

2.2.1 The Evolution of Industry Leading Up to Industry 4.0

Figure 2.1 shows how industry has changed because of several revolutions since the end of the 18th century. The First Industrial Revolution was marked using steam engines, which made it possible to move from handicraft to factory production. At the end of the 19th century, the Second Industrial Revolution brought electricity. With this new revolution, some of the steam engines were replaced by electricity and enabled faster and more efficient production. The Third Industrial Revolution, started at the beginning of the 20th century, did bring automation by the introduction of electronic components, informatic or computers and the first digital networks ([Michelsen, 2020](#)).

Currently we are experiencing the Fourth Industrial Revolution, and more recently the Fifth Industrial Revolution is starting to appear to deal with some complex challenges and limitations of the Fourth Industrial Revolution. This last Revolution will be discussed later in Section 2.3.

The fourth Industrial Revolution, or better known as Industry 4.0, is characterized by the digital transformation of manufacturing and production systems. Industry 4.0 appears from the advancement of digital technology with production systems, in Germany in 2011, and brought the emphasis of smart environments, Cyber Physical Systems (CPS), and IoT (Michelsen, 2020). These new principles have led to interoperability, virtualization, decentralization, real-time adaptation and flexible systems.

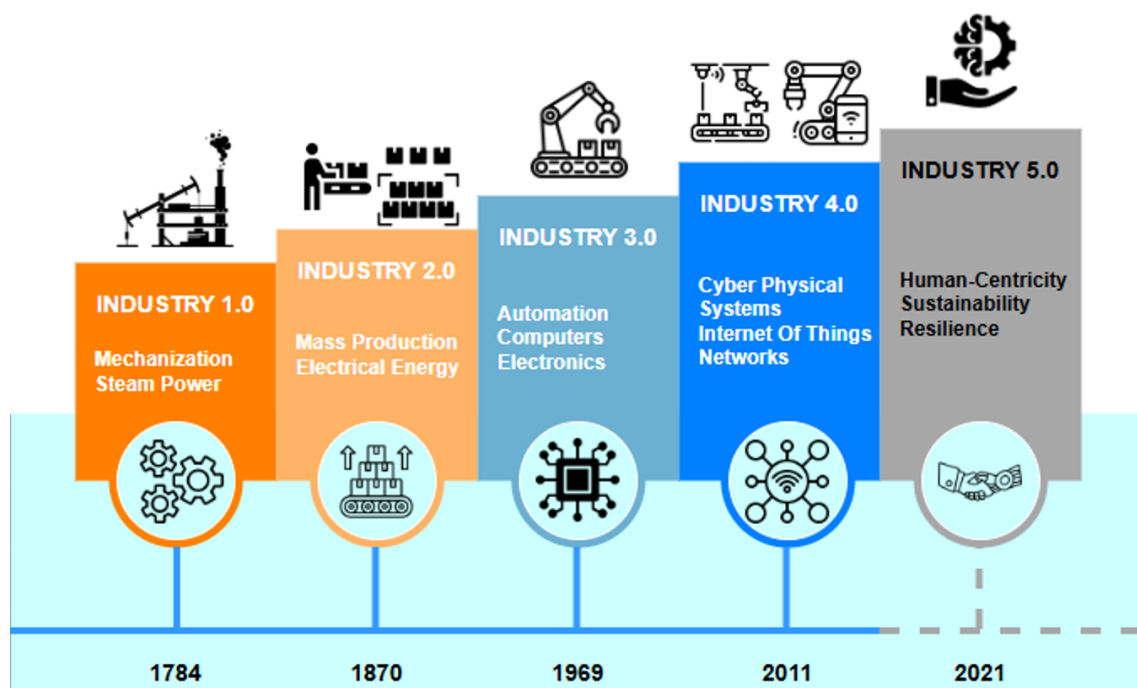


Figure 2.1: The Evolution of Industrial Revolution. Adapted. ((PDF) *INDUSTRY 4.0: THE FOURTH INDUSTRIAL REVOLUTION AND HOW IT RELATES TO THE APPLICATION OF INTERNET OF THINGS (IoT)* n.d.)

Table 2.1: Technologies of Industry 4.0 (Barton et al., 2022).

Technology	Description
Internet of Things (IoT)	The group of sensors, actuators and devices interconnected with machines, products, processes and the supply chain generate a heavy volume of data in real time. The IIoT specializes in industrial applications, and requires greater reliability, low latency, and more robust security.
Industrial Internet of Things (IIoT)	
Cyber-Physical Systems (CPS)	Integration of the physical world of machines with digital world of information. With this integration, the processes are interconnected and controlled by computer intelligence, which allows us to create more transparent systems that can adapt to emerging challenges.
Big Data & Analytics	Collecting data generated by IoT and IIoT requires analysis and processing techniques. The IoT and IIoT generate a heavy volume of data, so with these techniques it is possible to detect patterns and gain meaningful insights more effectively and accurately.
Cloud Computing	This pillar is responsible for storing the massive amount of data generated by IoT and IIoT. These platforms provide advanced analytical services, making them more accessible to companies.
Artificial Intelligence (AI)	Systems with continuous learning and improving from data.
Machine Learning (ML)	Subdomain of AI is responsible for analyze of data, detection of patterns and prediction to optimize operations.
Digital Twin	Virtual representations of reality or augmented reality of physical systems and/or products.
Additive Manufacturing (3D Printing)	Custom manufacturing of parts and prototypes.
Cybersecurity	Protecting networks, systems and data is a preventive measure against possible cyber threats.

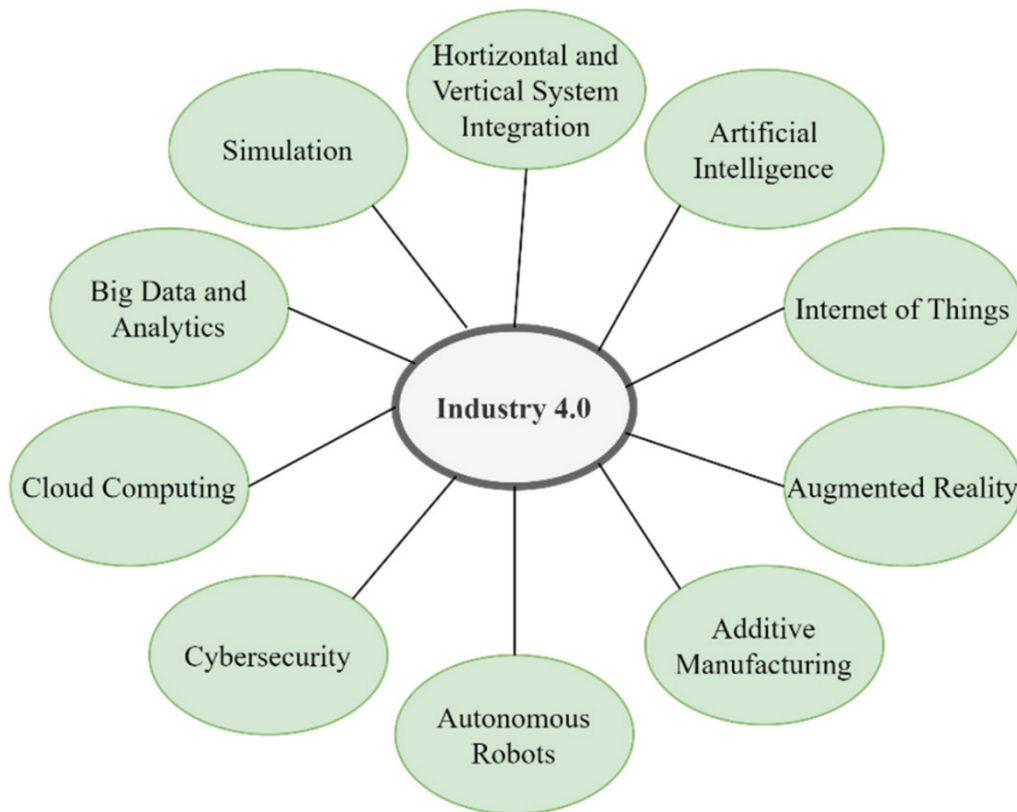


Figure 2.2: Industry 4.0 scheme of technologies adopted (Barton et al., 2022)

2.2.1.1 Advantages, Challenges and Limitations of Industry 4.0

The adoption of the Technologies of Industry 4.0 (Table 2.1), summarized in Figure 2.2, has introduced a new method of manufacturing control, through “smart” capabilities.

The main advantages identified were (Sony, 2020):

- Real-time monitoring and reporting of operational, product and system data.
- These technologies enable more detailed, faster and more accurate decision making.
- Predicting possible problems or necessary adjustments in production planning for the efficient management of operations.
- Reducing resources waste, downtime and operating costs.
- Increasing production efficiency by analyzing empirical data to identify opportunities for improvement.

However, not everything is perfect and wonderful. The new technologies that have emerged from Industry 4.0 have given rise to new barriers; the main ones are:

- **Implementation Costs:** The transition to Industry 4.0 requires an expensive investment for the integration of new technologies, systems, infrastructures and software. Unfortunately for small and medium-sized companies, this might be unavailable to them, due to their limited resources (Islam et al., 2025).
- **Cybersecurity risks:** With the increased connectivity of systems, platforms, devices and software comes a huge risk of cyber threats. So, it's necessary to protect interconnected systems and platforms from industrial espionage, cyber-attacks and malware, as well ensuring the protection of sensitive operational data and data privacy (Islam et al., 2025; Thames and Schaefer, 2017).
- **Need for new knowledge and skills:** The implementation of new Technologies from Industry 4.0 is causing problems in the selection of operators to operate these Technologies. It requires new digital and technical skills, which are lacking in many workers. Extensive re-skilling and training programs need to be carried out to get workers who can operate these new advanced technologies effectively (Nunes et al., 2022; Sony, 2020).
- **Data Management and Interoperability:** The new Technologies generate an enormous amount of data, known as Big Data. It is extremely important know how to effectively manage this data, as it is necessary to guarantee its interpretation, analysis, quality, accuracy, security and interoperability between different systems, machines, devices and supplier platforms. Given to the huge diversity of systems, platforms and devices, it is hard to have industrial standardization, which makes it more difficult to integrate digital systems (Islam et al., 2025).
- **Ethical issues:** The usage of Artificial Intelligence (one of biggest new technologies of Industry 4.0), is generating dilemmas related to transparency, accountability and data privacy in automated decision-making processes. It raises the issue of extreme technological dependence on digital infrastructure, which is subject to cybersecurity attacks and system failures (Islam et al., 2025; Thames and Schaefer, 2017).
- **Environmental Sustainability:** While Industry 4.0 technologies can improve resource efficiency, their large-scale adoption also introduces new environmental costs, including higher energy demand from data centers and computing infrastructure, and the accumulation of electronic waste from short-lived devices and hardware (Zhao et al., 2025).

Given all these limitations and challenges, it is possible to say that the implementation of Industry 4.0 is complex and generates new problems. The list of limitations and potential social impacts has led to an analysis of improvements and the development of new concept, Industry 5.0. This new concept will be addressed later in this study to better understand how it will try solving the limitations mentioned.

2.2.2 RAMI 4.0

With the introduction of Industry 4.0, there was a need to create a framework to support and structure the implementation of Industry 4.0. With the cooperation between the ZVEI (German Electrical and Electronic Manufacturers) and the government initiative Industry 4.0 Platform resulted in the development of three-dimensional reference architecture model.

This new Model operates as a map or coordinate system for Industry 4.0, based on existing standards (ISO/IEC 62264 and IEC 61512). The objective is to provide a common language and unified structure for Industry 4.0, as different industrial components, software and processes interact with each other. The interaction between components is carried out by a Service-Oriented Architecture (SOA), using network communication protocols to make services available to each other, emphasizing the independence of suppliers and specific technologies, as illustrated in Figure 2.3 (RAMI 4.0 - ISA n.d.; “The Internet of Things and Services Graphics © Bosch Rexroth AG” n.d.).

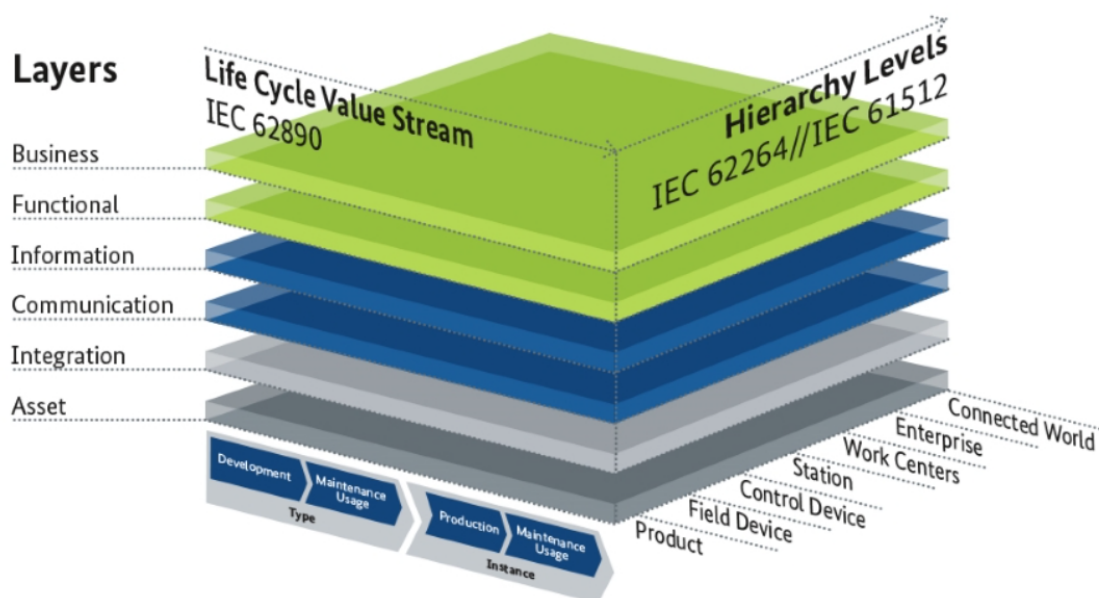


Figure 2.3: Three-Dimensional coordinate system showing the most important aspects of Industry 4.0 (RAMI 4.0 - ISA n.d.)

Its Three-Dimensional coordinate system is made up of three axes:

- “Hierarchy Levels” – Y-axis

Axis positioned horizontally to the right in Three-Dimensional coordinate system. Derived from ISO/IEC 62264, an international standard for the integration of enterprise control systems, known as International Society of Automation Standard 95 (ISA-95), this axis represents the levels of the automation systems industry. To meet the requirements of Industry 4.0, Reference Architecture Model Industry 4.0 (RAMI 4.0) complements the original ISA-95 structure to include work pieces, labeled as “Product” and the connection to the IoT, labeled as “Connected World”, as detailed in Table 2.2 (*RAMI 4.0 - ISA n.d.*).

Table 2.2: “Hierarchy Levels” shown by the Y-axis of the RAMI 4.0 Model.

Level	Description
Product	Individual Smart product or components (e.g. Sensors/Machines).
Field Devices	Devices such as sensors, actuators, controllers to interact with the Product.
Control Device	PLCs (Programmable Logic Controllers) and Devices for controlling processes.
Station	Group of Devices that form a workstation in manufacturing (e.g. Robots arms).
Work Centers	Group of stations working together in production.
Enterprise	The Company itself or the Company’s Industrial System.
Connected World	Integration with external ecosystems (e.g. cloud platforms, suppliers).

Regarding to Hierarchy Levels (Y-axis), the present thesis’s Artificial Intelligence-enhanced Materials Requirements Planning (AI-enhanced MRP system) focuses on perspective of planning perspective at the Enterprise Level, while extending into the Connected World Level, where company interacts with the external ecosystems to respond to supply chain disruptions and real-time demand changes. Figure 2.4 illustrates how this hierarchical structure differs between Industry 3.0 and Industry 4.0.

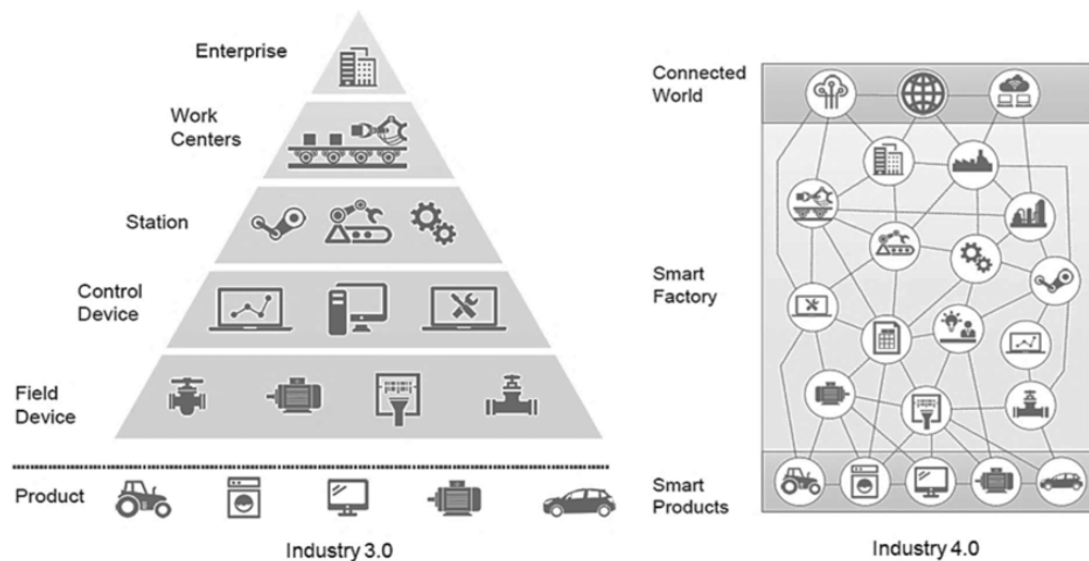


Figure 2.4: Comparison of the interaction between “Hierarchy Levels” in the Industry 3.0 and Industry 4.0 (Frère et al., 2018)

- **“Life Cycle & Value Stream” – X-axis**

This axis is positioned horizontally on the left side. This present axis is based on the IEC 62890 standard and maps the complete life cycle of an object (e.g. product, machine, system) and the associated flows, from its creation till recycling or disposal.

In this axis has also a fundamental distinction between:

- **“Type”**: Represents the design, prototype or model of the object, including phases such a Development and Maintenance.
- **“Instance”**: Represents the physical and individual copy of the “Type”, including the Production and Maintenance phases (of the “Instance”) (RAMI 4.0 - ISA n.d.).

- **“Layers” – Z-axis**

This axis is positioned vertically and shows the description of the Industry 4.0 component into its proprieties and representations, according to Model. This enables a step-by-step transition from the physical to the digital world of Industry 4.0 (RAMI 4.0 - ISA n.d.).

The axis in question has six flowing layers, from bottom to top, as shown in Table 2.3:

Table 2.3: “Layers” shown by the Z-axis of the RAMI 4.0 Model.

Layer	Description
Asset	The physical object in the real world (e.g. Component, Machine).
Integration	The interface between the physical and digital worlds (e.g. Sensors, RFID).
Communication	Standardization of communication and data access (e.g. Protocols, Networks).
Information	The relevant data and its processing (e.g. Production data, events, rules).
Functional	The functions that the assets perform, its horizontal integration platform (e.g. MES, MRP/ERP systems).
Business	The organizational perspective, business processes, rules.

The RAMI 4.0 Model can be seen as a map of Industry 4.0 solutions, since it has given rise to different user perspectives. RAMI 4.0 has provided a common framework for standards and use cases for new Industry 4.0 Technologies, where the requirements of sectors from manufacturing automation and mechanical till process engineering can be addressed through industry associations and standardization committees.

The Model makes it possible to identify the patterns applied and determine new ones, if necessary, to guarantee communication and integration without failures (*RAMI 4.0 - ISA n.d.*).

While AI-enhanced MRP system resides on Functional Layer working together with other systems — such as MES system — by providing support to real-time decision-making to operators, it contributes as well, on Information Layer, processing relevant data, events, rules using Machine Learning and transforming these insights into enhanced capabilities withing the Functional Layer.

2.2.3 ISA-95

The ISA-95 standard (also known as ISO/IEC 62264 in RAMI 4.0 Model) is an international standard developed by the International Society of Automation (ISA). This standard has proved crucial in the implementation of systems in the production environment, as it facilitates integration between business systems (e.g. MRP system or Enterprise Resource Planning (ERP system)) and production control and operation systems (e.g. MES system and control systems) (*ISA-95 framework and layers | Siemens Software n.d.*).

ISA-95 standard provides standardized terminology and structured models (hierarchical models, object models, activity models) between different levels, summarized in Table 2.4. With this standard, it is possible to reduce the complexity, costs, risk and errors associated with implementation of Information Technology (IT) and Operational Technology (OT) systems (*ISA-95 Standard: Enterprise-Control System Integration* n.d.).

Table 2.4: Five levels of ISA-95 Model (*ISA-95 Standard: Enterprise-Control System Integration* n.d.).

Level	Designation	Description
0	Physical production processes.	Represents the physical processes of production (e.g. Machines, Equipment).
1	Sensing and manipulating the production process.	Defines the activities of direct interaction with the physical process (e.g. read sensors values, act on valves and devices)
2	Monitoring and supervising control	Real-Time process supervision and control activities. (includes systems such as Programmable Logic Controllers (PLCs) and Supervisory Control and Data Acquisition (SCADA)
3	Manufacturing operations management	Crucial level. Defines the workflow activities needed to manage the production operations and transformation of raw materials into products. (The main system operating at this level is the MES system)
4	Business planning and logistics	Defines the higher-level business activities required to manage the production operation. (The primary system operating in this level is ERP system or MRP system)

Figure 2.5 illustrates how these five levels form a hierarchical structure between company management and physical production.

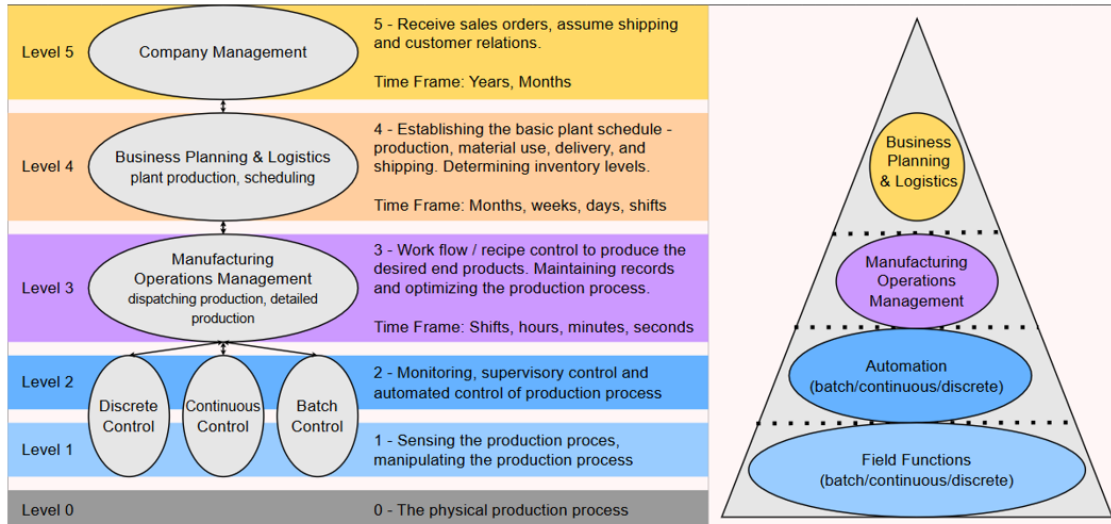


Figure 2.5: Hierarchical Structure of ISA-95 Model: Levels of Integration between Company and Production. Adapted. (Frère et al., 2018)

The ISA-95 standard contributes significantly to the detailed definition of the interface between information exchange system at Level 4 (e.g. ERP system) and production operations systems at Level 3 (e.g. MES system). This connection is crucial factor to aligning the more detailed and timely decisions of high-level planning with actual execution on the Shop Floor.

To achieve the integration, ISA-95 standard proposes:

- **Objects and Attributes (Parts 1, 2 and 4):**

Defines common information objects and their attributes, which are required for the exchange of information between Levels 3 (Manufacturing Operations Management) and 4 (Enterprise-Level Applications)

- **Activity Models of Manufacturing Operations Management Integration (Part 3):**

These models define the activity and information flow (e.g. production management, maintenance, quality and stock) to facilitate communication between various Levels of the production processes.

- **Business-to-Manufacturing Transactions (Part 5):**

Defines the communication between Levels 3 (Manufacturing Operations Management) and 4 (Enterprise-Level Applications) systems.

It is possible to see the complementary between the RAMI 4.0 Model and ISA-95 standard, where RAMI 4.0 Model implements the functional hierarchy defined by ISA-95 standard as part of its vertical axis and expand it. Many companies rely on this complementary, as it corresponds to Industry 4.0's objectives, such as flexibility, modularity and interoperability.

However, not everything is perfect: Industry 4.0 is facing the growing complexity of systems and the dependence of intelligent infrastructure (e.g. AI, and ML), which leads to exclusion of the Human as being the center of decision-making.

Within the ISA-95 hierarchy, AI-enhanced MRP system acts as real-time bridge between Level 4 (MRP system) and Level 3 (MES system). By applying Machine Learning to the activity models and information objects defined by the standard, the system transforms static planning into a predictive decision-support tool.

In practice, this thesis's AI-enhanced MRP system operates at Level 4 of the ISA-95 hierarchy, implemented through the CentralWebservice and Analytical Computation Engine described in Chapter 4. Compliance with the standard's data structure is achieved indirectly: the system's Storage Infrastructure ingests Level 3 Shop Floor data — such as active production schedules and stock movements — that would, in a full industrial deployment, be supplied by a MES system through middleware platforms such as E2E.pt (see Section 2.5). The prototype does not implement direct communication with Level 0/1/2 field devices (Programmable Logic Controller (PLC), Supervisory Control and Data Acquisition (SCADA)); this integration boundary is intentional as the platform is designed to consume standardized, aggregate Level 3 outputs rather than raw device signals, consistent with ISA-95's own principle that each level should interface only with its immediate neighbors.

2.3 Industry 5.0

The analysis of Industry 4.0 integration revealed significant limitations that led to a new evolution: Industry 5.0. The new evolution did not aim to replace the Industry 4.0, but to complement some of Industry 4.0 weaknesses. While Industry 4.0 was marked by the advance of technology and the search for efficiency, Industry 5.0 reinforces the importance of some values, with the goal to make the industry more Human-Centric, Sustainable and Resilient. With this new point of view, it allows the alignment with social and environmental objectives (Grabowska et al., 2022).

2.3.1 Objectives and Developments

Industry 5.0 represents a new development for industry, characterized by the collaboration between human beings (human knowledge and human intelligence) and smart and autonomous systems. The goal is to generate more prosperity, not only the possibility of generating more jobs and economic growth, but also to achieve broader social objectives, within the limits of the raw materials the planet can produce and prioritize the well-being of operators as the center of production process.

Industry 5.0 aims to align technological progress with social needs and ecological responsibility, recognizing the power of industry shape a fairer and more lasting future, with the introduction of new pillars: Human-Centric, Sustainability and Resilience (Grabowska et al., 2022).

Human-Centric

As analyzed in the Industry 4.0's study, which places importance on automation and reducing operators' costs, the transition to Industry 5.0 has made a notable difference by prioritizing the human being, well-being needs, creativity, skills and human ethics in industrial systems.

Industry 5.0 reinforces the use of new technologies, collaborative robots (Cobot), Smart Systems (e.g. AI and ML), Digital Twin, and augmented and virtual reality technologies, but in different approaches, instead of replacing the human skills, it will be used as tools to support tasks. Cobot, collaborative robots, for instance, are specifically developed to work alongside humans in processes, assisting in their tasks instead of replacing them. In this way, the technologies will be responsible for repetitive, dangerous or physical demanding tasks, while the Operator will be responsible for activities with greater added value, such as innovation, solving complex problems, customization and quality control of materials and/or products (Demir et al., 2019; Grabowska et al., 2022).

Sustainability

Industry 5.0 imposes sustainability as a pillar, allowing proactive integration of resource efficiency, optimized energy management, promotion of the circular economy and social responsibility as central points of industrial practice. In this way it is possible to align industrial progress with environmental and respect for the planet's limit, with raw material generation.

Circular economy is a strategy to encourage the reuse, remanufacturing and recycling of materials and products. This strategy allows for a reduction in the amount of waste generated and a reduction in the use of raw materials.

Energy efficiency is also a new focus, with the adoption of renewable energy, the development of energy efficiency technologies and systems and the integration of smart systems with the aim of monitoring and optimizing energy consumption (Akundi et al., 2022; Grabowska et al., 2022).

Resilience

Industry 5.0 aims to respond to and adapt to disruptions, uncertainties or crises that may occur in the industrial world (e.g. COVID-19). This gives rise to the third pillar proposed by Industry 5.0, Resilience, defined by the industrial system's ability to respond and adapt to unforeseen circumstances (Akundi et al., 2022; Grabowska et al., 2022).

To achieve the new goal, it is necessary to include some approaches, such as:

- Flexible and Reconfigure Manufacturing Systems: the integration of smart technologies such as AI and ML, make it possible to make real-time adjustments to production processes, thus enabling rapid adaptation to fluctuations in demand or resource availability.
- Robust and Agile Supply Chains: the implementation of Digital Twin technology makes it possible to create virtual systems or scenarios to simulate and analyze. This approach makes it possible to anticipate possible disturbances.
- Crisis Adaptation Capacity: this consists of the industry's ability to operate as a critical infrastructure in times of crisis, a crucial aspect to produce essential goods.

Table 2.5 summarizes the key difference between Industry 4.0 and Industry 5.0 across these dimensions.

Table 2.5: Comparison between Industry 4.0 and Industry 5.0 (Bazel et al., 2024).

Characteristic Dimension	Industry 4.0	Industry 5.0
Focus	Digitalization, Automation, Connectivity, Efficiency	Human-Machine, Human-Centric, Sustainability, Resilience
Objective	Optimizing productivity and Efficiency	Prosperity (Human, Social, Environmental, Economic), Well-being
Human Role	Viewed as a cost/factor to optimize or replace	Central, Collaborative, Creative, Focused, Skills Development, Well-being
Technology Role	Allow automation and efficiency	Allow collaboration, sustainability and resilience; used as tool to human's services
Sustainability	Secondary	Core pillar
Resilience	Not a focus	Core pillar
Production Type	Optimized for Mass production and customization	Optimized for mass personalization, agile and flexible production
Approach	Technology-oriented	Value-oriented (Human, Sustainability, Resilience)
Key Technologies	IoT, Cloud, Big Data, Automation, AI (Optimization)	Cobot, AI (Collaborative, Explainable), Digital Twin, IoT
Main challenges	Human-Centric, Sustainability, Resilience	HMC integration, Measurement of New Values, AI Ethics, Industry 5.0 skill requirements, Transition Costs

2.3.2 Advantages, Challenges and Limitations

The transition from Industry 4.0 to Industry 5.0 has resulted in a more balanced model that combines technological advances with human and sustainable values. Among the benefits that have stood out are (Akundi et al., 2022; Rame et al., 2024):

- Human-Centric: Valuing collaboration between smart systems and operators, rather than replacing them in tasks. In this way it is possible to promote well-being, motivation and innovation on the part of operators.
- Mass customization: Allows customized production on large scale. It aims to align products with customer preferences and increase customer satisfaction.
- Environmental sustainability: an essential pillar, integrating circular economy practices, energy efficiency and environmental monitoring with IoT technologies and data analysis.
- Resilience and agility: Collaboration and unity between smart systems and operators ensures quick and efficient responses to crises and disruptions, reinforcing industrial flexibility.
- Ethics, Transparency and Accountability: The humanized approach reinforces traceability, while ethical processes and the trust of consumers and employees reinforce the reputation of organizations.

However, the integration of Industry 5.0 is also subject to challenges and limitations:

- Technological acceptance: a crucial aspect, as it implies adapting to and accepting collaborative technologies. It requires transparency, explanation and human involvement in the development of solutions.
- Training and qualification: The transition requires operators to have digital and human skills, which implies continuous investment in training and education.
- Cybersecurity and privacy: The increase in Human-Machine collaboration and the use of sensitive data leads to the introduction of new vulnerabilities.
- Implementation costs: The transition requires investments in technologies, infrastructures and organizational restructuring. These investments can be high, especially to small and medium-sized companies, this makes adoption more difficult.
- Lack of standardization and clarity: New framework standards need to be established to guide the development of Industry 5.0 (Akundi et al., 2022).

In summary, Industry 5.0 is in a consolidation phase, requiring technical, human and political efforts to overcome structural obstacles. It is essential to analyze and acknowledge these limitations to guarantee the effective adoption and sustainable success of the new paradigm.

This thesis's own conceptual architecture (Chapter 3) represents a modest contribution toward addressing this gap: by explicitly mapping its layered design — the Support Infrastructure, Knowledge, and System Functionalities layers — onto the established ISA-95 and RAMI 4.0 hierarchies, the proposed system offers a concrete, standards-aligned reference pattern for integrating AI-enhanced decision-support tools into existing Level 4 planning system, specifically within the food manufacturing domain, where such reference architectures remain scarce.

2.4 Material Requirements Planning System

2.4.1 The Evolution

Material Requirements Planning System (MRP system) emerged in the 1960s as a computer solution for planning material requirements in manufacturing. Initially, this system was designed to only calculate the quantities and order dates for raw materials based on expected demand and product structure (such as a list of materials needed for one product). Over the years, the system evolved, resulting in the acquisition of new functionalities such as automated inventory replenishment, Master Production Scheduling, MPS, (Dolgui and Prodhon, 2007; Thevenin et al., n.d.[a]).

During the 1970s, MRP system was consolidated and began to be implemented in companies that accepted the risk of using the system. However, this system had some limitations. The most alarming limitation was the following one: The system ignored the limitations of machines and labor, in other words, it considered that there was infinite production capacity (Dolgui and Prodhon, 2007).

During the 1980s, the system evolved with the aim of surpassing the alarming limitation recorded in the 1970s. This gave rise to Material Requirements Planning system II (MRP II systems). This new version included the processing of materials planning and the planning of manufacturing resources (in other words, production capacity, labor and financial integration). This version of the system moved closer to an integrated factory management system, with the introduction of a closed planning cycle, incorporating capacity planning and feeding execution information back into the production plan (Dolgui and Prodhon, 2007).

During the 1990s, the MRP II systems were extended to corporate management, giving rise to Enterprise Resource Planning (ERP system). This new system preserves all functionalities of the MRP II systems, and includes the business management modules (Dolgui and Prodhon, 2007).

As a result, the ERP system has the following functions:

- Production Planning
- Stock/Inventory Management
- Purchasing Management
- Sales Management
- Financial Management

The MRP system have changed to meet the demands proposed by Industry 4.0, with the emergence of Material Requirements Planning 4.0 (MRP 4.0 systems) associated with Digital Twin. The new versions aim to overcome the limitations of traditional MRP system by introducing real-time data, synchronous planning and advanced algorithms. With the integration of Big Data Analytics and Artificial Intelligence, MRP 4.0 systems take on a dynamic and adaptive role, where production plans can be adjusted based on continuous feedback from the systems and operators on the Shop Floor.

Industry 4.0 has brought about a significant transformation in MRP system, and MES system which are now operating collaboratively in continuous optimization cycle, instead of working in isolation. MRP system is now a real-time control system, not just a periodic planning tool (Javaid et al., 2023).

However, this evolution still presents challenges. Effective data management becomes critical, since these systems depend on the quality of the data received. The evolution of MRP system therefore depend not only on technological developments, but also on robust management and integration strategies.

In the transition between Industry 4.0 and Industry 5.0, MRP system must introduce human-centered interfaces. Interfaces must be more intuitive, collaborative and aligned with well-being, sustainability and resilience. Collaboration between human knowledge and AI is considered essential, promoting collaboration between the operation and the machine, coinciding with concepts of Industry 5.0 (Jiang et al., 2024).

As mentioned above, Industry 5.0 focuses on positive social, environmental and economic impact. The MRP system acquire the ability to use the data provided by the MES system to calculate the carbon footprint or adapt working conditions according to human preferences (Jiang et al., 2024).

Ensuring that operators understand and trust the automated decisions generated by AI models is one of the biggest challenges of the transition (Jiang et al., 2024).

2.4.2 Advantages, Challenges and Limitations

The introduction of MRP system in the industry context has brought several benefits. While these benefits were established by traditional systems, MRP 4.0 systems enhance them by integrating real-time data to ensure even greater precision. The primary benefits include (Thevenin et al., n.d.[b]):

- **Guarantee the materials needed for production:** It is possible to guarantee the raw materials needed to produce a product, avoiding production stoppages due to lack of materials.
- **Reducing stock and waste:** This system provides a precise calculation of production requirements. This makes it possible to reduce overstock and waste of materials (e.g. due to the expiry date of raw materials in the food sector).

- **Improved compliance with deadlines and reduced delays:** With the plans generated by the system for production and purchasing control, companies can be more flexible to meet delivery deadlines, reducing delays by the dates stipulated in the plan.
- **Improved internal coordination:** Through this system, it is possible for all company's departments to have the same up-to-date information, with all detailed information needed for the various areas (such as purchasing, sales, stock, production). In this way, the main departments, purchasing and production, are aligned with the company's needs.

However, despite all the benefits the traditional MRP system provide, it does have some limitations. Identifying these traditional gaps is essential to understand the necessity for implementation of Artificial Intelligence and integration of Real-Time data, advantages from MRP 4.0 systems in modern environments ([Thevenin et al., n.d.\[b\]](#)):

- **It assumes infinite production capacity:** The system only considerate the materials needed for production and does not include the company's production capacity.
- **Lack of adaptability in real time:** The traditional MRP system create calculation cycles (e.g. daily or weekly) based on static events. However, the industrial sector has many unforeseen events, and the system is unable to adapt in real time to these events to create a plan that is adaptable to the circumstances.
- **Depending on data quality:** MRP system require the data to be extremely accurate and viable. If the data has an error, the system will generate an incorrect plan. This requires a continuous effort to update the data.

2.4.3 Models in the Material Requirements Planning Context

Determining the ideal size of planned order entries is a fundamental element of MRP system calculations logic. The techniques implemented in the MRP system aim to determine the quality and timing of orders to meet estimated net requirements, minimizing the costs associated with placing orders and maintenance costs. Some MRP system models stand out based on cost optimization ([Iskandar and Ananda, 2024](#)):

- **Economic Order Quantity (EOQ):**

The Economic Order Quantity, EOQ, model aims to determine the ideal order quantity that minimizes the total cost associated with inventory or stock control. The total value of this cost is the sum of the ordering costs and the stock costs. The model assumes constant demand and allows a predetermined quantity to be ordered periodically, thus contributing to efficient stock management.

- **Period Order Quantity (POQ):**

The Period Order Quantity (POQ) model is an adaptation of the Economic Order Quantity (EOQ) model for contexts in which demand fluctuates over time. Unlike the EOQ model, the POQ model determines the ideal time interval between orders, this providing greater flexibility and accuracy in the face of seasonal variations or fluctuations in demand.

- **Lot For Lot (LFL):**

The Lot For Lot, LFL, model demonstrates a high capacity to respond to actual production needs. This model provides detailed information on the exact quantity required to meet net demand for a single period. This method aims to minimize stock maintenance costs as much as possible. The ideal goal would be to reduce costs to a minimum or even eliminate them altogether. On the other hand, this model is likely to generate high numbers of orders, which increases administrative costs in the absence of adequate control.

The approach described in this thesis adopts the POQ model as its foundation logic for managing demand volatility. However, traditional POQ model typically calculates orders based on historical averages, resulting in a static schedule. This thesis proposes a solution that integrates ML into the model to provide a dynamic scheduling capability. By feeding the model with real-time data, such as actual stock levels and production status, at the moment a request is made, the system ensures that schedules are grounded in the “actual” reality of the manufacturing environment rather than theoretical forecasts alone. With this approach, the system can react instantly to immediate fluctuations.

2.5 Role of Manufacturing Execution System in AI-enhanced MRP system

The Manufacturing Execution System (MES system) emerged in the 1990s to fill the gap between Material Requirements Planning System (MRP system) planning and Shop Floor execution, centralizing real-time production information and coordinating the execution of manufacturing orders ([Shojaeinasab et al., 2022](#); [Tariq et al., 2024](#)). Although the focus of this thesis is the enhancement of MRP system through AI, MES system is a necessary component of this work: it is the system responsible for supplying the real-time, Shop Floor level data — stock levels, production status, order execution — that any predictive or adaptive MRP system require to remain accurate and actionable.

MES-MRP Integration. Following the introduction of the ISA-95 standard, and formalized within the RAMI 4.0 model, MES system is positioned at Level 3 of the automation hierarchy, acting as the bridge between business planning (Level 4, e.g. MRP system/ERP system) and Shop Floor automation (Level 2, e.g. Programmable Logic Controller (PLC)/Supervisory Control and Data Acquisition (SCADA)). In practice, this Level 4-3 integration is commonly realized through dedicated middleware platforms. One applied example is E2E.pt, a Portuguese integration platform that connects ERP system/MRP system (such as SAP, Sage X3, Primavera, and PHC) to Shop Floor and business data sources through a unified, OpenAPI-based interface ([E2E.pt, 2026](#); [OpenAPI Initiative, 2026](#)). Rather than requiring custom point-to-point connections between every system, E2E.pt provides a single secured channel — using an on-premise agent, encrypted communication, and tunnel-based firewall protection — through which stock levels, production status, and transactional data can be read from and written to enterprise systems. This kind of middleware illustrates, in applied form, how the abstract Level 4-3 integration defined by ISA-95 and RAMI 4.0 is realized in real industrial settings.

Data Provided by MES system. For the purposes of this thesis, the data most directly enabled by MES system systems are updated stock levels, real-time production and order status, and Shop Floor execution timestamps. This information allows an AI-enhanced MRP system to move from static, periodic planning toward dynamic, continuously re-evaluated production schedules.

Integration Challenges. This integration is not without limitations. Data mapping and interoperability between MES system and MRP system/ERP system systems remain technically challenging, particularly when connecting to legacy Shop Floor equipment, which often requires additional IoT gateways ([Engelbrecht, 2007](#); [Gomes et al., 2024](#)). Implementation and integration costs can also be prohibitive for small and medium-sized manufacturers.

In summary, this thesis does not implement or study a complete MES system; MES system is treated as an external, complementary data source. The core contribution of this work lies in the analytical layer built on top of that data: predictive demand models, Bill of Materials (BOM)/First-Expired-First-Out (FEFO) based planning logic, and the decision-support interface presented to human operators.

2.6 Artificial Intelligence and Machine Learning

Artificial Intelligence (AI) is a technology whose purpose is to automate processes that require human intelligence and create intelligent systems. Machine Learning (ML) technology is a fundamental sub-technology of AI. The initial ideas for this technology emerged in the mid-20th century, with AI evolving from theoretical discourse to practical application.

2.6.1 The Development of Artificial Intelligence and Machine Learning

The development of AI and ML can be outlined in four phases ([Hiremath and Kenchakkanavar, 2025](#)):

Initial foundations (early 1950s and 1960s)

Philosophers such as Aristotle and mathematicians such as Bayes and Boole contributed to the initial foundations. In 1950, Alan Turing proposed the Turing Test and the following year, in 1951, Minsky and Edmonds developed the first artificial neural network, SNARC. The self-learning checkers program was designed by Arthur Samuel in 1952. Three years later, in 1955, John McCarthy coined the term AI. Subsequently, in 1956, the official birth of AI was formalized at Dartmouth Conference.

“AI Winters” (1970s-1980s)

At this stage, AI went from a symbolic methodology and knowledge modeling to machine learning and advanced applications. During this period, the development of expert systems and advances in neural networks were observed. However, in the 1970s and 1980s, this area was affected by reduced funding and skepticism, characterized by the “AI Winters”. Despite these challenges, the introduction of backpropagation, in 1969, by Bryson (an advance in deep learning) and the commercialization of Lisp machines in 1980 represent significant milestones.

Renaissance and Expansion (1990s-2010s)

This period was marked by significant growth and rebirth. Technological advances and corporate applications were driven by the third AI “boom” in 2013. The increase in computing power during this period and the introduction of new tools were key factors in this development. Technologies such as Machine Learning, natural language processing, and deep learning gained popularity.

The Deep Learning Revolution (2010s-Present)

Deep Learning has become a fundamental technique. This technique has provided remarkable accuracy in some tasks, namely image and voice recognition. This advance has been driven by reinforcement learning, computational resources, and the development of generative models.

The evolution of learning models is evidenced by notable examples:

- In 2017, Google introduced the Transformers model, a breakthrough that revolutionized large language models, LLMs ([Vaswani et al., 2017](#)).
- OpenAI played a prominent role in this evolution:
 - In 2018, GPT was launched ([Radford et al., 2018](#)).
 - In 2020, the GPT-3 model was introduced, setting a new benchmark in terms of capacity, with 175 billion parameters ([Brown et al., 2020](#)).
 - In 2021, with the launch of DALL-E, an AI application, a model capable of creating images from textual descriptions was introduced ([Ramesh et al., 2021](#)).
 - In 2022, ChatGPT, a natural language model designed to be accessible to the public and businesses, was launched ([OpenAI, 2022](#)).
 - In 2023, OpenAI introduced GPT-4. This model features significant improvements in reasoning, enabling it to solve more complex problems with greater accuracy, with the help of broader general knowledge. With its new problem-solving capabilities, it can use images as data input and create analyses, classifications, and generate descriptions ([OpenAI, 2023](#)).
- Other major AI labs released increasingly capable frontier models over the following years:
 - Google DeepMind introduced the Gemini family of multimodal models, later iterated into Gemini 2.5 and Gemini 3 Pro, emphasizing advanced reasoning and long-context capabilities ([Google DeepMind, 2025](#)).
 - Anthropic released the Claude family of models, including Claude 3 and subsequent iterations, and later Claude Code, an agentic coding assistant ([Anthropic, 2024](#)).
 - In late 2024 and early 2025, the Chinese AI lab DeepSeek released DeepSeek-V3 and DeepSeek-R1, open-weight models that achieved performance comparable to leading closed-source models at significantly lower training cost, drawing attention to the growing competitiveness of open-source and non-US AI development ([DeepSeek-AI, 2024, 2025](#)).

2.6.2 Advantages, Challenges and Limitations

AI has demonstrated remarkable capabilities in solving complex problems and innovating in various fields.

A sub-technology of AI known as ML is a key advantage. Its ability to learn from experience and adapt without explicit programming are some of its main factors. ML technology has developed a system capable of identifying patterns, making predictions, and automating tasks with increasing sophistication. This technology features natural language processing, which allows machines to interpret and analyze images with high accuracy, using systems that can understand and generate text like that of human beings (Oliveira and Figueiredo, 2024).

Deep Learning has also driven systems that excel at complex games. In addition, AI systems using ML have taken on central role in Industry 4.0 due to the exploitation and extraction of significant value from large data sets.

AI has also become an engine of automation, enabling robots and computer systems to replace or augment human effort in wide range of tasks. AI is also present on the internet today, particularly in search engines, which use AI to retrieve information and quickly locate and rank relevant information (Oliveira and Figueiredo, 2024).

However, despite all the advantages provided by AI there are still some challenges and limitations that require attention.

One of the main concerns lies in ethical issues and inherent biases, since AI, and ML systems can reflect the biases present in the data used for analysis and training, resulting in unfair and/or discriminatory results.

The robustness of ML models is determined by the quantity and quality of data. Training these models requires large amounts of high-quality data to improve their accuracy. However, there are contexts where it is not possible to obtain the necessary amount of data, thus compromising the effectiveness and reliability of the models.

Interpretability and transparency represent another obstacle. These models use “Black Boxes” in their deep learning process, making it difficult to understand the underlying reasoning behind their decisions and predictions. This obstacle leads to a lack of trust in AI systems in critical contexts such as the financial and healthcare sectors (Hiremath and Kenchakkanavar, 2025).

The increased use and sophistication of AI models in systems raises concerns about security and cyber threats. AI systems are vulnerable to attacks by adversaries, which can enable the creation of more powerful cyber threats, such as deepfakes or automated hacking tools.

The use of advanced AI, and ML models require high computing power and consumes significant amounts of energy. This raises concerns about the environment impact and sustainability of large-scale AI implementations.

Finally, there are challenges related to collaboration between humans and AI models, as well as the potential impact of job losses. The initial implementation of these systems was intended to replace humans in some tasks to automate them, which raised some concerns about potential job losses in certain sectors. However, Industry 5.0 emerged as an attempt to solve this challenge by placing humans at center of the decision-making process and AI, and ML models as decision support, promoting human-machine collaboration.

2.6.3 Applications for Artificial Intelligence and Existing Models

AI is getting used more in complex challenges, industry sectors and human-centered fields.

One big area is using AI to understand, simulate, and improve parts of human thinking. Researchers are looking into how AI models can be developed as tools to support psychological well-being and offer ways to help people with therapy ([Harika et al., 2022](#)).

Another significant area of application is the optimization of demand forecasting, which is a key factor for efficient production and optimized resource management. For example, in the food industry, product quality is a concern and demand for these products is inconsistent, so forecasting supported by AI can offer greater accuracy. In the food industry, studies have focused on techniques to improve the MRP system. Statistical models, such as Autoregressive Integrated Moving Average (ARIMA), ETS (Error, Trend, and Seasonality), and Prophet are some of the statistical approaches compared with Machine Learning techniques. The Prophet is an approach that includes several models, namely Linear Regression (LR), K-Nearest Neighborhood (KNN), Support Vector Regression (SVR), Random Forest (RF), and eXtreme Gradient Boosting (XGBoost). In this industry, these models have been studied with the aim of improving forecast accuracy, using time series data at various levels of aggregation. The results of this study showed that the Random Forest model with Ordinary Least Squares (OLS) reconciliation performed best in improving demand forecasting ([Perera et al., 2024](#)).

These application examples demonstrate the potential for using sophisticated Machine Learning models, which can have significant positive impact on predictive accuracy in complex environments.

2.6.4 Impact of AI and ML Integration on MRP and ERP Systems

The two new technologies, AI, and ML are technologies that can increase the capabilities of MRP system namely advanced analytical, predictive, and automation capabilities, which are fundamental for real-time decision support.

The integration of the two technologies into the MRP system affect the following concepts ([Mhaskey, 2024](#); [Peng et al., 2024](#)):

Methodologies and Architectures

ML is responsible for natural language processing and predictive analysis to optimize interactions between users/operators and systems, as well as robotic process automation for performing tasks in ERP system. In ERP system that use Cloud technology, the integration of ML is facilitated due to the accessibility of data.

The proposed architecture uses AI to process real-time data streams provided by MES system on the Shop Floor. By analyzing the data collected from sensors and machinery by MES system the AI can perform advanced forecasting, which one will directly inform the MRP system.

Opportunities and Benefits (Improving, optimizing, automating) ([Aslam, 2023](#); [Makin et al., 2024](#))

Optimization of demand forecasting through the application of ML algorithms that analyze historical data, seasonality, and external factors for more accurate demand forecasts. The information is then sent to the MRP system.

Production optimization and the implementation of an intelligent scheduling system are fundamental to production planning, resource allocation, and operational efficiency in the MES system. Optimization will be performed by AI.

Predictive maintenance, defined by equipment failures and reduced downtime, can be predicted from an analysis performed by AI using data collected by sensors and/or machines in the MES system.

Decision support is optimized by AI through recommendations and automation of decision-making processes. AI provides real-time insights.

The optimization of resource management, which covers the management of materials, labor, and equipment, is a particularly important aspect.

Challenges (data quality, integration, knowledge) (Espina-Romero et al., 2024)

Data integrity is an important aspect of AI. This technology requires high quality and large amounts of data, where any corruption in the data compromise system performance.

The complexity of the integration process is a factor that influences companies. The integration of AI into MRP system, and others business systems is a significant challenge.

The use of AI requires a knowledge base. Acquiring this knowledge can be interpreted as resistance to change. The implementation of these technologies requires high costs for companies.

The interpretation of ML reasoning is a barrier to critical decision making. ML models rely on “Black Box”.

This thesis aims to demonstrate how the AI usage more precise the ML contributes for the transformation of the MRP system into dynamic system rather than a static support tool. A significant contribution of integration of ML on these systems over traditional models like POQ (which rely on rigid historical averages) is the capability to process the real-time stock levels and production status from MES system.

2.7 Comparative Analysis of Machine Learning Applications in Supply Chain Forecasting

To provide a foundational support for the proposed solution of this thesis, it is essential to analyze how others ML architectures address the unique challenges of modern supply chain. This section addresses some studies that demonstrate the evolution of forecasting from traditional statistical models to advanced algorithmic approaches. The studies will show the importance of some factors such as managing high noise-to-pattern ratios in demand signals and necessity of integrating external environmental variables to improve model performance in complex industrial sectors.

2.7.1 Machine Learning for Production Planning in the Food Industry Under Uncertainty

This study analyzed historical production data from a food company to model the discrepancy between planned and actual output, driven by uncertainty in yield, demand, and raw material variability. The results showed that Machine Learning algorithms outperformed classical statistical methods and expert-defined decision rules in predicting this discrepancy, providing production managers with actionable insight into process deficiencies and reducing uncertainty in food waste generation ([Garre et al., 2020](#)).

This research aligns closely with the objectives of this thesis, as it directly demonstrates that Machine Learning applied to food production planning can reduce the gap between planned and actual output — the same core problem this thesis’s AI-enhanced MRP system aims to address.

2.7.2 Demand Forecasting for the Food Industry Using Machine Learning

This study compared forecasting models for confectionery products sold by a large food distribution company, evaluating Machine Learning approaches against traditional statistical methods for short-term and long-term demand prediction. The results showed that accurate demand forecasting directly reduces stored and expired inventory, a critical concern for perishable food products ([Nassibi et al., 2023](#)).

This research supports this thesis’s objectives by validating that reliable demand forecasting in the food sector, when supported by Machine Learning, has a direct and measurable impact on reducing food waste through better-aligned production and stock levels.

2.7.3 Integration of Demand-Driven MRP and Machine Learning in Food Industry Logistics

This study proposed an integrated logistics model for the food industry, combining Demand-Driven MRP system with ARIMA-based Machine Learning forecasting and RFID-enabled traceability. The model was designed to dynamically respond to demand fluctuations and correct operational deviations through real-time feedback, addressing low order-delivery service levels caused by inaccurate demand planning and inventory control errors ([Rodriguez et al., 2026](#)).

This research is the most directly aligned with the scope of this thesis: it demonstrates, in a real food-industry setting, that combining MRP system-based planning logic with Machine Learning forecasting improves stock accuracy, order responsiveness, and adaptability — the same combination of planning method and predictive modeling this thesis proposes.

CONCEPTUAL APPROACH

3.1 From Problem to Design

Food manufacturing presents production planning challenges that are not fully addressed by generic Material Requirements Planning System (MRP system) logic: demand is strongly seasonal, raw materials and finished products are perishable, and production schedules must absorb sudden shifts in market demand without generating excess waste. Addressing these challenges requires more than a general-purpose Artificial Intelligence (AI) layer added to an existing MRP system — it requires a set of requirements and a system design conceived specifically around the constraints of the food sector.

This chapter translates that problem into a concrete system design. Section 3.2 defines the functional requirements that drive the proposed solution, each grounded in a specific limitation of traditional MRP system when applied to food manufacturing, such as the need for waste-aware stock rotation (First-Expired-First-Out (FEFO)), recipe-based demand decomposition through Bill of Materials (BOM) explosion, and human-in-the-loop override mechanisms aligned with Industry 5.0 principles. Section 3.3 then models how these requirements translate into user interactions across the system's operational roles — management, inbound/outbound logistics, and Shop Floor — before Section 3.4 presents the overall conceptual architecture that satisfies them.

3.2 Requirements

The functional requirements of this system were established to form a direct, reliable bridge between raw market history and daily food manufacturing operations. To ensure system stability, the framework first audits historical sales records for dead or low-volume inventory and flagging these anomalies with system warnings to prevent wasteful computation on unreliable forecasts. Once data quality is verified, the system combines the outputs of the forecasting models into a single demand projection with a 90% prediction interval, and allows the manager to override the result manually to incorporate market knowledge that the historical data does not capture.

Finally, the framework translates these digital demand targets into physical reality on the Shop Floor by analyzing the BOM, optimizing shelf-life consumption via FEFO rules to actively prevent food waste, and simulating how global supply chain shocks alter material timelines.

This complete end-to-end workflow is mapped across separate predictive and operational modules in Tables [3.1](#) and [3.2](#) to provide clear, role-based visibility across the factory floor.

Table 3.1: System Requirements — Forecasting and Analytical Layer (Req-01 to Req-05)

Req ID	Core System Objective		Operational Description
Req-01	Historical Pattern Aggregation		The system must be able to ingest and structure multi-year transactional records into a reliable baseline time-series suitable for seasonal analysis.
Req-02	Data Quality Density Audit		The system must be able to detect inventory items with insufficient or inactive historical data and flag them accordingly, preventing unreliable forecasts from being generated.
Req-03	Heterogeneous Modeling	Ensemble	The system must be able to combine multiple complementary forecasting approaches to produce a more robust demand outlook than any single method could achieve alone.
Req-04	Statistical Safety Customization		The system must be able to maintain a dynamically adjustable safety margin around the base forecast, sized according to observed market volatility, to absorb localized demand fluctuations.
Req-05	Human-in-the-Loop Integration		The platform must incorporate dedicated override mechanisms, allowing management to manually update operational values or adjust algorithm weights based on qualitative market changes.

Table 3.2: System Requirements — Execution and Interface Layer (Req-06 to Req-09)

Req ID	Core System Objective	Operational Description
Req-06	Recursive Recipe Explosion	The planning service must execute an automated BOM decomposition, translating finalized finished product forecasts into accurate net dependent raw material targets.
Req-07	Waste-Aware Stock Attrition	The system must be able to track material shelf-life over time and proactively flag expiration risks before waste occurs, prioritizing consumption of the oldest available stock first (FEFO).
Req-08	Macro Scenario Simulation	At a manager's request, the planner must activate a probabilistic stress-test mode, scaling demand and extending supplier lead times based on historical precedents to evaluate supply chain resilience.
Req-09	Role-Based Operational Views	The user interface layer must partition system outputs into separate specialized views, synchronizing inbound and outbound timelines across strategy offices, loading docks, and floor workstations.

3.3 Use Case Diagram

At an architectural level, the system's operational model comprises three primary life-cycles, each responsible for orchestrating the transformation of raw data into progressively refined levels of business value. These sequential phases function as a cohesive pipeline, enabling the system to progress from exploratory forecasting to operationalized production planning.

Operational Environment: Office Room (Strategy & Administration)

The Office Room serves as the central strategic hub where high-level management and automated system maintenance occur. Unlike the physical Shop Floor, this environment functions as the Digital Twin of the factory: a continuously updated virtual representation of production data, predictive models, and simulated risk scenarios, used to inform decisions before they are executed on the physical Shop Floor.

As illustrated in Figure 3.1, this environment is governed by two distinct actors:

- **Manager and/or Admin:** The primary human actor responsible for qualitative decision-making, such as overriding AI forecasts based on market experience or initiating "What-if" crisis scenarios.
- **System Timer:** A secondary automated actor that ensures the system's "heartbeat". It independently triggers routine requests for training and scheduling to ensure the factory always operates on the most current data without requiring constant human intervention.

The use cases within this station are organized into three pillars:

- **Data Management:** The manager and admin can manually Add Stock or Add Product Recipe(s) (BOM) to ensure the system has the correct product compositions and inventory levels needed for accurate planning.
- **Predictive Oversight:** This pillar allows the manager to audit the Training Process and view Demand Predictions. Through the "Update Demand Forecast" relationship, the manager can "harden" the machine's output by incorporating manual adjustments.
- **Resilience Planning:** Users can view the final Demand Schedule to understand upcoming procurement needs. This includes the ability to request Scenarios of External Events, simulating how crises (e.g. wars, pandemics) would influence the current supply chain.

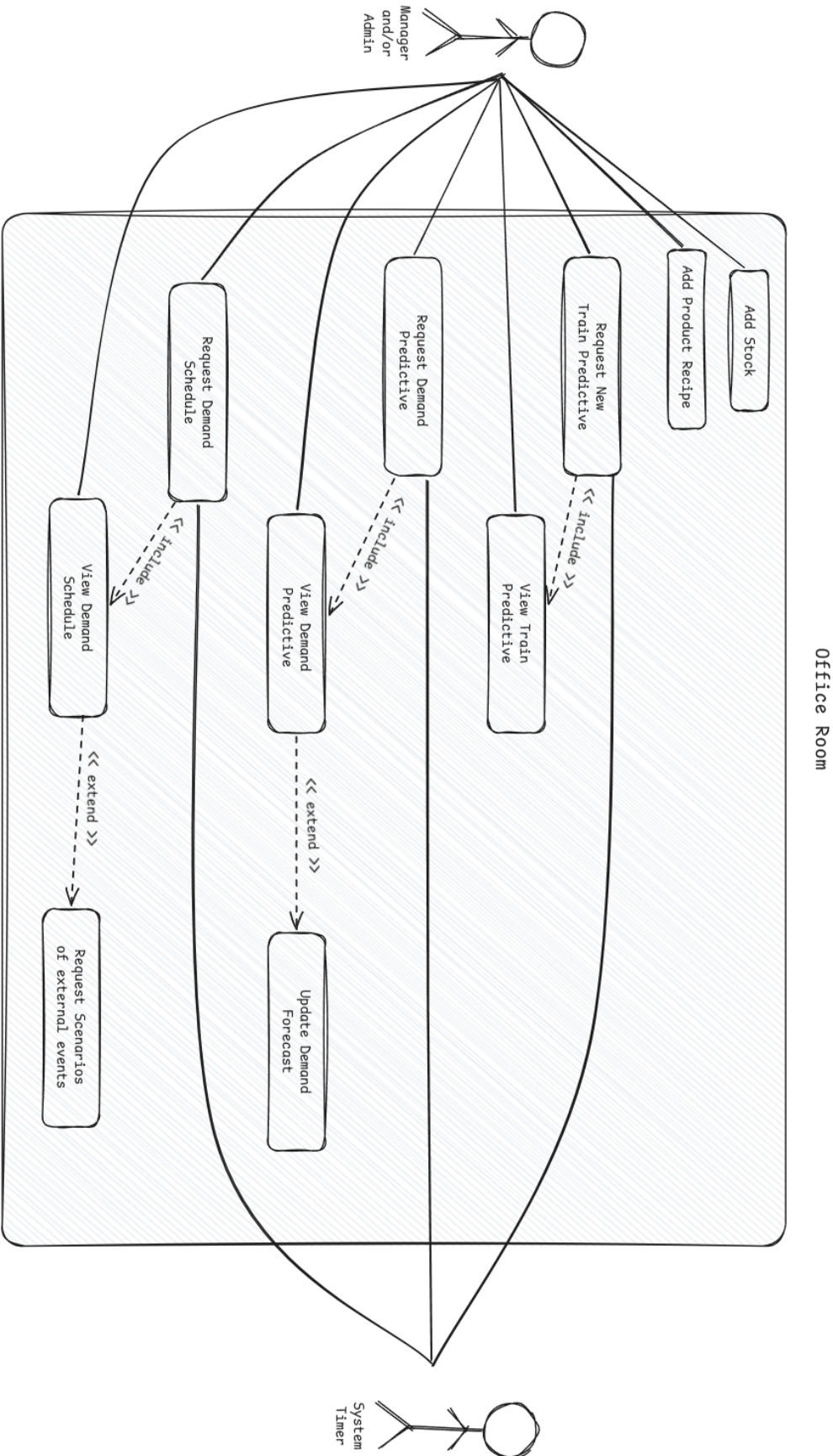


Figure 3.1: UML Diagram of Office Room Environment.

Operational Environment: DockStation (Logistics & Material Flow)

The DockStation serves as the primary operational hub for managing the movement of goods into and out of the factory. Within this environment, the priority is ensuring that physical inventory levels remain synchronized with the digital planning environment.

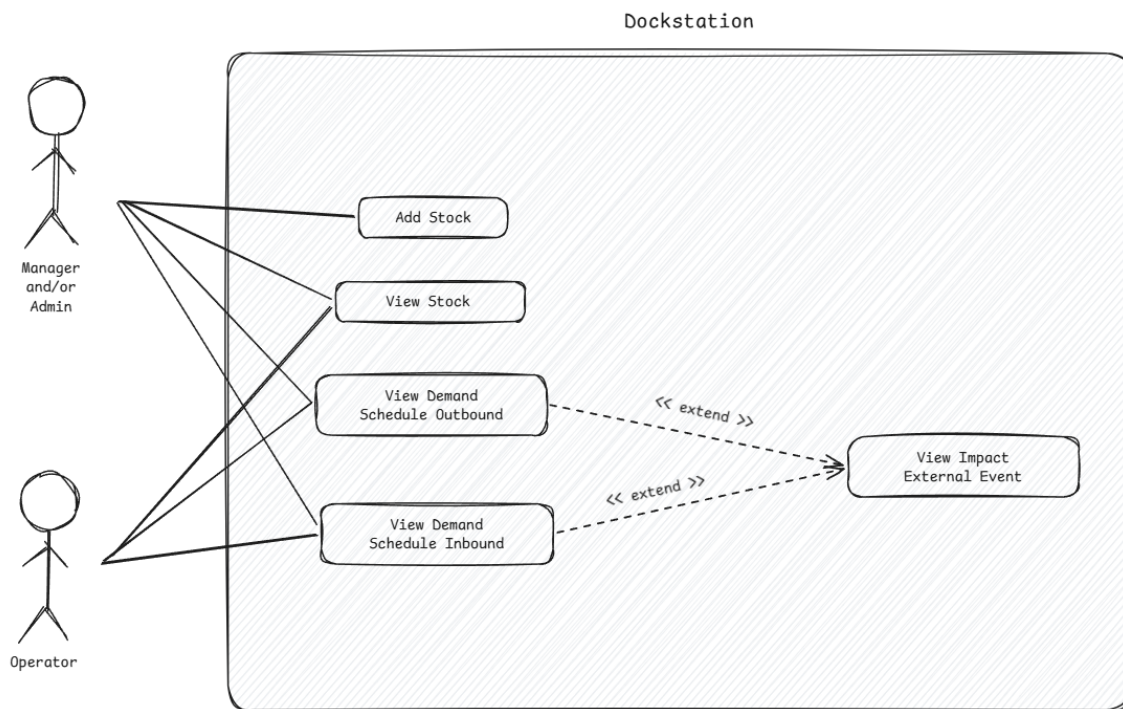


Figure 3.2: UML Diagram of DockStation Environment.

As illustrated in Figure 3.2, this environment is governed by two distinct actors:

- **Operator:** The primary actor responsible for the physical handling of goods. Their digital responsibility is limited to real-time visibility into stock levels, allowing them to confirm physical inventory against the system without altering records directly.
- **Manager:** Responsible for supervising the logistics flow. The manager utilizes the DockStation to ensure that inbound raw materials and outbound finished products are aligned with the broader production goals.

The use cases in this station are organized into three pillars to support seamless material handling and logistics visibility:

- **Supply Chain Visibility:** Users can monitor both the Inbound and Outbound demand schedules. This ensures that logistics personnel can prepare for arriving raw materials or scheduled shipments of finished goods based on the system's optimized planning signals.
- **Inventory Synchronization:** This pillar focuses on maintaining Inventory Fidelity. Stock entry restricted to Manager, while the Operator retains View Stock access from the loading dock. This separation keeps inventory write access accountable to a single role while still giving floor personnel the visibility they need to confirm physical reality against the system. This role separation reflects a scoping assumption made during prototype design and was not validated against the partner organization's actual operational policy.
- **Logistics Resilience:** Through the «extend» relationship with the "View Impact External Event", the system provides immediate context for logistics disruptions. If a geopolitical or external crisis occurs, logistics actors can visualize how these events influence delivery windows and shipment priorities.

Operational Environment: WorkStation (Production Floor & Execution)

The WorkStation represents the tactical execution center of the manufacturing facility. In this environment, the WorkStation is focused entirely on the Shop Floor. It connects digital predictive models and physical production by providing real-time operational visibility to the team transforming raw materials into finished products.

As illustrated in Figure 3.3, this environment is governed by two distinct actors:

- **Operator:** The primary floor actor who utilizes the station to check product recipes, verify available ingredient quantities, and execute the daily production run.
- **Manager:** The production supervisor who monitors operations to ensure that daily floor outputs perfectly match the broader inbound and outbound constraints dictated by the planning system.

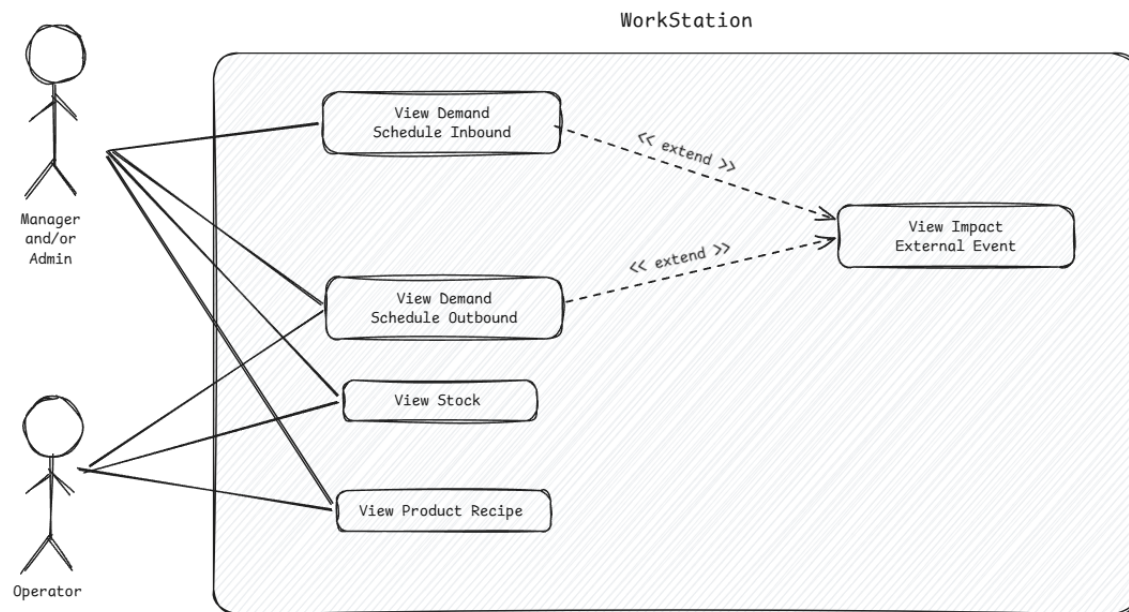


Figure 3.3: UML Diagram of WorkStation Environment.

The use cases in this station are organized into three pillars to streamline manufacturing operations:

- Manufacturing Transparency:** Through the View Product Recipe and View Stock capabilities, personnel can immediately access the precise product bills of materials (recipes) and monitor current stock locations. This ensures that physical raw materials on the line are utilized according to the exact structural guidelines of the factory.
- Time Synchronization:** Shop Floor actors can interact with both the Inbound and Outbound demand schedules. This visibility ensures production teams know exactly what materials are arriving from the warehouse and what finished volumes must be packaged and transferred to shipping.
- Floor-Level Resilience:** Both the inbound and outbound schedule views contain an «extend» relationship with the View Impact External Event use case. If an external supply chain shock or crisis occurs, personnel can instantly see how these macro disruptions modify their active manufacturing timelines or ingredient priorities.

3.4 Conceptual Architecture

To define the system’s structural logic without binding it to specific coding implementations, a high-level layered architecture is established. As illustrated in Figure 3.4, this framework organizes the platform into four progressive conceptual pillars, moving from foundational technical utilities up to the user interaction layers. This design ensures complete modularity, allowing future iterations to upgrade individual modules without disrupting the broader platform.

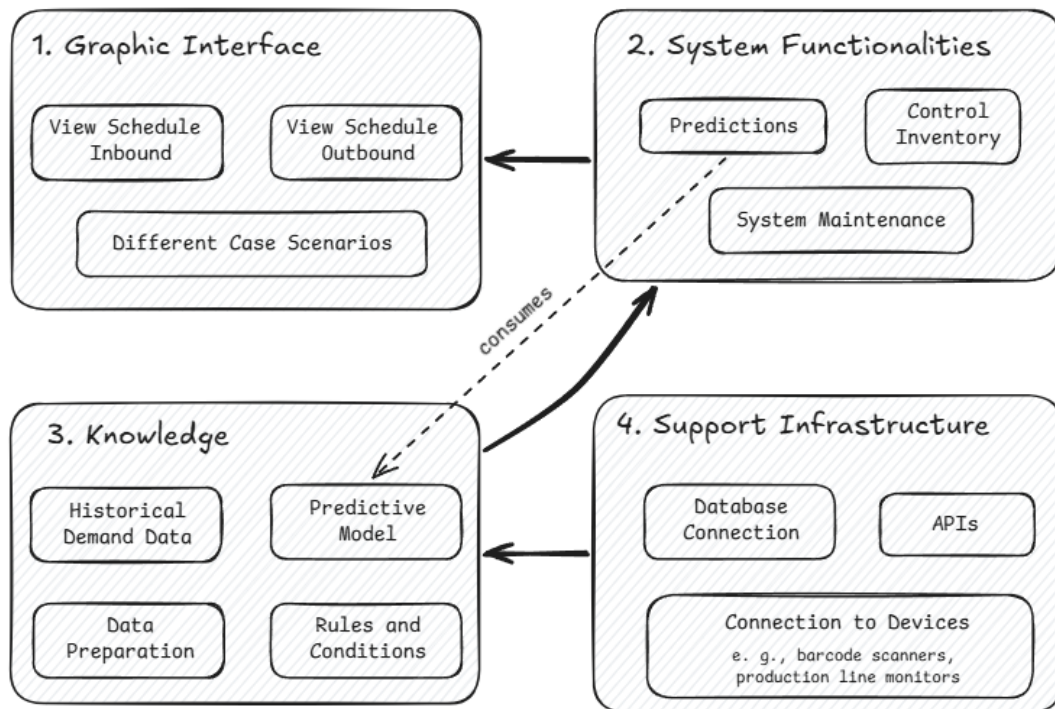


Figure 3.4: High-Level System’s Architecture.

The **Support Infrastructure Layer** serves as the foundational technical baseline of the system, responsible for securing core resource pipelines and providing reliable external access points. Within this foundational layer, **Database Connections** establish secure operational channels to transactional tables, ensuring that both raw historical parameters and intermediate processed states are safely preserved. Standardized **Application Programming Interface (API)** complement this setup by offering robust, reliable communication pathways that let decoupled services exchange without tight software dependencies. To ensure physical integration with factory operations, a dedicated **Connection to Devices** service provides a clear abstraction layer for external data touch points, effectively accommodating hardware tracking mechanisms such as warehouse barcode scanners or active production line monitors.

Directly above the base infrastructure sits the **Knowledge Layer**, which organizes and maintains the semantic and quantitative memory required to guide the factory's active planning decisions. This layer manages **Historical Demand Data & Data Preparation** by systematically ingesting raw sales records and cleansing anomalies to form a reliable, continuous timeline matrix. Additionally, it preserves the initialized **Predictive Models**, housing the underlying mathematical behaviors and validation weights derived during background learning cycles. To ground these mathematical forecasts in actual factory constraints, the layer incorporates operational **Rules and Conditions** that dictate explicit manufacturing guardrails, including detailed material recipes through BOM and strict stock freshness boundaries.

The primary computational engine of the architecture is defined by the **System Functionalities Layer**, which contains the core execution logic responsible for processing live information state. The execution loop coordinates independent **Predictions**, running background optimization routines that decode seasonal consumption habits and macro market trends. At the same time, the **Control Inventory** function executes time-phased attrition tracking simulations governed by FEFO rules, which maximizes raw material utilization on the floor and cuts down on food expiration waste. Supporting these main workflows, a **System Maintenance** service handles background scheduling tasks, evaluates statistical safety cushions to mitigate unforeseen market shifts, and securely orchestrates non-blocking asynchronous operations across system's backend services.

Sitting above the System Functionalities Layer, the **Graphic Interface Layer** exposes the platform's outputs directly to the system's operational roles. It presents the **Schedule Inbound** and **Schedule Outbound** views that drive day-to-day logistics visibility, alongside the **Different Case Scenarios** view that lets users explore how external disruptions alter these schedules. This layer consumes, but does not compute, the results produced by the System Functionalities Layer below it, keeping the interface strictly presentational and decoupled from the underlying business logic.

IMPLEMENTATION APPROACH

4.1 Architecture

4.1.1 System Architecture

The technical implementation of the planning platform is structured as decoupled, three-tier system architecture. As illustrated in Figure 4.1, the system segregates its operational responsibilities into three processing columns: Client-Side Interaction, Server-Side Orchestration and Storage Infrastructure. This modular approach ensures that intense data processing operations and backend algorithmic computations run independently from the user interfaces, preserving stability and responsiveness across the various station screens.

Client-Side Interaction

The presentation layer is deployed across three dedicated graphic user interfaces, each tailored to a specific operational station within the manufacturing environment. The Office Room Screen serves as the management and administrative console, featuring a restricted security layer that requires administrative authentication. Through this interface, managers are able to audit macro demand forecasts, configure manual adjustments, and execute complex "What-If" disruption scenarios — the mechanism behind the war stress tests evaluated in Scenario 2 (Section 5.3).

In contrast, the DockStation Screen is deployed directly at logistics warehouse bays and operates under a localized operator-level authentication layer. This console provides logistics personnel with real-time visibility into inbound and outbound shipment windows while accommodating direct physical inventory logging.

Finally, the WorkStation Screen is situated directly on the active assembly floor to display live product recipes and material tracking schedules for manufacturing teams. It features a prominent notification layer designed to instantly highlight raw material shelf-life warnings or active line alerts for floor operators.

Communication between these client screens and the backend infrastructure is entirely asynchronous, meaning the user interfaces send structural actions and request active status updates using standard RESTful HTTPS requests that carry lightweight JSON payloads over the network.

Server-Side Orchestration

The backend computational logic is split into two independent web engines to safely isolate transactional traffic from background analytical calculations.

The **Central Webservice** is implemented as the primary system gateway, managing the unified entry point for all incoming client data streams. This engine hosts the Application Programming Interface (API) gateways to process incoming REST requests, enforces role-based access security, houses the core business logic layer, and translates object-relational schema for persistence. Working alongside it, the **Analytical Computational Engine** serves as the specialized calculation core of the backend framework. It independently processes background data cleaning routines, manages predictive modeling algorithms, evaluates optimization filters, and runs the recursive recipe calculations required to split finished product quotas into raw material targets.

Instead of binding these two technical environments together tightly, they interact through a decoupled Internal REST interface loop. When a user requests a complex calculation from a specific screen, the **Central Webservice** quickly hands off the processing task to the **Analytical Computation Engine** and immediately frees its own resources up to continue serving independent incoming data requests from other stations.

Storage Infrastructure

At the base of the system lies the **Central Database**, which serves as the single source of the truth for the entire platform. The backend servers do not store persistent application states locally, instead, they remain synchronized by executing transactional database queries to save and retrieve records. The database schema contains three key functional data models to preserve system history and state.

First, the **Transactional Sales Logs** capture cleaned historical records that supply the baseline training data for demand patterns. Second, the **Model Performance Metadata** acts as an evaluation repository, storing metrics, evaluation history, and error tracking values captured during each candidate model's execution. Third, the **Active Production Schedules** maintain the current state of manufacturing orders, materials requirements, and station timelines, remaining ready to be pulled and displayed instantly by the client-side screens whenever a user logs in.

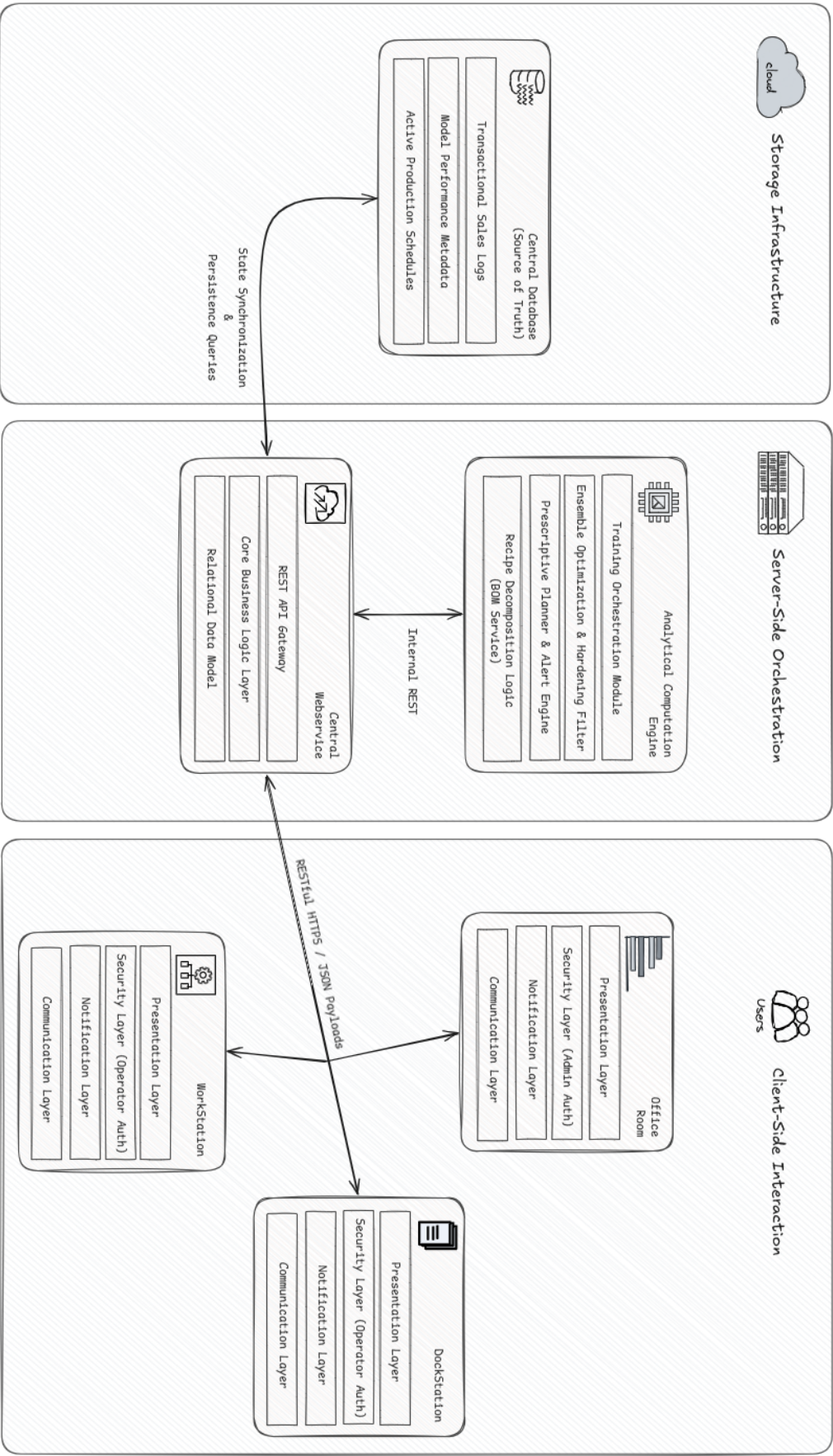


Figure 4.1: Three-Tier General Architecture Diagram of system.

4.1.2 System Communication

The initialization of the forecasting lifecycle, as illustrated in Figure 4.2, begins with the automated background training routine. When a user or system timer triggers an execution request from the station interface, the payload bypasses temporary front-end storage structures to signal the backend orchestration layer directly. The core service verifies the user context, queries the database infrastructure for multi-year historical sales records, and streams these continuous data matrices back to the analytical engine.

To achieve high forecasting precision, this engine does not rely on a single algorithm, instead it executes a diverse library of predictive components simultaneously, generating multiple demand output variants that represent different mathematical perspectives. The engine formats the respective accuracy metrics and evaluation parameters for each candidate model, writing this model performance data to the database.

A raw set of six independent model outputs is not, by itself, something a manager can act on — six competing forecasts for the same item raise the question of which one to trust, not how much to order. For this reason, the *Server-Side Orchestration* automatically triggers the refinement process as soon as training completes, as shown in Figure 4.2, so that requesting a training run always produces an updated, merged forecast rather than leaving the manager with six disconnected charts and no consolidated view to act on. The two processes remain independently callable, however: refinement can also be re-run on its own, without triggering a fresh training cycle, whenever a manager only needs to adjust how the existing model outputs are weighted.

Once the six forecasts are saved, the *Server-Side Orchestration* starts the baseline refinement loop shown in Figure 4.3. It retrieves the stored forecasts together with the performance metadata recorded during training and combines them through Inverse Error Weighting. The error used is the root mean squared error (RMSE) obtained by walk-forward validation: the cut-off date starts twelve months before the target month and advances one month at a time, the model is retrained on the data before each cut-off, and its error is measured on the months after it. Every error is therefore measured on data the model did not see during training. When walk-forward results are not available for a model, the RMSE of its single date-based hold-out split is used instead. Each model receives a weight inversely proportional to its error:

$$w_{i,t} = \frac{1/\text{RMSE}_i}{\sum_{j \in M_t} 1/\text{RMSE}_j}, \quad \hat{y}_t = \sum_{i \in M_t} w_{i,t} \hat{y}_{i,t} \quad (4.1)$$

where M_t is the set of models that produced a forecast for month t , RMSE_i is the average walk-forward error of model i across all folds, bounded below by 0.001 to avoid division by zero, $\hat{y}_{i,t}$ is the forecast of model i for month t , and $\hat{y}_t$ is the combined forecast. Because the weights are renormalized over M_t , the combined forecast remains valid when some models fail, as in Scenario 3 (Section 5.4). When the manager supplies weights manually, these replace the computed $w_{i,t}$ and are normalized in the same way.

The same weighting is applied in all three scenarios of Chapter 5, and this combined forecast is computed first so that the manager has a baseline to review before making manual adjustments. Section 5.1 discusses how this setup affects the interpretation of those results.

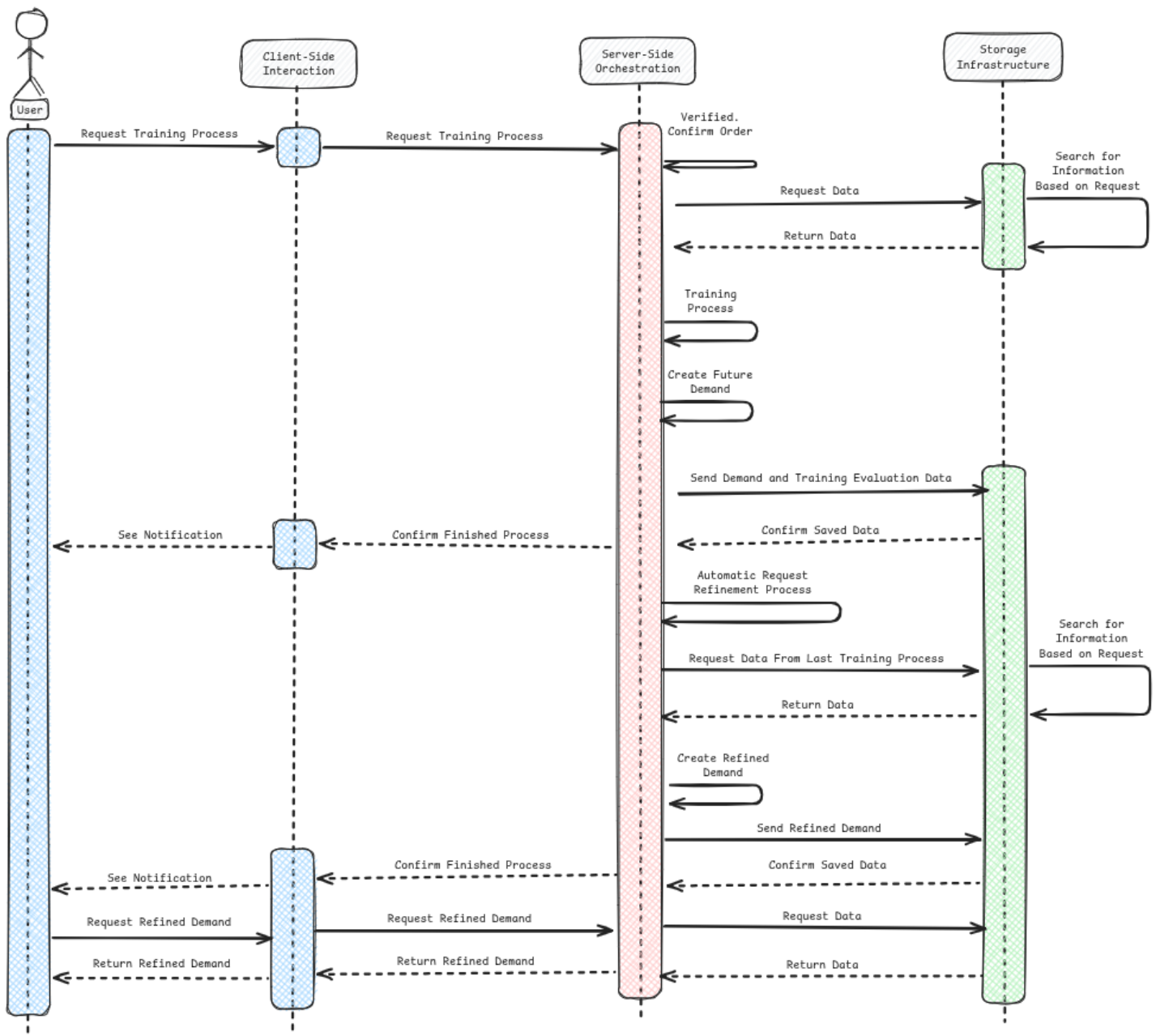


Figure 4.2: Sequence Diagram: Training Process.

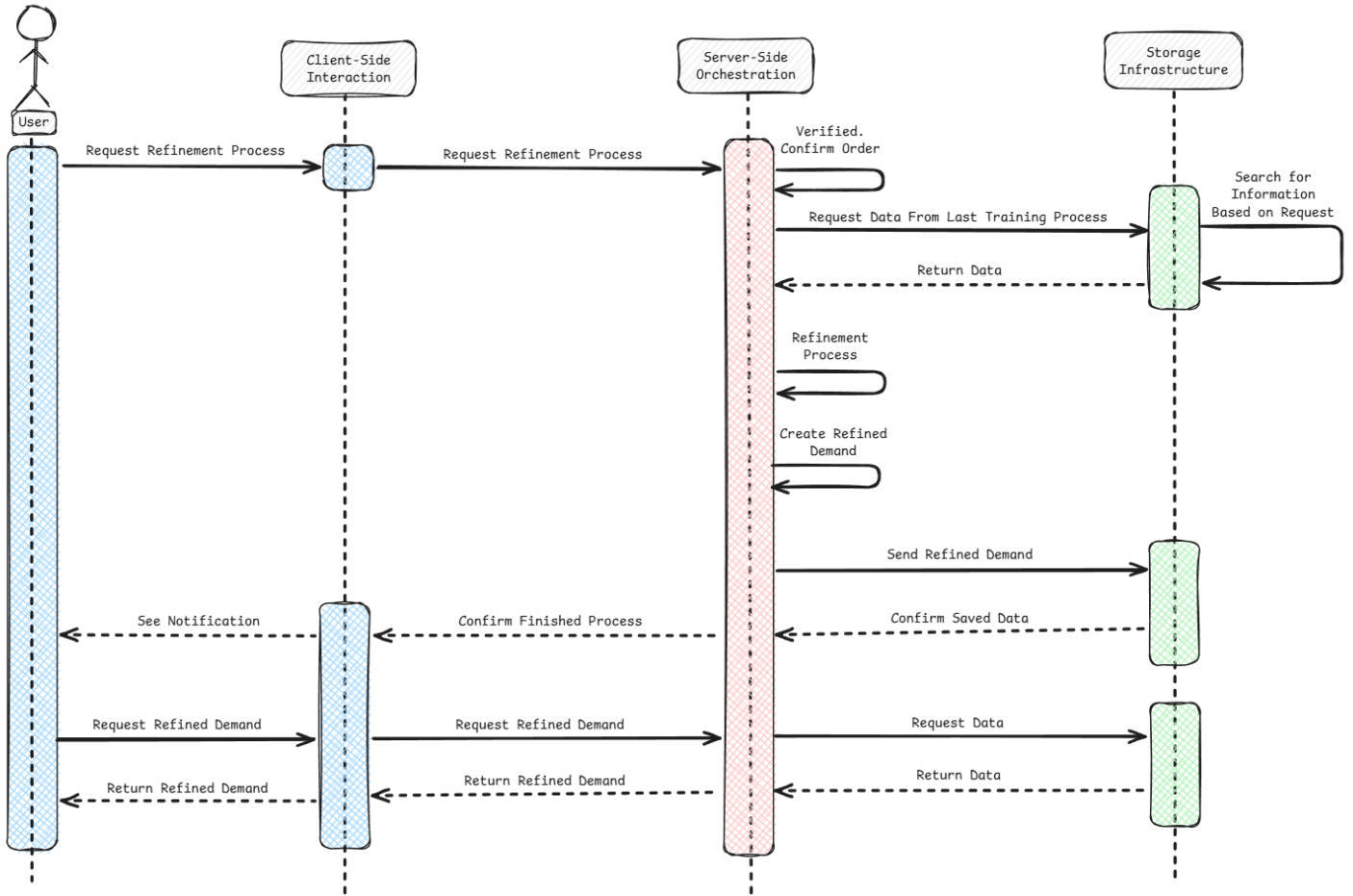


Figure 4.3: Sequence Diagram: Refinement Process.

If the manager reviews this machine-generated curve and disagrees with the automated blending ratios, they can actively switch to the manual override sequence mapped in Figure 4.4. During this interaction, the user utilizes an interactive human-in-the-loop filter to adjust the explicit configuration weights for each demand stream. The backend gateway intercepts these custom scaling choices, recalculates the mathematical combination across the candidate predictions within the analytical core, and replaces the baseline with a human-hardened demand output. Regardless of whether the final curve was derived automatically or shaped via manual overrides, this refinement block compresses all data perspectives into a single unified demand output, which acts as the immutable input for the subsequent manufacturing scheduling.

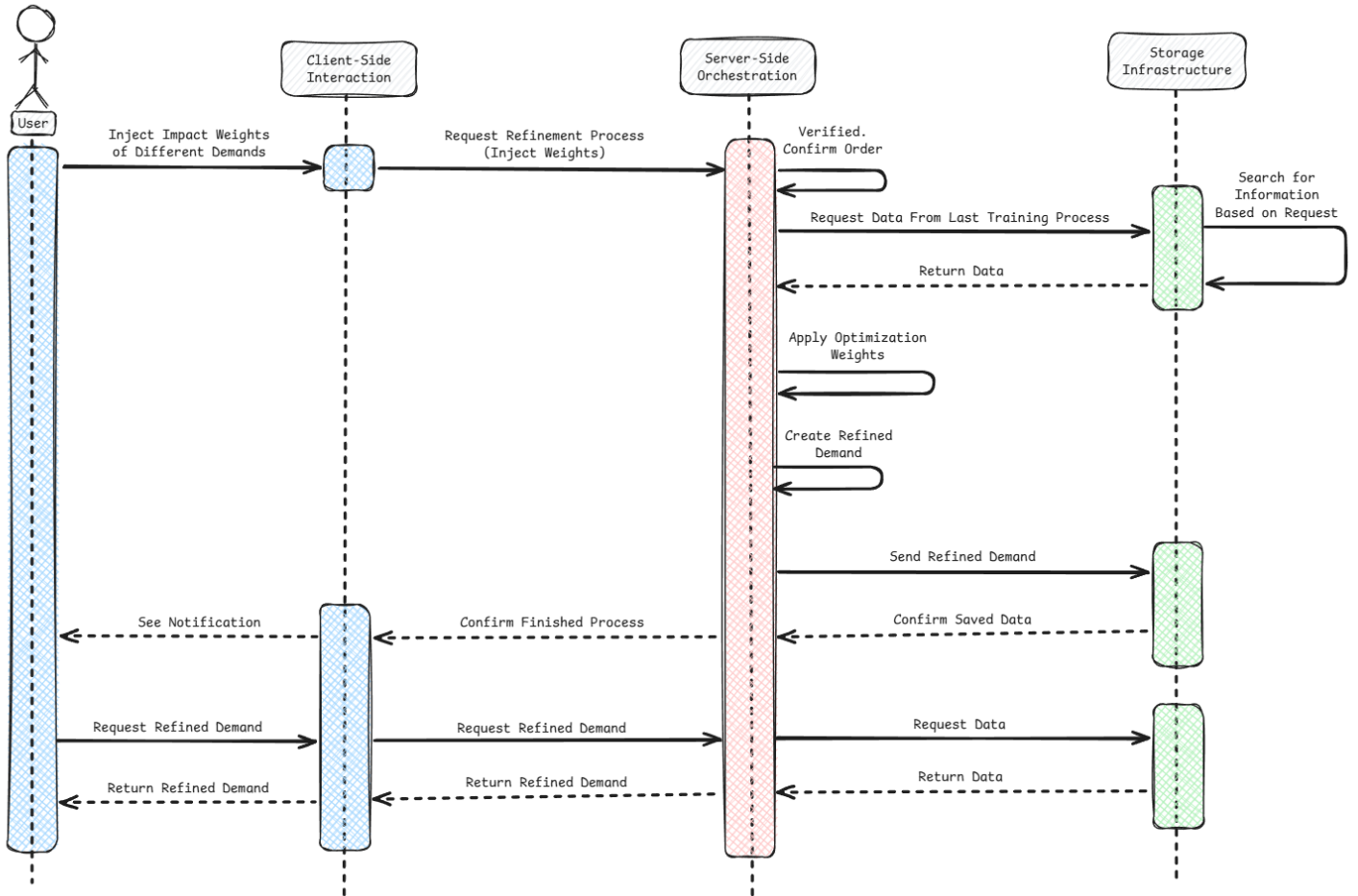


Figure 4.4: Sequence Diagram: Refinement with User Impact Process.

The final system layer, as mapped in Figure 4.5, handles physical plan generation, drawing the singular unified output from the refinement process to drive all down-stream Shop Floor execution. In a standard baseline planning sequence, the server queries this finalized demand record, processes a recursive recipe decomposition via the Bill of Materials (BOM) service, tracks physical lot expiration limits, and calculates the baseline manufacturing and procurement timelines.

Alternatively, if an administrator needs to evaluate factory resilience under extreme changes, they can actively inject a scenario-based disruption package directly through the client interface, as illustrated in Figure 4.6. The backend captures these stress-test arguments and matches them against historical crisis coefficients stored in the database.

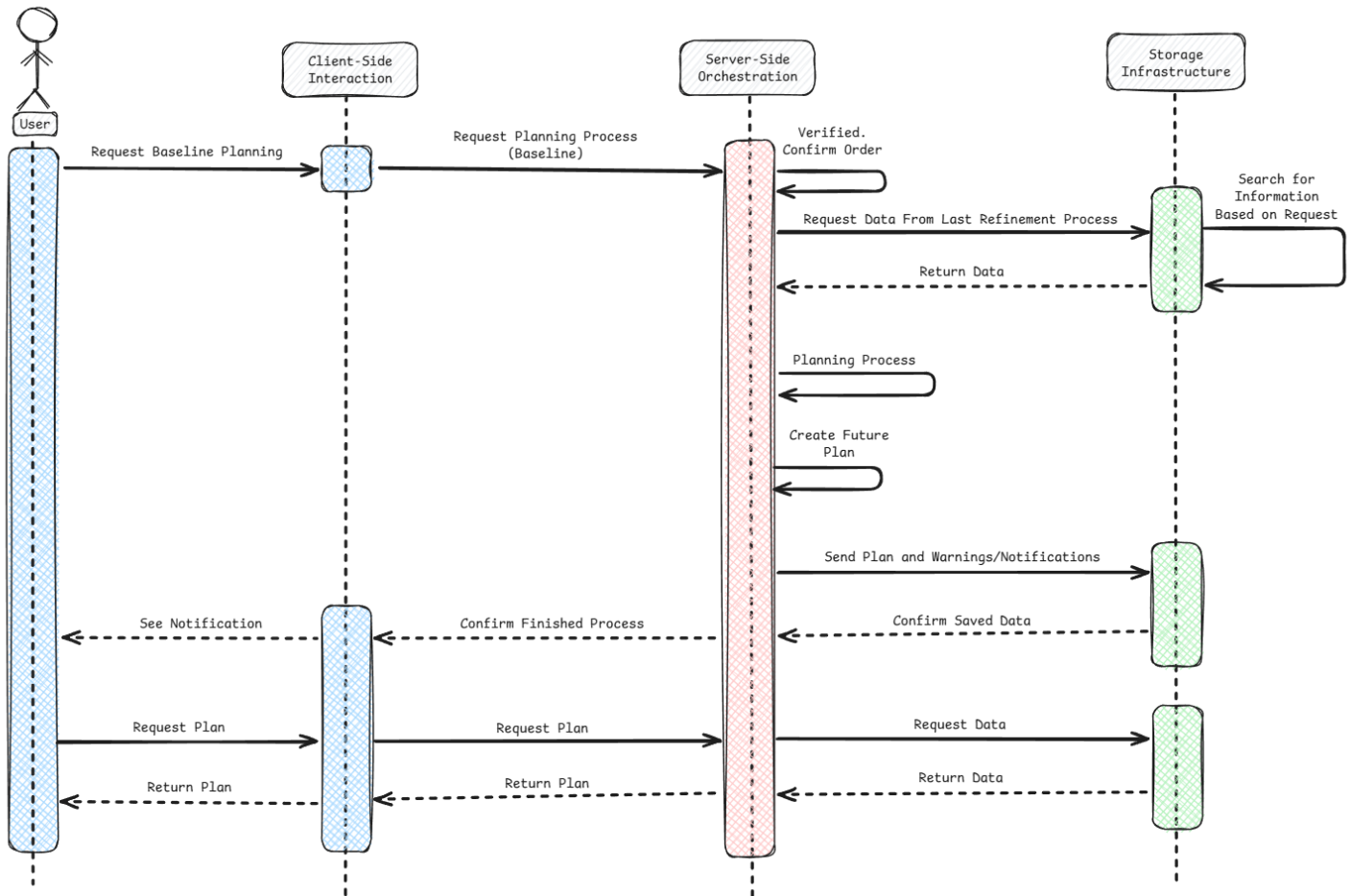


Figure 4.5: Sequence Diagram: Planning Process.

By applying these macro scaling parameters to the inventory models, the planner flags raw material shortages or buffer safety violations. The resulting plan, updated with warning notices and altered material deadlines, is written to the database infrastructure, immediately refreshing the connected workstations and loading docks for real-time risk assessment.

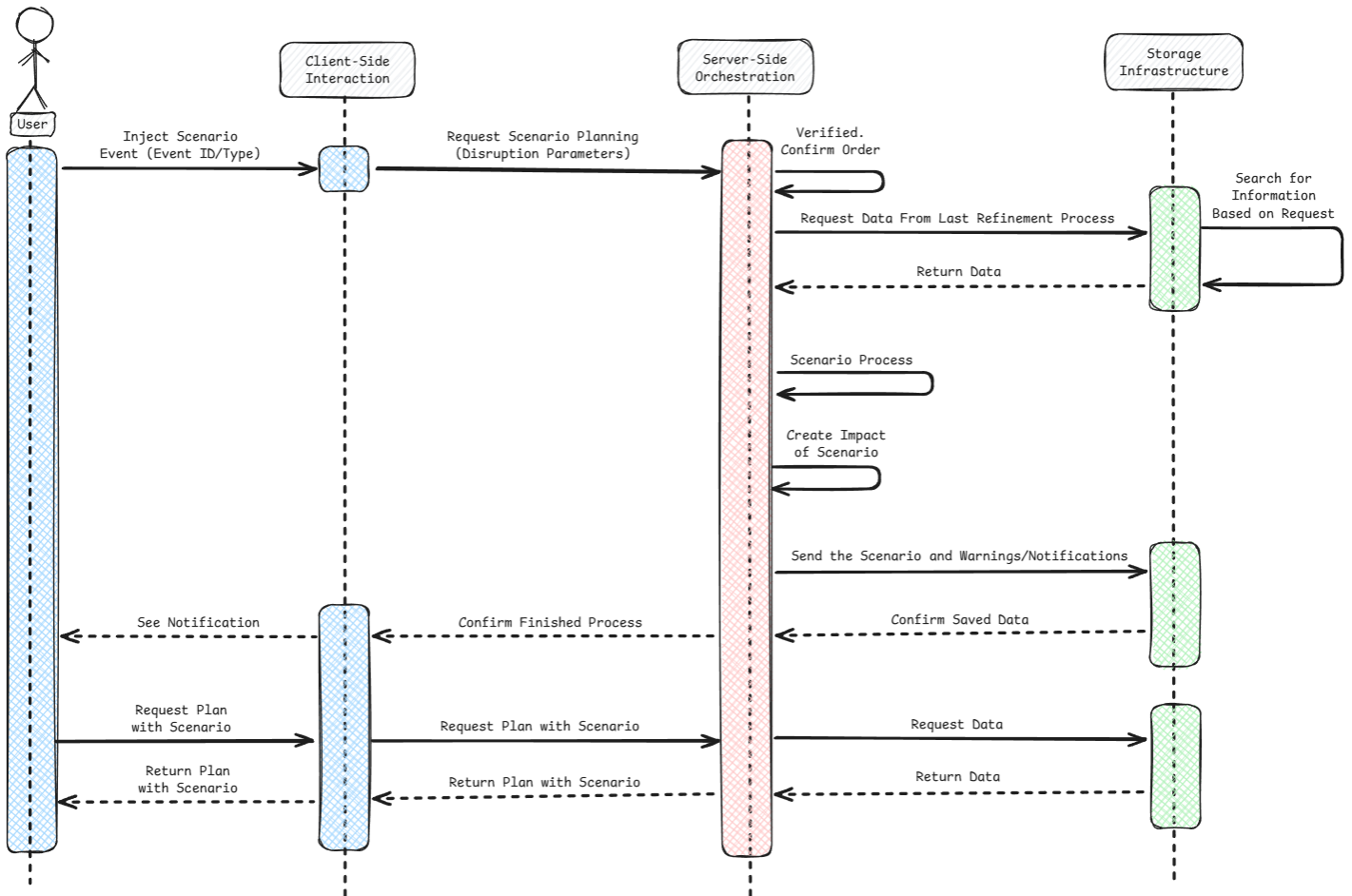


Figure 4.6: Sequence Diagram: Scenario Impact on Planning Process.

4.1.3 Implementation Scope and System Maturity

To provide full transparency before evaluating the system's performance, it is necessary to explicitly map out the current implementation boundaries of the functional prototype. While the system architecture and dynamic sequences detailed throughout this chapter represent the complete design for the production-ready platform, the practical execution of this first phase faced evolving technical challenges that required strategic scoping. As the development progressed, unforeseen data complexities emerged, significantly increasing the development time required to stabilize the backend. To successfully conclude this foundational phase of the project, a strict prioritization framework was established to fully code and validate the high-value computational core, while leaving auxiliary communication pipelines and granular interface screens as high-fidelity conceptual blueprints for subsequent deployment phases.

The initial development phase prioritized building scalable, general-mode Analytical Core, deliberately decoupled from any specific, static algorithm. Because a primary project requirement was to evaluate multiple concurrent predictive models, the core architecture was designed to be modular: adding, removing, or upgrading a machine learning model in future phase does not break the active system execution, and the framework recognizes new model streams and handles individual processing errors without failing outright.

To demonstrate this modularity in Phase 1, the core was integrated with six distinct modeling approaches - Random Forest, Prophet, LightGBM, XGBoost, Chronos, and Granite - as illustrated in Figure 4.7. This same structural logic extends across both the planning and refinement engines.

The resilience of this modular design is tested directly in Chapter 5: all six models execute successfully against the controlled 2025 dataset in Scenario 1, while only three complete execution under the unsupervised 2026 data in Scenario 3 — without the platform failure, a direct consequence of the decoupling described here.

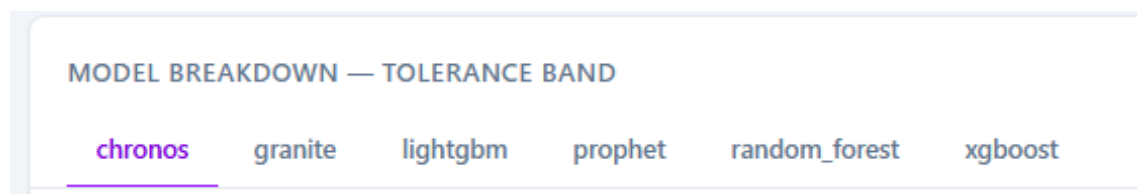


Figure 4.7: Different predictive models used.

To maintain focus on this core analytical execution, certain trade-offs were made regarding real-time system telemetry and network messaging. In this current prototype phase, all operational messages, execution warning logs, and process progress indicators are written directly to, and managed by, the central database infrastructure. Building a dedicated, live websocket gateway or an event-driven tunnel to stream real-time progress bars directly from the deep backend execution threads to the frontend interface was explicitly classified as a lower priority. The focus was firmly directed at proving the algorithmic integrity of the system first, utilizing the relational database as the reliable intermediary state synchronization layer.

This prioritization also shaped the current state of the front-end application interfaces. The prototype currently features a streamlined, high-level user interface designed to validate core roles rather than host the completely detailed workstation zones. At this stage, the system implements Role-Based Access Control (RBAC) restricted to primary Manager and Operator credentials, locking specific features based on the active session. This front-end environment delivers four prioritized, centralized views that directly validate the backend’s capabilities: a training results viewer to audit machine learning outputs, a refinement console to review combined demand curves, a scheduling screen to visualize time-phased planning requirements, and a comprehensive unified warning panel.

These four views structure the evaluation presented in Chapter 5: Scenario 1 exercises the training and refinement console against controlled 2025 data, Scenarios 2 and 3 extend the scheduling screen and warning panel to disruption and live-data conditions, respectively.

Rather than acting as static dashboards, these views explicitly support human-in-the-loop parameter manipulation to evaluate different execution results. Within the refinement console, the user is empowered to manually alter and choose the specific impact weights of each underlying forecasting model, as shown in Figure 4.8, dynamically merging the different machine learning outputs into a finalized, human-hardened demand configuration that directly feeds the planning process.

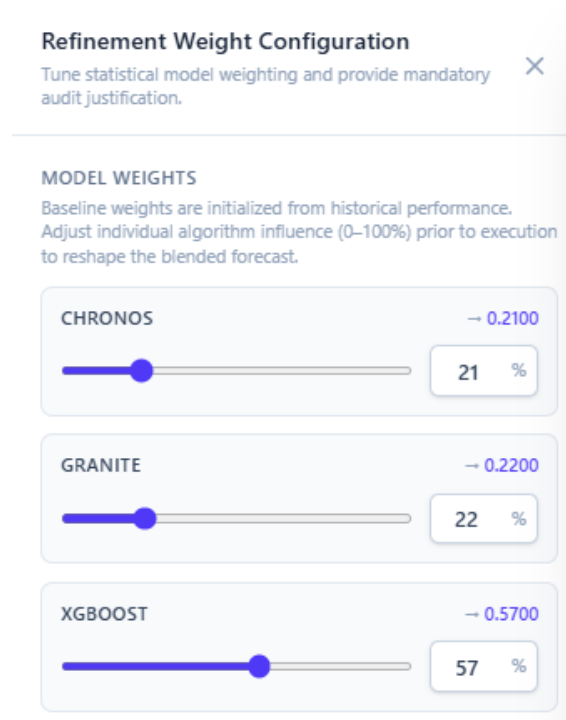


Figure 4.8: Prototype Interface showing Interactive Refinement Weights Inputs.

Following this data flow into the scheduling screen, the operator can interactively submit simulation requests for macro-environmental disruptions, such as wars or pandemics as illustrated in Figure 4.9. The interface captures these custom disruption arguments, passes them to the backend computation models, and renders modified production schedules so managers can visually analyze the immediate structural impact of these crises on material availability and factory timelines.

Scenario Configuration

Configure disruption scenarios and constraints. Executing a new planning cycle will recalculate master schedules across all active items using these parameters. ✕

Parameter Overrides — User-defined scenario configurations strictly constrain the ML planning engine. All adjustments are permanently logged in the system audit trail.

SCENARIO / EVENT TYPE

Select a predefined disruption model to stress-test current supply chain resilience.

Geopolitical Crisis ▼

GEOPOLITICAL CRISIS PARAMETERS

Historical Reference

Ukraine War (2022) ▼

The planning engine will apply statistical disruption patterns mapped from this historical event to evaluate current inventory risk.

Figure 4.9: Prototype Interface Scenario Disruption Simulation Inputs.

Finally, the comprehensive warning view consolidates all automated system alerts, as presented in Figure 4.10, data density errors, and SKU inactivity flags generated across the entire processing lifecycle into a single operational interface designed specifically for managers. Developing the fully isolated, room-specific dashboards for individual floor workers remains a task for future operational expansion.

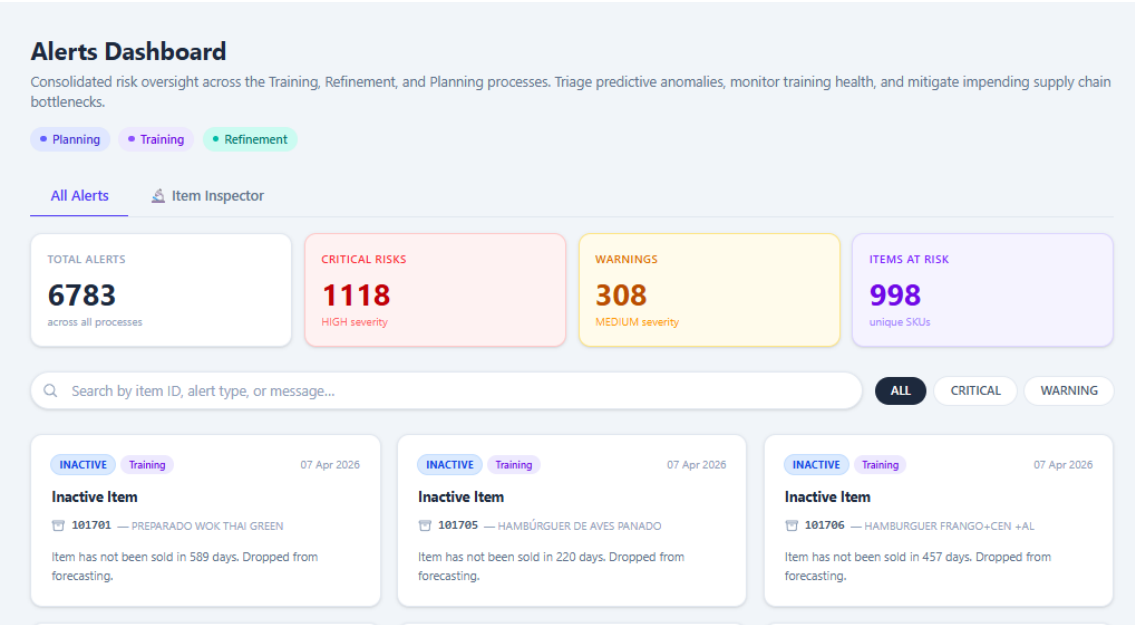


Figure 4.10: Prototype Interface Warnings View.

To thoroughly test and prepare this architecture for future deployment, the prototype was deployed across two distinct execution environments. The first is a highly controlled development and verification environment running on historical data. This testing ground allows for the deep validation of models, the optimization of data preparation inputs, and the close auditing of database interactions. This controlled environment functions as a simulation of the past, re-creating factory plans for the entire year of 2025. This allows the system’s prescriptive outputs to be compared directly against the actual, real-world historical results of 2025 to prove the validity of the software. The year 2025 was also chosen because it was the most recent complete year of historical data available at the start of the project, giving the development environment a known outcome to validate the prototype’s early modeling decisions against. The second deployment is a non-controlled, live environment representing the active year of 2026. In this environment, the system operates without the benefit of hindsight, ingesting data day-by-day as time progresses to simulate true production conditions.

Because enterprise database structures vary significantly across industrial clients, the prototype had to adapt to a complex, real-world data environment. This work was made possible through a partnership with Tecnicamente, the company with which the author collaborates, which authorized and facilitated access to a production database belonging to one of its client companies, a Portuguese food manufacturing organization, containing transactional records from 2014 to the present year, 2026. Beyond granting data access, Tecnicamente’s team supported the project by clarifying the semantics of the client’s internal database schema, helping bridge the gap between raw records and the structured data models required by the prototype. This raw data was extracted, cleaned, and refined by the author to construct the reliable baseline datasets used to train and validate the

system’s predictive models. Because a live factory database cannot be modified and the raw data provided did not contain an explicit, structured table detailing product recipes, a dedicated data-mining process was constructed to reconstruct this information, as illustrated in Figure 4.11.

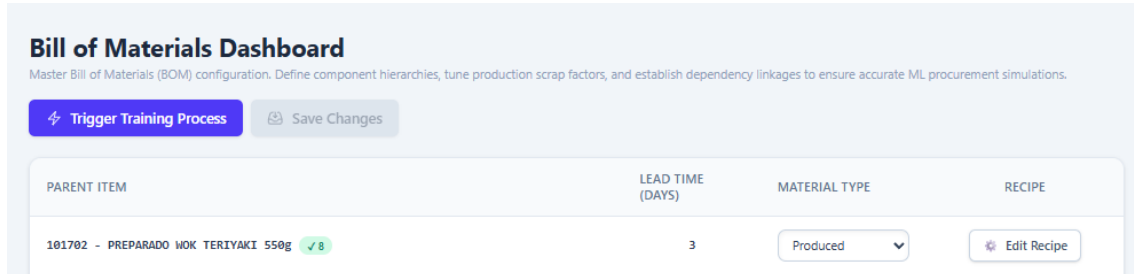


Figure 4.11: Prototype Interface Bill of Materials View.

Because the source data came directly from a live factory database rather than a pre-structured benchmark dataset, most of the preparation effort went into making the raw records usable at all, not just accurate. Relevant fields were cross-referenced across the available transactional tables, joining sales records, inventory logs, and production entries where a common key allowed it, so that the training pipeline could draw on a more complete picture of each item than any single table provided on its own.

During execution, the training process retrieves this data through the required backend services and applies an initial filtering pass that discards columns carrying no predictive value — administrative metadata, notes, and other fields irrelevant to demand behavior —, reducing the noise present in the raw export before any model sees it. This general cleaning pass is deliberately broad: since the same underlying dataset feeds six structurally different models, columns removed at this stage are columns none of them could use, rather than ones specific to any single algorithm.

A second, model-specific pass then prepares the data in the form each model expects, as summarized in Table 4.1. All models start from the same monthly series per item, in which months without recorded sales are filled with zero so that every item covers the same continuous calendar. From that point the two model families diverge. The tree-based models do not read a time series directly; for each month they receive a row of features describing the recent past and learn to predict the following month’s demand. The sequence models (Prophet, Chronos, and Granite) receive the monthly demand series itself.

Table 4.1: Model-specific data preparation.

Model	Input	Model-specific preparation
Random Forest, XGBoost, LightGBM	Feature table per item and month; target is next month's demand	Demand capped at the item's 95th percentile; demand of the previous one to three months; three-month moving average; exponential moving averages over three and six months and their ratio as a trend indicator; calendar features; holiday flag (on monthly data, active only for January); first months without complete lag values removed
Prophet	Date and demand per item	Fitted separately for each item; multiplicative yearly seasonality; Portuguese public holidays; items with fewer than two months of training data skipped
Chronos	Monthly demand history per item	Pre-trained model used without further training; history scaled to the 0–1 range using training data only; forecast taken as the 70th percentile of the sampled paths; negative values set to zero
Granite	Monthly demand history per item	Pre-trained model used without further training; fixed context of 512 periods, with shorter histories padded by repeating the oldest value; scaled to the 0–1 range using training data only

This component reconstructs each recipe from the client's production records, as detailed in Figure 4.12. It reads the production documents recorded since January 2024 and groups their lines by document, session, and workstation, so that each group corresponds to one production event. Within a group, products entering stock are treated as parents, and raw materials and consumed intermediate products as components. For each parent–component pair, the ratio between the quantity consumed and the quantity produced is recorded, and QTY REQ is the average of these ratios across all production events in which the pair appears. To avoid implausible pairings when several products are made in the same event, fish-based components are not linked to meat-based products, and vice versa, based on keywords in the item descriptions. SCRAP represents the expected loss factor for a component during production, such as trimming or processing waste; in the prototype it is set to a uniform 2% for all components, as no per-component estimate was implemented. These per-component values allow the planning engine to expand a single finished-product demand figure into net raw-material requirements downstream.

Recipe Builder
Parent: 181782 - PREPARADO WOK TERIYAKI 550g — check components and set their quantities

COMPONENT	QTY REQ	SCRAP
☒ FRANGO TRIMMING COM MARINADA ID: 117003	0.491699	0.019999
☒ FRANGO PEITO CONGELADO ID: 117004	0.599300	0.019999
☒ ADITIVO ALIMENTAR SACO CO ID: 121917	0.005100	0.019999
☒ ADITIVO ALIMENTAR SACO CO ID: 121922	0.006099	0.019999
☒ SAL FINO ID: 123911	0.001799	0.019999
☒ MISTURA CONDIMENTO WINZER ID: 123924	0.014899	0.019999
☒ ÓLEO GIRASOL ID: 126501	0.051899	0.019999
☒ MISTURA VEGETAIS P/ WOK TERIYAKI ID: 128002	0.375999	0.019999

Figure 4.12: Prototype Interface Recipe of Item View.

The reconstructed quantities are therefore estimates derived from real production records rather than simulated values, but they were not validated against the manufacturer's actual recipes. When a production event contains several products, each component is attributed to all of them, which can overstate some ratios. Products for which no component could be identified receive a placeholder component with a one-to-one ratio, marked as mock data so that the planner can still process them. Lead times, whether for receiving raw materials, producing, or delivering, were not available in the client's records. The values used by the planner, shown in Figure 4.11, were therefore assigned randomly between two and seven days so that its scheduling logic could be exercised. The reconstructed BOM serves to demonstrate the recursive decomposition and scheduling logic rather than to provide exact material quantities.

Following this extraction, a separate inventory simulation component generates the lot-level stock data required by the planning engine. For each item, it estimates the typical shelf life as the median interval between production and expiration dates in the client's inventory records, using seven days when no valid dates exist. It then creates two synthetic lots relative to the simulation date of 15 February 2025: one expiring two days later and one expiring after the estimated shelf life, both with a physical quantity taken from the item's inventory records. These lots are plausible rather than real, so the waste and shortage results that depend on them in Chapter 5 illustrate the First-Expired-First-Out (FEFO) logic rather than the client's actual stock position. Keeping recipe reconstruction and inventory simulation in the infrastructure layer isolates the analytical engines from the structure of the client's database, so a future deployment could replace both with real recipe and lot data without changing the forecasting or planning logic.

4.2 Technologies

The selection of specific technologies for this system was driven by two primary decision criteria: alignment with the industrial requirements and operational constraints of the partner organization, and the technical capability to run six structurally different machine learning models concurrently against multi-year transactional data. The design concentrates on two strategic technology domains, each selected for specific operational responsibilities.

API Management: Java and Quarkus

The choice of Quarkus as the primary framework for web services and orchestration functionality reflects deliberate strategic considerations.

Industrial Alignment and Organizational Capability Development: The partner organization explicitly requested the adoption of Quarkus to build internal technical knowledge and evaluate its suitability for future production initiatives. This strategic selection recognizes that technology decisions must balance immediate project requirements with longer-term organizational learning objectives.

Lightweight Cloud-Native Architecture: Quarkus is a modern Java framework engineered specifically for cloud-native microservices deployment. In contrast to traditional Java enterprise frameworks characterized by substantial memory footprints and initialization overhead, Quarkus employs compile-time dependency injection and native image compilation techniques, resulting in minimal runtime overhead ([Red Hat, 2026](#)).

Computational Intelligence: Python 3.12 and FastAPI

While the orchestration and state management responsibilities are concentrated within the Java technology domain, the intelligence and analytical computation layer is implemented exclusively in Python 3.12.

Model Compatibility and Machine Learning Ecosystem: Python 3.12 was selected as the core language for analytical computation because it offers the highest compatibility with contemporary machine learning libraries and modern deep learning frameworks. The Python ecosystem has emerged as the de facto standard environment for machine learning model development, with comprehensive libraries for statistical modeling, ensemble techniques, and neural architectures ([Python Software Foundation, 2023](#)).

Asynchronous Job Dispatching: FastAPI enables the computational layer to handle asynchronous job dispatching, ensuring that the user-facing API remains responsive even during execution of computationally intensive training and optimization tasks. Using Python's asynchronous I/O ecosystem, the system distributes heavy computational burden across multiple CPU cores without introducing thread-safety complexity or blocking the orchestration interface ([Ramírez, n.d.](#)).

Predictive Modeling Methodologies

Rather than relying on a single algorithm, the prototype combines six forecasting models from three families, so that the combined forecast does not depend on one modeling assumption. Their inputs are listed in Table 4.1 and their configuration in Table 4.2; parameters not listed were left at the library defaults. The configurations were selected during development by comparing forecasts with 2025 demand across all items (Section 5.1).

Tree-based models: Random Forest (Breiman, 2001), XGBoost (Chen and Guestrin, 2016), and LightGBM (Ke et al., 2017) are each trained as a single model across all items, using the feature table described in Table 4.1. The item identifier is excluded from the features, so these models learn how demand relates to its recent history in general rather than memorizing individual items. Random Forest averages many deep trees grown independently, while XGBoost and LightGBM add shallower trees one at a time, each correcting the errors of the previous ones; their low learning rates (0.01 and 0.05) combined with 500 trees favor gradual, stable fits over training speed.

Prophet: Prophet (Taylor and Letham, 2018) is fitted separately for each item and decomposes its monthly series into a trend and a yearly seasonal component. Seasonality is multiplicative, so seasonal peaks scale with the item’s overall volume, and the changepoint prior scale was raised from its default of 0.05 to 0.1 so that the trend can react faster to structural breaks. Weekly and daily seasonality are disabled, as they cannot be observed in monthly data.

Foundation models: Chronos (Ansari et al., 2024) and Granite (Ekambaram et al., 2024) are pre-trained on large collections of time series and are used without further training (zero-shot): each receives an item’s history and returns a forecast directly. Chronos, a transformer-based model, produces a set of sampled future paths, from which the prototype takes the 70th percentile rather than the median, which biases its forecasts upward. Granite TinyTimeMixer is a lightweight, non-transformer architecture that requires a fixed input context of 512 periods.

Combining the three families means that a weakness specific to one approach, such as the foundation models’ difficulty with long runs of zero demand, is diluted in the combined forecast rather than passed directly to the planner.

Table 4.2: Model configuration used in the prototype.

Model	Implementation	Configuration
Random Forest	scikit-learn RandomForestRegressor	300 trees; maximum depth 30; random seed 42
XGBoost	XGBRegressor	500 trees; learning rate 0.01; maximum depth 6; row and column subsampling 0.8; minimum child weight 1; random seed 42
LightGBM	LGBMRegressor	500 trees; learning rate 0.05; maximum depth 5; 31 leaves; minimum 10 samples per leaf; column subsampling 0.8; row subsampling set to 0.8 but inactive, as the bagging frequency was left at its default of 0; random seed 42
Prophet	Prophet, one model per item	Multiplicative yearly seasonality; weekly and daily seasonality disabled; changepoint prior scale 0.1; seasonality prior scale 10; Portuguese public holidays
Chronos	amazon/ chronos-t5-small	Zero-shot; full item history as context; 70th-percentile point forecast; CPU inference
Granite	ibm-granite/ granite-timeseries- ttm-v1	Zero-shot; context length 512; CPU inference

VALIDATION & RESULTS

The prototype is evaluated in three scenarios, each testing a different part of the system. Scenario 1 (Section 5.2) tests the forecasting pipeline under normal conditions: models trained on data up to January 2025 forecast the following months of 2025, and the forecasts are compared with the demand actually recorded in that period. Scenario 2 (Section 5.3) tests how the planning layer responds to an external shock: a demand multiplier derived from the 2022 outbreak of the war in Ukraine is applied to the forecast, and the resulting changes in material requirements and alerts are examined. Scenario 3 (Section 5.4) runs the unchanged pipeline on uncontrolled 2026 data from March onward, to observe its behavior on data that played no part in its development. Section 5.1 first sets out the limitations that apply to all three.

5.1 Evaluation Setup and Limitations

Two aspects of the weighting described in Section 4.1.2 limit how far the model weights can be read as a fair comparison between models. First, the tree-based models forecast one month ahead from observed lag values, while Prophet, Chronos, and Granite forecast the whole evaluation window from the end of the training data; the tree-based models are therefore evaluated on an easier task and tend to receive larger weights. Second, the outlier clipping applied during feature engineering computes its threshold over the full series, including the evaluation months, which makes the RMSE of the tree-based models optimistic.

In Scenario 1 (Section 5.2) the training data ended in January 2025, so every fold was evaluated on months up to that date, and the weights were not derived from the February–December 2025 period used to assess the combined forecast. The model hyperparameters, however, were selected during development by comparing the forecasts for all items with actual 2025 demand. A single parameter set was chosen per model for the whole catalog rather than per item, which limits overfitting to individual items, but the Scenario 1 results still reflect some tuning on that period. Scenario 3 (Section 5.4) reused the same parameters without further adjustment, so its 2026 data played no part in tuning, making it the more independent test of the approach.

Together with the reconstructed recipes, simulated lead times, and synthetic inventory lots used by the planning layer, these limitations mean that the three scenarios provide a functional validation of the prototype: they show that its mechanisms work as designed, not how much they improve planning in practice, as discussed in Chapter 6.

5.2 Baseline Performance Analysis (Scenario 1 Execution)

The objective of Scenario 1 is to establish a performance reference for the system under normal market conditions. The models are trained on monthly demand from 2014 to January 2025 and forecast the remaining months of 2025, a period without major external disruptions in the client's demand. The scenario examines whether the individual models and their combination reproduce the yearly seasonal pattern of food demand. Two uses of the RMSE should be distinguished here: each model's walk-forward RMSE was computed and determines its weight in the combined forecast (Section 4.1.2), but the error of the combined forecast itself was not computed across the product catalog, so its accuracy is assessed visually for a representative item.

The initial phase of validation focuses entirely on setting this performance reference. As the primary engine of predictive intelligence, the background training process avoids relying on a single algorithm. Instead, it generates a heterogeneous ensemble comprising six distinct modeling approaches to capture a multidimensional risk map of future demand. Figures 5.1 through 5.6 illustrate this multi-model generation for a representative SKU. The models' inputs and configuration are described in Section 4.2 (Tables 4.1 and 4.2).

In each of the following per-model charts (Figures 5.1 through 5.6), the orange line shows actual historical demand for the sampled SKU, while the purple line and shaded band show that individual model's forecast and its 90% prediction interval — allowing each model's raw output to be inspected before all six are combined into a single baseline.

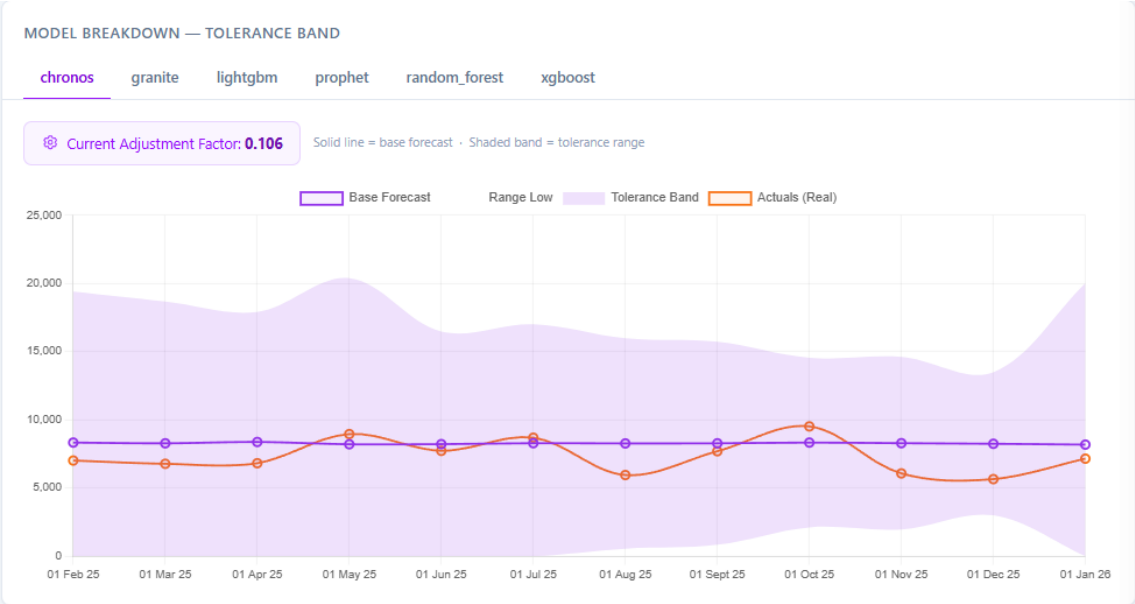


Figure 5.1: Model Output: Chronos.

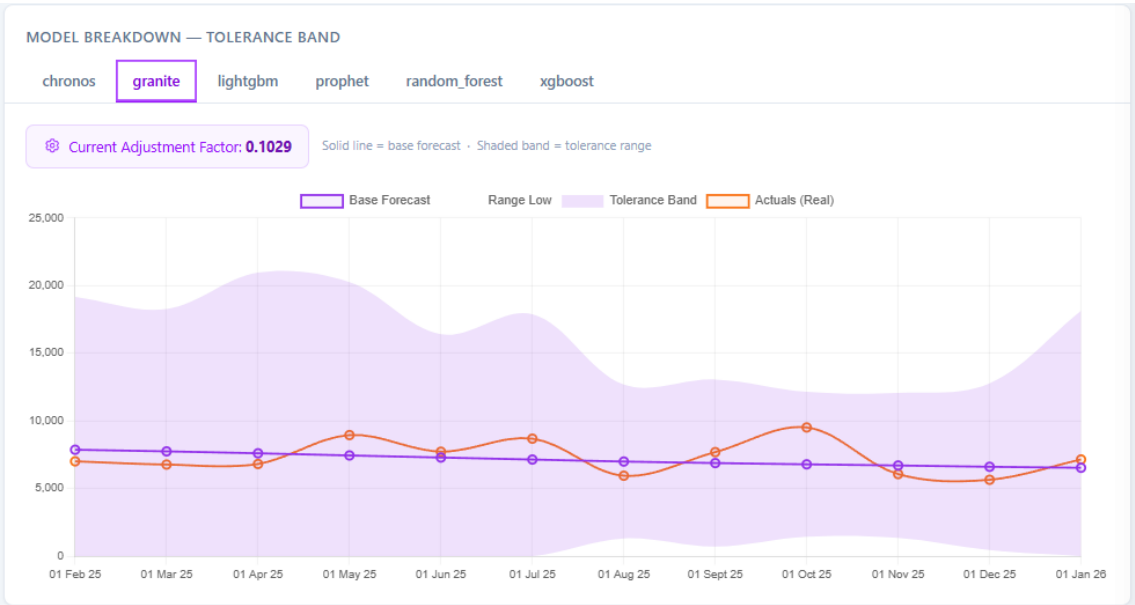


Figure 5.2: Model Output: Granite.

5.2. BASELINE PERFORMANCE ANALYSIS (SCENARIO 1 EXECUTION)

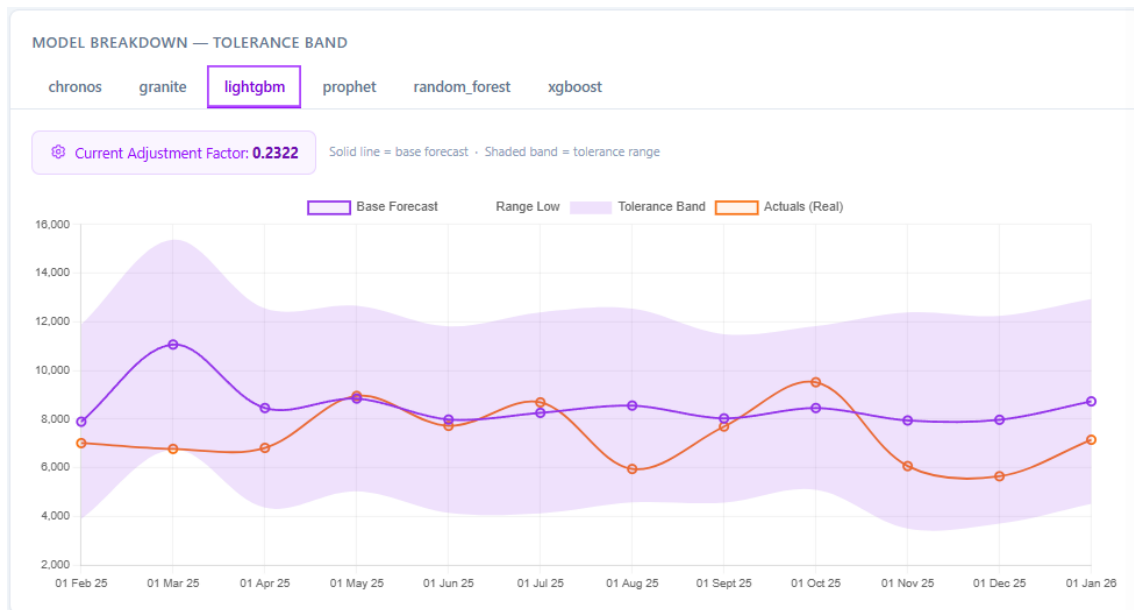


Figure 5.3: Model Output: LightGBM.

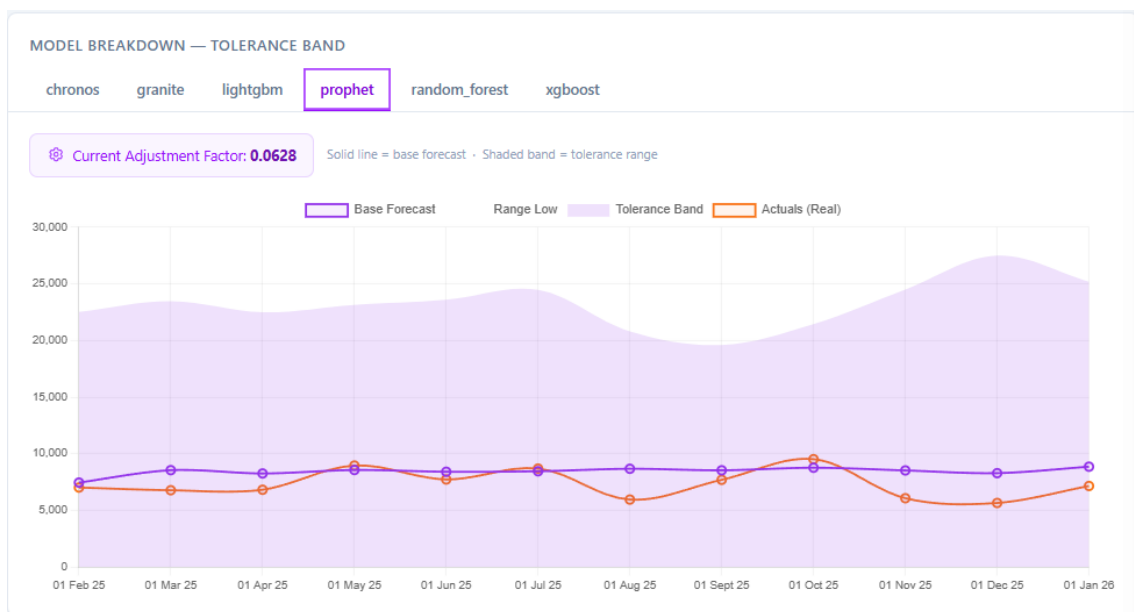


Figure 5.4: Model Output: Prophet.

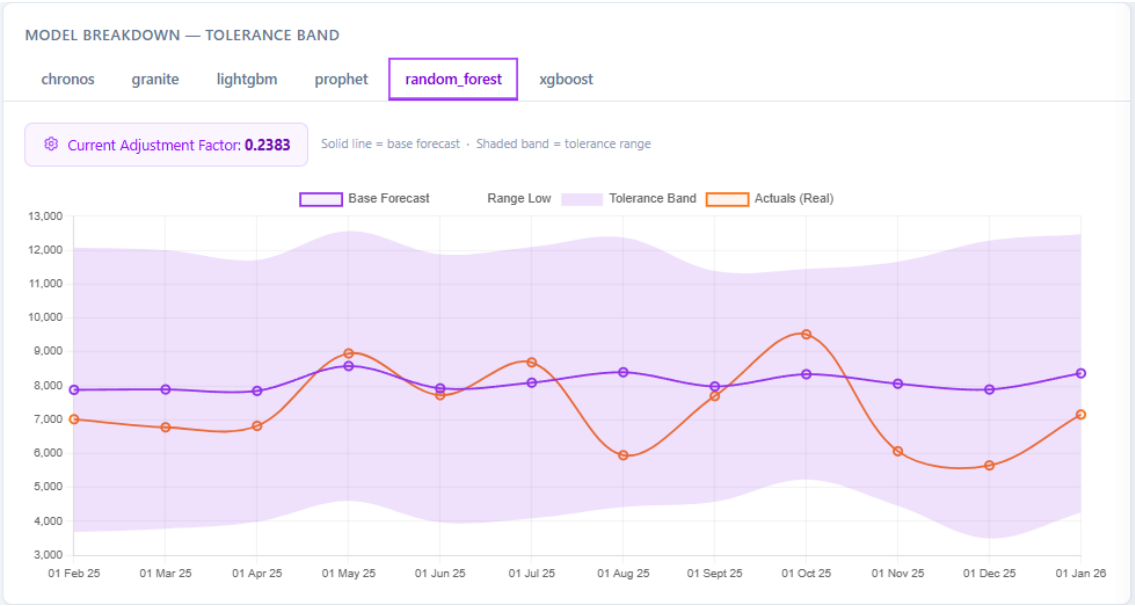


Figure 5.5: Model Output: Random Forest.

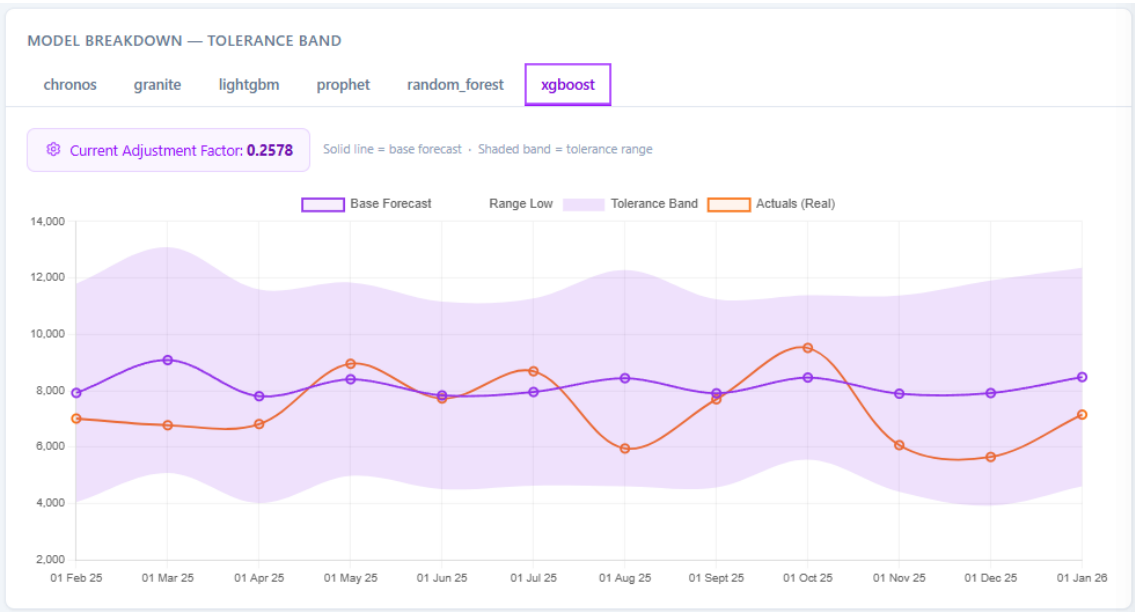
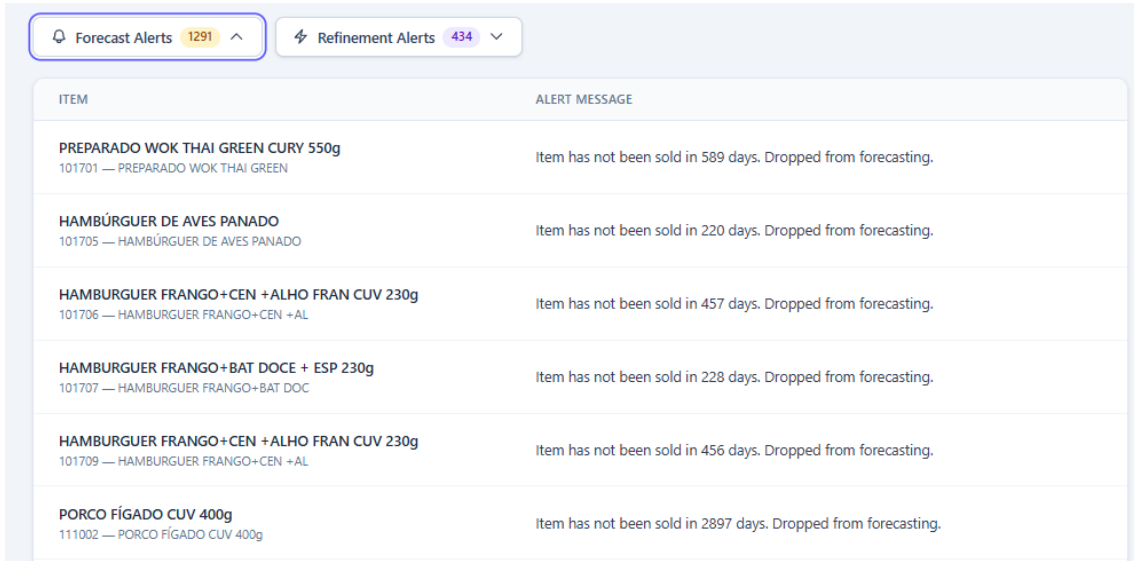


Figure 5.6: Model Output: XGBoost.

System validation extends beyond raw statistical tracking to evaluate data integrity and automated safety constraints. The background training process features an embedded fail-safe data validation gate that monitors product states in real-time. As demonstrated in Figure 5.7, the engine runs an automated inactive item detection script during the ingestion cycle, immediately flagging SKUs that have ceased market movement. This self-aware data audit stops the system from allocating valuable computational database resources to dead inventory profiles, ensuring processing efficiency remains optimized.



ITEM	ALERT MESSAGE
PREPARADO WOK THAI GREEN CURRY 550g 101701 — PREPARADO WOK THAI GREEN	Item has not been sold in 589 days. Dropped from forecasting.
HAMBÚRGUER DE AVES PANADO 101705 — HAMBÚRGUER DE AVES PANADO	Item has not been sold in 220 days. Dropped from forecasting.
HAMBURGUER FRANGO+CEN +ALHO FRAN CUV 230g 101706 — HAMBURGUER FRANGO+CEN +AL	Item has not been sold in 457 days. Dropped from forecasting.
HAMBURGUER FRANGO+BAT DOCE + ESP 230g 101707 — HAMBURGUER FRANGO+BAT DOC	Item has not been sold in 228 days. Dropped from forecasting.
HAMBURGUER FRANGO+CEN +ALHO FRAN CUV 230g 101709 — HAMBURGUER FRANGO+CEN +AL	Item has not been sold in 456 days. Dropped from forecasting.
PORCO FÍGADO CUV 400g 111002 — PORCO FÍGADO CUV 400g	Item has not been sold in 2897 days. Dropped from forecasting.

Figure 5.7: Alerts Inactive generated by Training Process.

To construct this unified projection, the refinement logic applies the Inverse Error Weighting protocol introduced in Section 4.1.2: each of the six models is assigned a weight inversely proportional to its walk-forward error, so models with lower historical error receive a larger share of influence over the combined forecast, while less accurate models are down-weighted rather than discarded. This lets the system favor the more accurate models without requiring manual model selection.

Rather than a single point forecast, the refinement block produces a range. It adds and subtracts 1.65 times the combined error around the blended forecast, $\hat{y}_t \pm 1.65 \overline{\text{RMSE}}_t$, where $\overline{\text{RMSE}}_t$ is the weighted average of the six models' walk-forward RMSE, using, when available, the error of the walk-forward fold whose cut-off falls in the same calendar month as the forecast. Assuming approximately normal forecast errors, this band corresponds to a 90% prediction interval, and its upper bound is a one-sided 95% limit: actual demand is expected to exceed it in about 5% of months. The bounds are stored as Range_Up and Range_Down, as shown for Item 101703 in Figure 5.8. The planning engine uses the distance between Range_Up and the forecast as safety stock, so the material planning phase receives each forecast together with its expected error range.

As shown in Figure 5.8, the purple per-model lines used in the individual charts above are replaced here by a single blue line — the combined output of the Inverse Error Weighting protocol — while the orange line continues to represent actual demand and the shaded band now reflects the 90% prediction interval computed for this blended forecast.

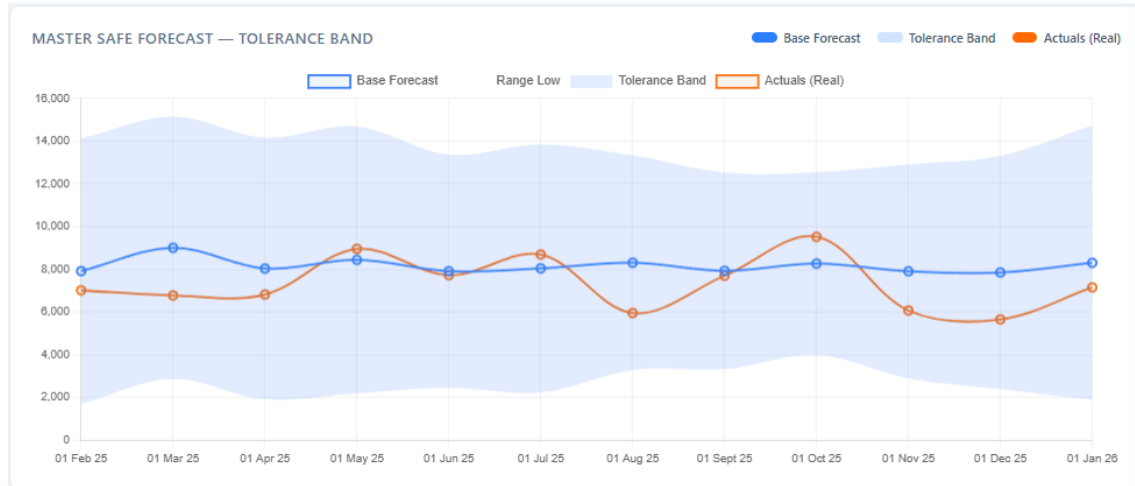


Figure 5.8: Final output from Refinement Process.

The refinement process also functions as an automated heuristic auditor to preserve operational safety on the floor. While the initial training algorithms optimize purely for mathematical error reduction, the refinement engine monitors data density parameters to flag commercial risk. During experimental runs, the system generates targeted data history warning logs whenever an active SKU contains less than one full year of transactional records. Because the item has not yet completed a complete seasonal business cycle, the optimization logic attaches a medium-certainty flag to the entry. This visual alert warns the material planning service to apply a conservative inventory procurement strategy to these young SKUs, mitigating the financial risk of over-purchasing unverified items, as illustrated in Figures 5.9 through 5.10.

5.2. BASELINE PERFORMANCE ANALYSIS (SCENARIO 1 EXECUTION)

ITEM	ALERT MESSAGE
HAMBURGUER FRANGO+BAT DOCE + ESP 230g 101708 — HAMBURGUER FRANGO+BAT DOC	Models predict 46 units, but item has had ZERO sales in 44 days. Review before purchasing.
SALMÃO "SALMO SALAR" LOMBOS SKINPACK 250g 116148	Models predict 1017 units, but item has had ZERO sales in 9999 days. Review before purchasing.
FILETE ROBALO "DICENTRARCHUS LABRAX" 119207	Models predict 26 units, but item has had ZERO sales in 561 days. Review before purchasing.
ENTRECOSTO TEMPERADO PD TALHO kg 112049	Models predict 72 units, but item has had ZERO sales in 9999 days. Review before purchasing.
FILETE DOURADA "SPARUS AURATA" 119206	Models predict 110 units, but item has had ZERO sales in 9999 days. Review before purchasing.
FILETES ROBALO "DICENTRARCHUS LABRAX" SKINPACK 220G 116149	Models predict 2967 units, but item has had ZERO sales in 9999 days. Review before purchasing.

Figure 5.9: Alerts generated by Refinement Process.

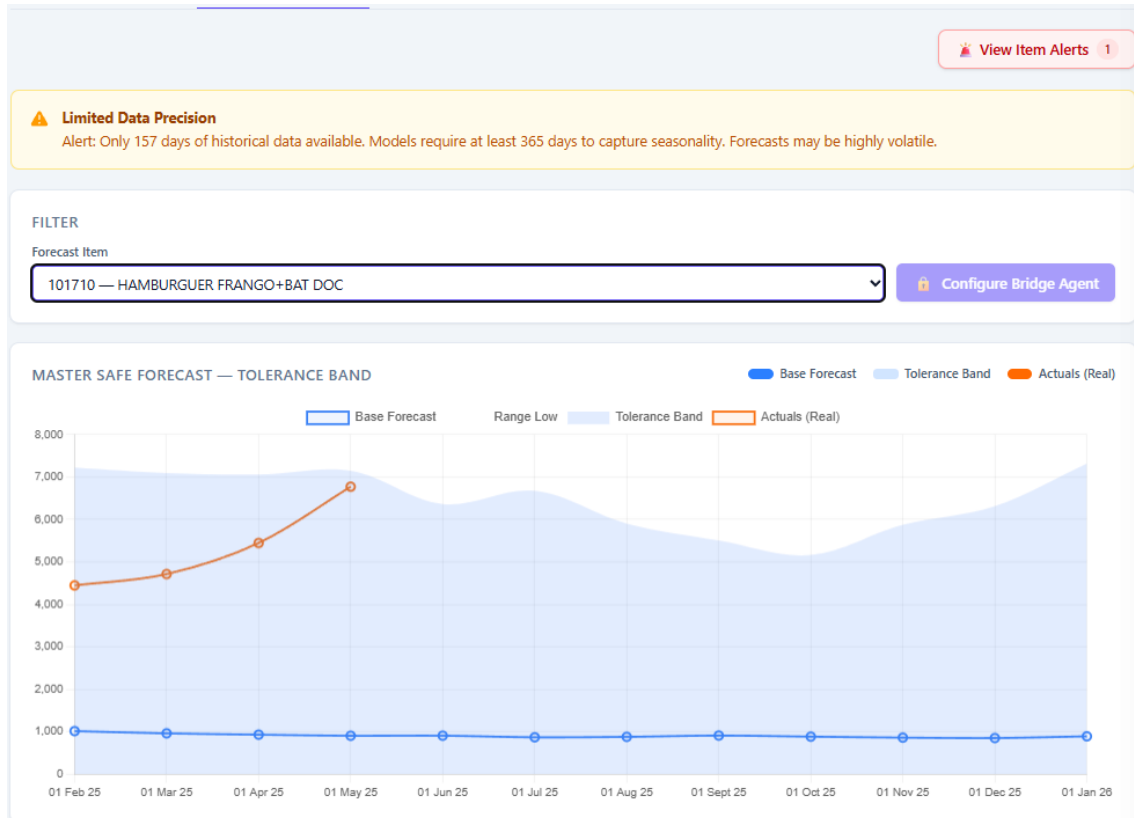
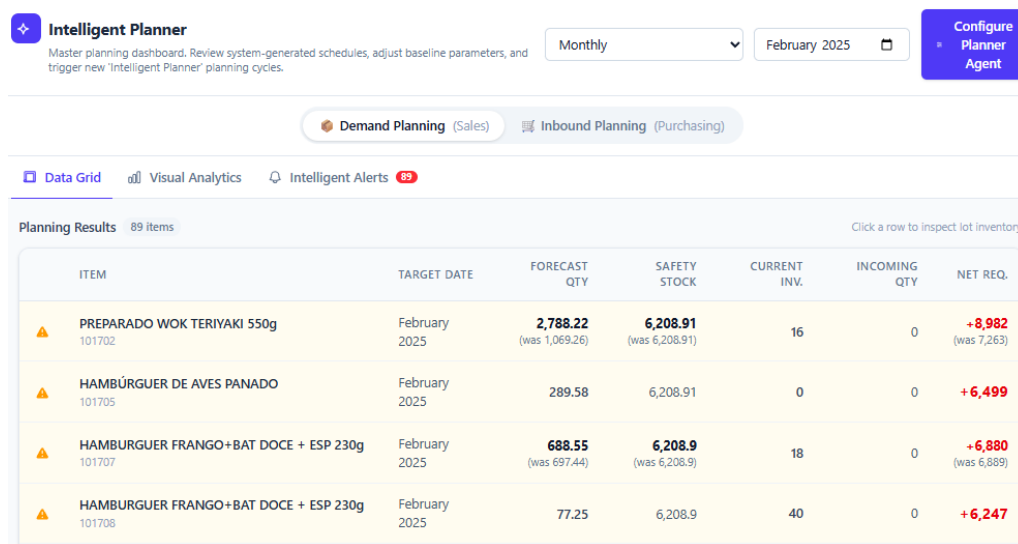


Figure 5.10: Alert 'low data' generated by Refinement Process.

While the refinement console manages the boundaries of statistical safety, the manufacturing planning engine serves as the execution engine that converts these secure forecasts into actionable procurement and production directives, marking the transition from demand planning to inbound inventory routing. The planning engine ingests the calculated safety bounds and baseline forecast to execute a time-phased inventory attrition simulation. The primary mission of this layer is the calculation of precise net requirements, as displayed in the data grid interface in Figures 5.11 through 5.12. This workflow executes a recursive Bill of Materials (BOM) decomposition across the manufacturing hierarchy, automatically expanding finished product demand into dependent target requirements for raw ingredients, drawing on the recipe data reconstructed through the data-mining process described in Section 4.1.3



Intelligent Planner
Master planning dashboard. Review system-generated schedules, adjust baseline parameters, and trigger new 'Intelligent Planner' planning cycles.

Monthly February 2025 [Configure Planner Agent](#)

Demand Planning (Sales) **Inbound Planning (Purchasing)**

[Data Grid](#) [Visual Analytics](#) [Intelligent Alerts 89](#)

Planning Results 89 items [Click a row to inspect lot inventory.](#)

ITEM	TARGET DATE	FORECAST QTY	SAFETY STOCK	CURRENT INV.	INCOMING QTY	NET REQ.
PREPARADO WOK TERIYAKI 550g 101702	February 2025	2,788.22 (was 1,069.26)	6,208.91 (was 6,208.91)	16	0	-8,982 (was 7,263)
HAMBÚRGUER DE AVES PANADO 101705	February 2025	289.58	6,208.91	0	0	+6,499
HAMBÚRGUER FRANGO+BAT DOCE + ESP 230g 101707	February 2025	688.55 (was 697.44)	6,208.9 (was 6,208.9)	18	0	+6,880 (was 6,889)
HAMBÚRGUER FRANGO+BAT DOCE + ESP 230g 101708	February 2025	77.25	6,208.9	40	0	+6,247

Figure 5.11: Demand Planning generated by Planning process.

5.2. BASELINE PERFORMANCE ANALYSIS (SCENARIO 1 EXECUTION)

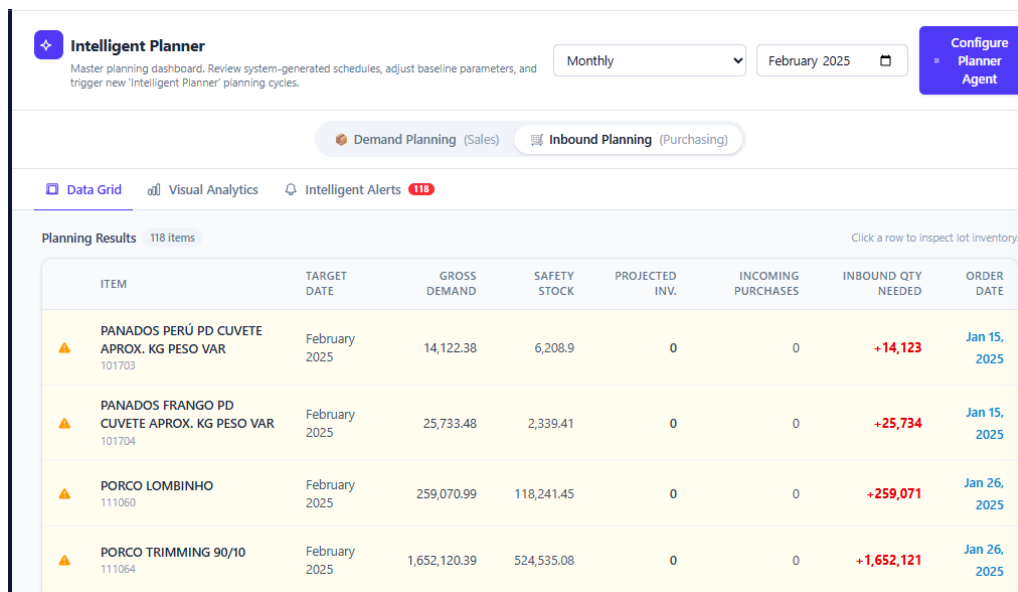


Figure 5.12: Inbound Planning generated by Planning process.

In alignment with modern Industry 5.0 supply chain sustainability objectives, the planning engine tracks material assets dynamically over time instead of treating stock as a static constant. The inventory simulation runs on a strict First-Expired-First-Out (FEFO) consumption logic, cross-referencing forecasted manufacturing draws directly against specific warehouse lot expiration parameters, as illustrated in Figure 5.13.

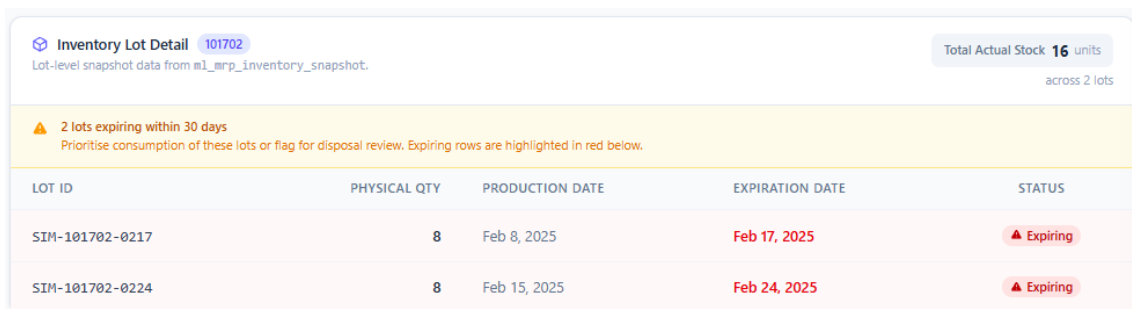


Figure 5.13: Detail information about expiration lots.

While interactive overrides are not supported for this block in the current prototype phase, the system provides operational safety by consolidating demand metrics, calculated net requirements, and stock-expiry tracking into a single view. As shown in Figure 5.14 for Item 101708, the interface automatically calculates a baseline forecast of 42.11 units and a safety stock of 4,028.53 units, resulting in a net requirement of +4,071 units. Directly below, the system flags critical shelf-life risks, warning the operator that lot SIM-101708-0217 is set to expire on February 17, 2025, providing clear operational visibility to prevent physical stock waste.

While automated backend rescheduling loops were out of scope for this initial prototype phase, the planning engine still connects the forecast to actionable execution by exposing, multi-tiered operational warnings. Instead of relying on manual data inspection, the system runs continuous checks against current stock levels and inbound supply timelines to generate explicit risk alerts for the manager. As demonstrated in Figure 5.15, the system populates a consolidated alert panel whenever an active SKU faces an immediate deficit during the 2025 execution window. Because lead times and inventory lots were simulated, these alerts demonstrate the planning logic rather than the client’s actual stock position.

These results support the Scenario 1 benchmark qualitatively: as shown in Figure 5.8, actual demand for the sampled SKU remained within or close to the prediction interval for most of the year, with the clearest deviation occurring between August and December, where demand tracked near the band’s lower boundary before recovering. The error of the combined forecast was not measured across the product catalog in this prototype phase, and doing so is the first step proposed for future work (Chapter 6); this baseline nonetheless gives Scenarios 2 and 3 (Sections 5.3 and 5.4) a reference point against which the impact of external disruption and live, uncontrolled data can be observed.

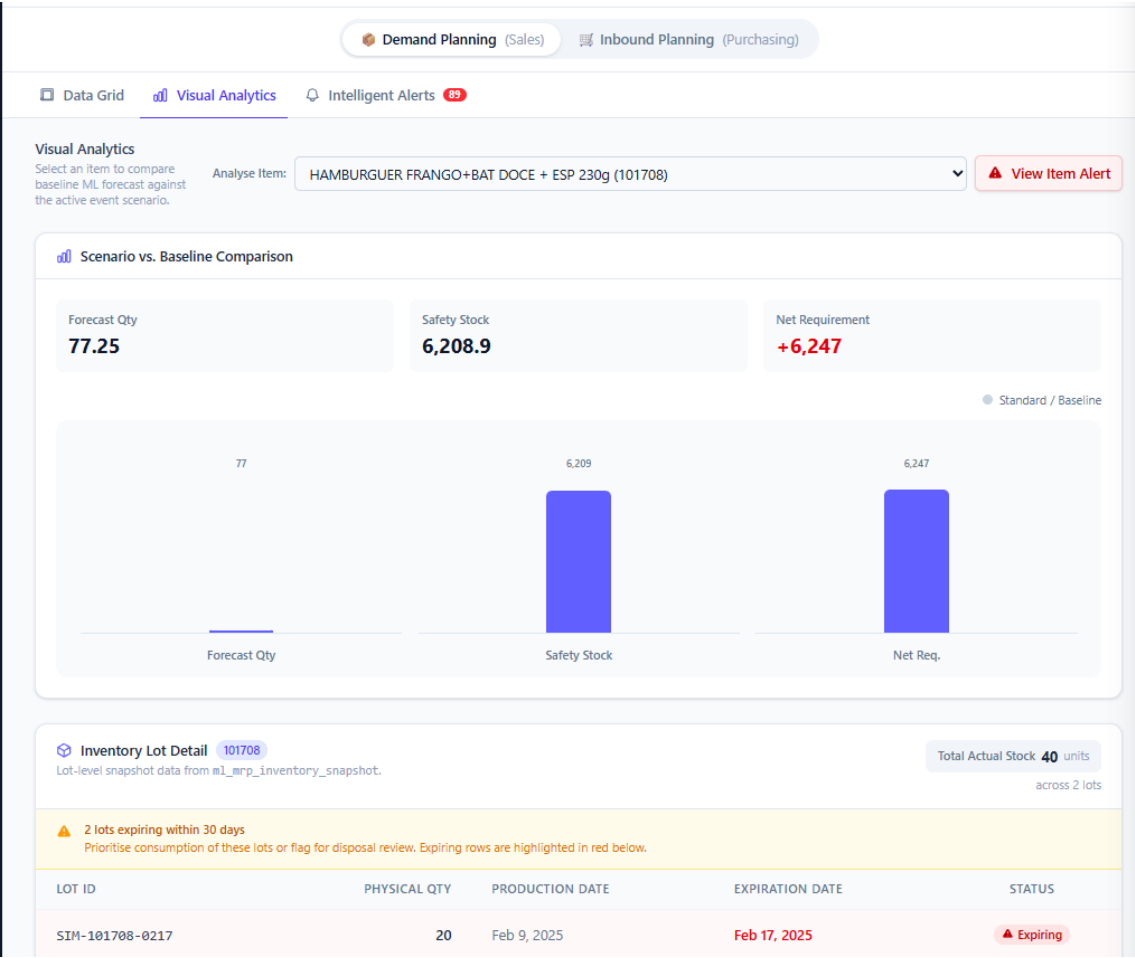


Figure 5.14: Visual Analytics Interface Displaying Forecast Metrics, Calculated Net Requirements, and FEFO Stock Expiration Alerts for SKU 101708.

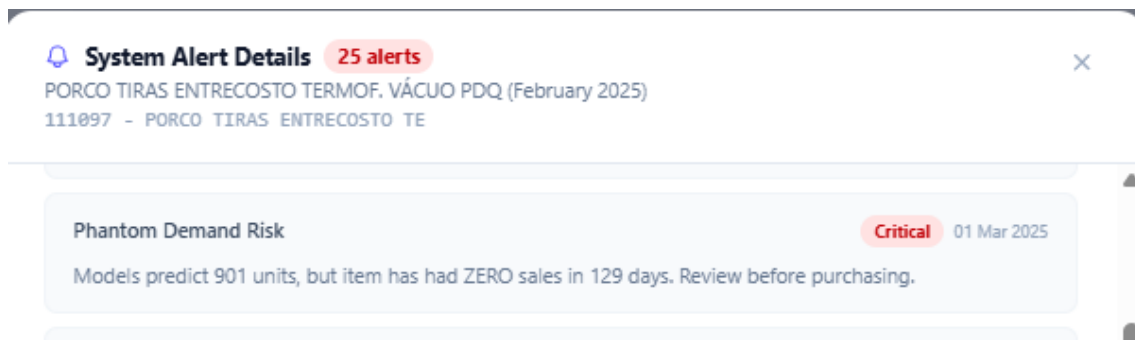


Figure 5.15: System Alert Details view displaying baseline material shortage warnings.

5.3 External Shock Analysis (Scenario 2 Execution)

Building on the 2025 baseline established in Scenario 1, Scenario 2 evaluates system resilience under abnormal market conditions, testing how the platform would react to sudden crisis by addressing a specific stress-test question: *"If a war occurs during this planning cycle, what is the immediate structural impact on the manufacturing and inbound plan?"*. This scenario utilizes the exact same baseline data and environmental constraints as Scenario 1, but activates the system's scenario simulation engine to process an active disruption package.

Simulation Logic and Reference Anchoring

The core logic behind the simulation involves re-generating the time-phased production schedule by anchoring current trends against localized historical precedents. For the war scenario, the system uses the 2022 outbreak of the Ukraine conflict as its mathematical reference anchor.

The engine extracts historical data from the selected crisis window, systematically comparing month-by-month volumetric shifts between the pre-disruption baseline and the active shock period to calculate a specific percentage multiplier. For example, if the reference historical data reveals a demand spike of +160.8%, a corresponding 2.61× multiplier is instantly computed and scaled across the forecast. Conversely, if no historical records are detected for a given month within that reference timeline, the engine applies a protective 0.00× default modifier, zeroing out the demand for that specific period to flag the data visibility gap for human evaluation.

Visual Verification and Operational Impacts

The visual validation of this crisis simulation is captured across the user interface views. Within the refinement domain illustrated in Figure 5.16, toggling the scenario overlay reveals a direct comparison between the *Master Forecast* line and the *Machine Learning Scenario* trend, which is rendered as a distinct dot line inside the 90% prediction interval to isolate visual volatility.

Figure 5.16 shows three lines: the orange line represents real 2025 demand, the blue line represents the prototype’s baseline prediction, and the new dotted line — absent from Figure 5.8 — represents the scenario’s projected impact.

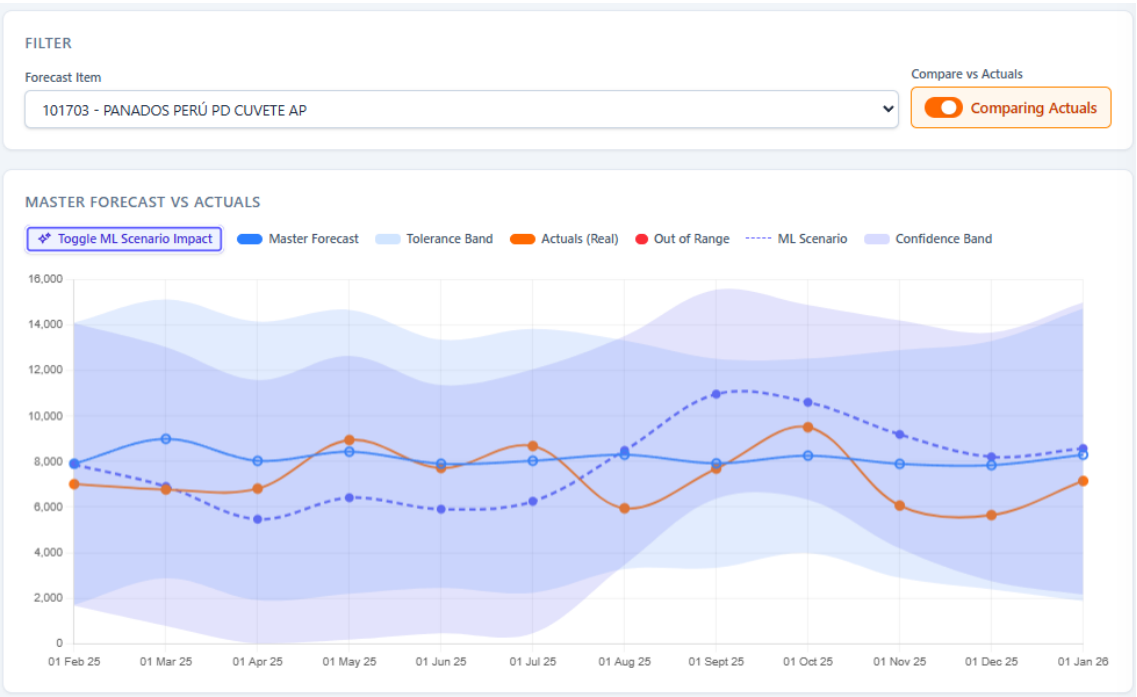


Figure 5.16: Refinement Chart View displaying Master Forecast vs. Dashed Machine Learning Scenario Impact Overlay example.

When this structural demand shift flows into the final planning grid, the interface presents a dual-layer analysis. The grid displays the newly calculated scenario targets in prominent, bold primary typography, while keeping the original baseline numbers immediately visible underneath in smaller, muted parentheses, as shown in Figure 5.17.

5.3. EXTERNAL SHOCK ANALYSIS (SCENARIO 2 EXECUTION)

ITEM	TARGET DATE	FORECAST QTY	SAFETY STOCK	CURRENT INV.	INCOMING QTY	NET REQ.
PREPARADO WOK TERIYAKI 550g 101702	February 2025	2,788.22 (was 1,069.26)	6,208.91 (was 6,208.91)	16	0	+8,982 (was 7,263)
HAMBÚRGUER DE AVES PANADO 101705	February 2025	289.58	6,208.91	0	0	+6,499
HAMBÚRGUER FRANGO+BAT DOCE + ESP 230g 101707	February 2025	688.55 (was 697.44)	6,208.9 (was 6,208.9)	18	0	+6,880 (was 6,889)
HAMBÚRGUER FRANGO+BAT DOCE + ESP 230g 101708	February 2025	77.25	6,208.9	40	0	+6,247

Figure 5.17: Planning Grid View presenting Primary Scenario Targets with Small Baseline References in Parentheses.

Simultaneously, as documented in the alerts panels in Figures 5.18 and 5.19, the consolidated warning center generates critical notifications summarizing the exact modifications. These warnings can be of two types: Critical and Information. As demonstrated in Figure 5.18 it generates a Critical message to warn the manager about the percentage impact of requested scenario based on reference, also it informs about the multiplier applied on new demand calculation with a default information related to *extended lead times*. Looking at Figure 5.19 we have the other warning type, Information, this message is to explain to manager why the item requested has no scenario demand generated after the scenario request, explaining that no solid historical data could be found to build a reference impact for that item.

These results confirm that the scenario engine responds coherently to an injected shock: the Critical and Information warnings correctly distinguished items with a usable historical reference from those without one, and the resulting demand multipliers propagated through to the planning grid without requiring manual reconciliation. This confirms the platform's capacity to convert a macro-level disruption into item-level, actionable guidance — the resilience behavior Scenario 2 set out to rest.

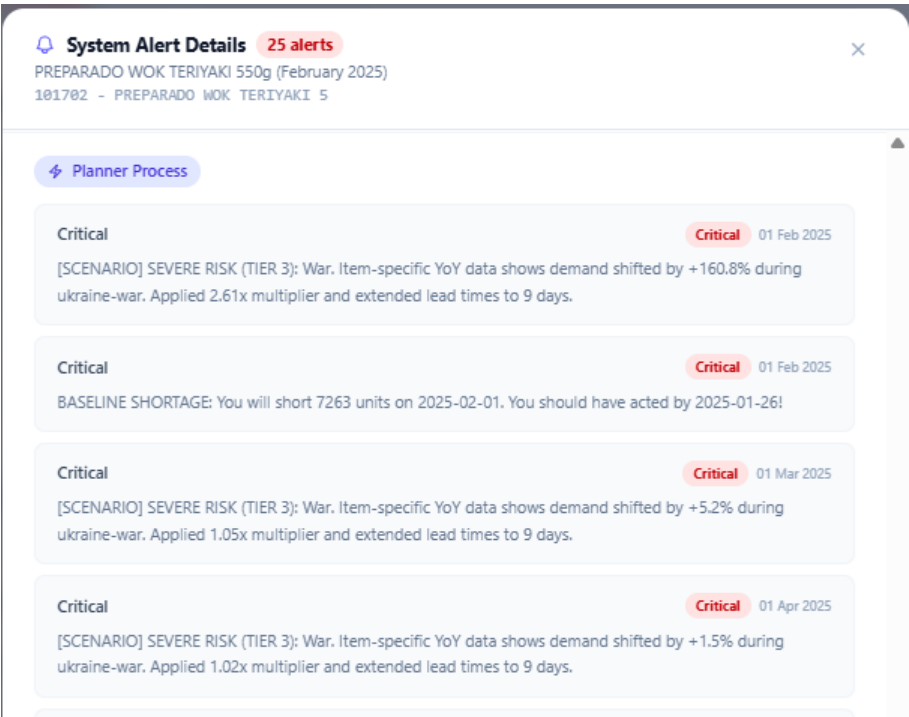


Figure 5.18: Scenario Warning Center: Critical Warnings generated during scenario process.

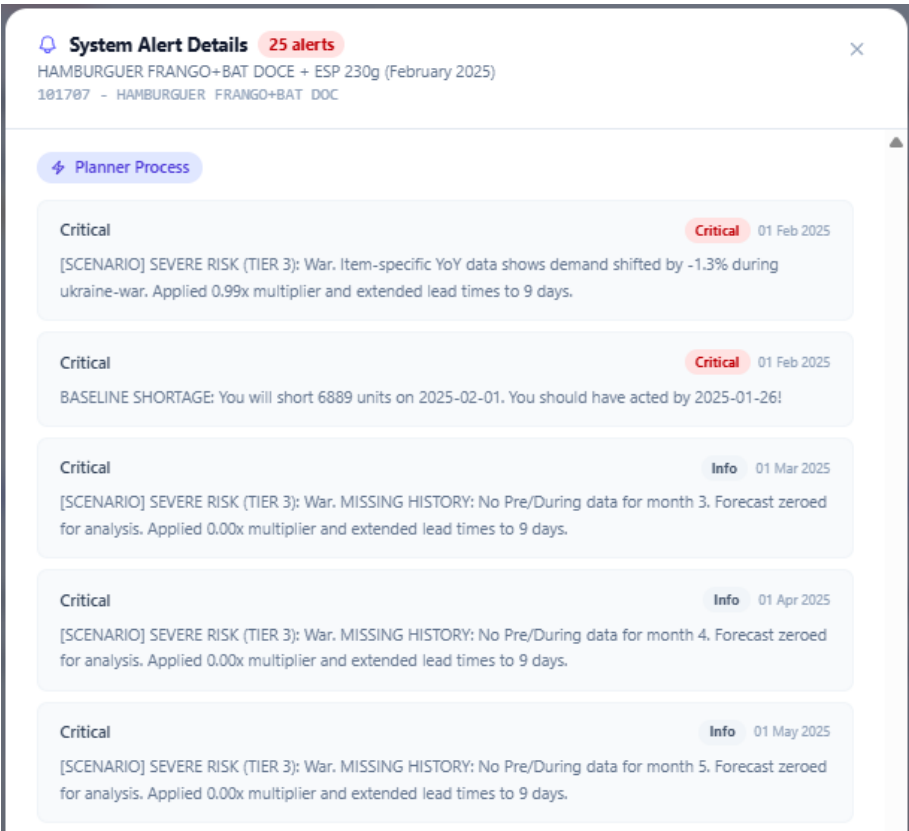


Figure 5.19: Scenario Warning Center: Information Warnings generated during scenario process.

Scenario 2 should therefore be read as a stress test, or what-if analysis, rather than as a forecast of future events. The demand shock is anchored on a historical precedent, the 2022 outbreak of the war in Ukraine, to simulate how a similar event could affect demand and production planning; it does not predict whether or when such an event will occur, nor its actual magnitude. Its value lies in showing, before a disruption happens, which materials and items would be affected and which alerts the manager would receive.

5.4 Live Real-Time Verification (Scenario 3 Execution)

Scenario 1 established the platform's behavior under controlled conditions, where restrictions and manual interventions applied during model testing, debugging, and selection kept the environment predictable. Scenario 3 removes those constraints: its goal is to let the prototype run fully automatically on raw, uncontrolled 2026 data, exactly as it would in an unsupervised production setting, and observe whether the architecture built and validated in Scenario 1 holds up without manual correction. "Deployment" here refers to this operational mode — the system executing end-to-end without intervention — rather than deployment, since no separate infrastructure was available for this phase. Deployed to predict demand timelines moving beyond March 2026, this stress test presents a dual narrative regarding the platform's core architectural resilience and individual algorithm vulnerabilities under live data conditions.

Architecture Resilience and Multi-Model Fault Tolerance

The live deployment showed the structural integrity of the decoupled three-tier system architecture introduced in Section 4.1.1. Faced with an unverified, rolling data horizon, the underlying backend orchestration layer never suffered a system crash or state synchronization failure. Instead, it successfully generated the requisite demand outputs, calculated target requirements, and populated operational exception logs exactly as verified during the controlled 2025 runs.

However, the real-world deployment exposed technical limitations within the predictive framework. As illustrated in Figures 5.20 through 5.22, only three of the six core predictive models — Chronos, Granite and XGBoost — executed successfully and streamed calculations to the interface. The remaining three failed under live conditions, most likely because constraints introduced during Scenario 1's controlled testing and model-selection phase — validation gates, data-density thresholds, or filters tuned specifically to the 2025 dataset — were never fully removed before this unsupervised run. This is itself a useful finding: it confirms that the two scenarios exercise the same underlying pipeline under different degrees of restriction, and that Scenario 3's value lies precisely in surfacing constraints that fully controlled test would never expose.

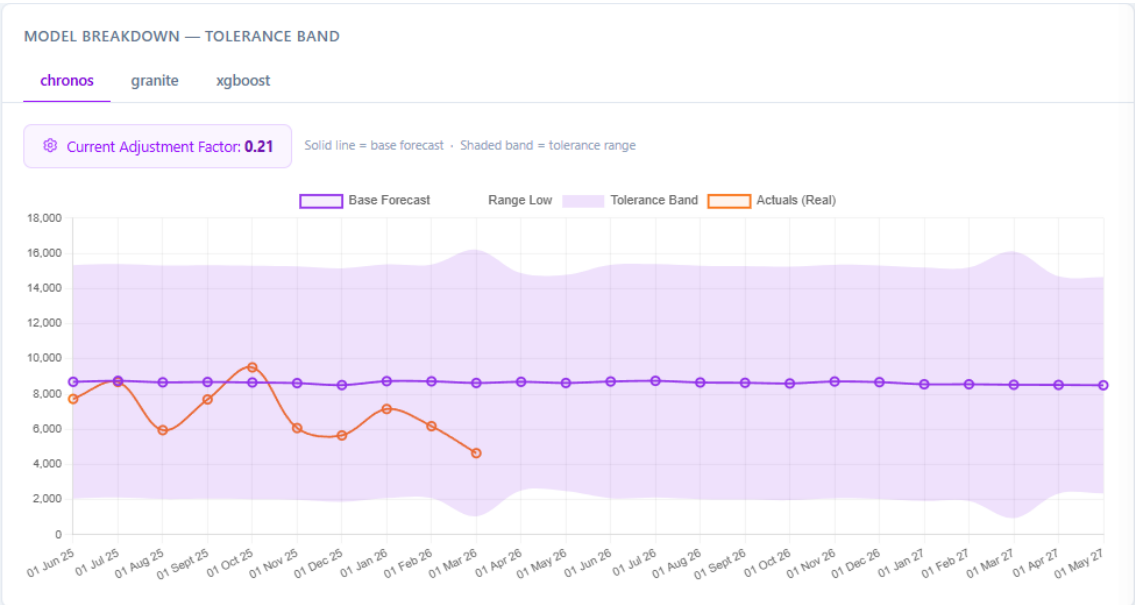


Figure 5.20: Active Chronos Forecast and Prediction Interval for one year demand after February 2026.

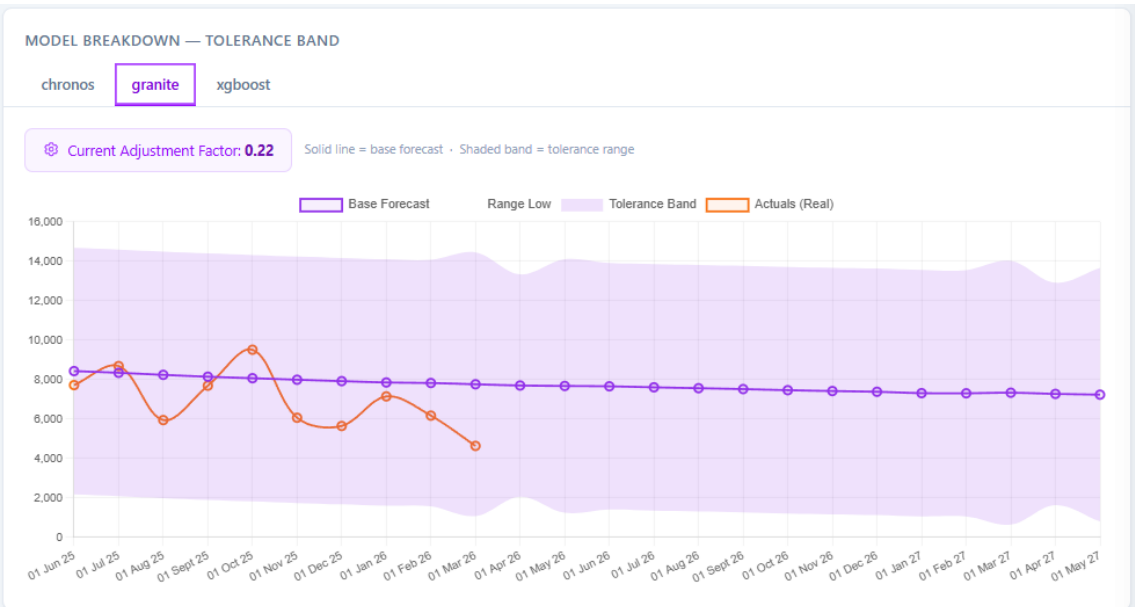


Figure 5.21: Active Granite Forecast and Prediction Interval for one year demand after February 2026.

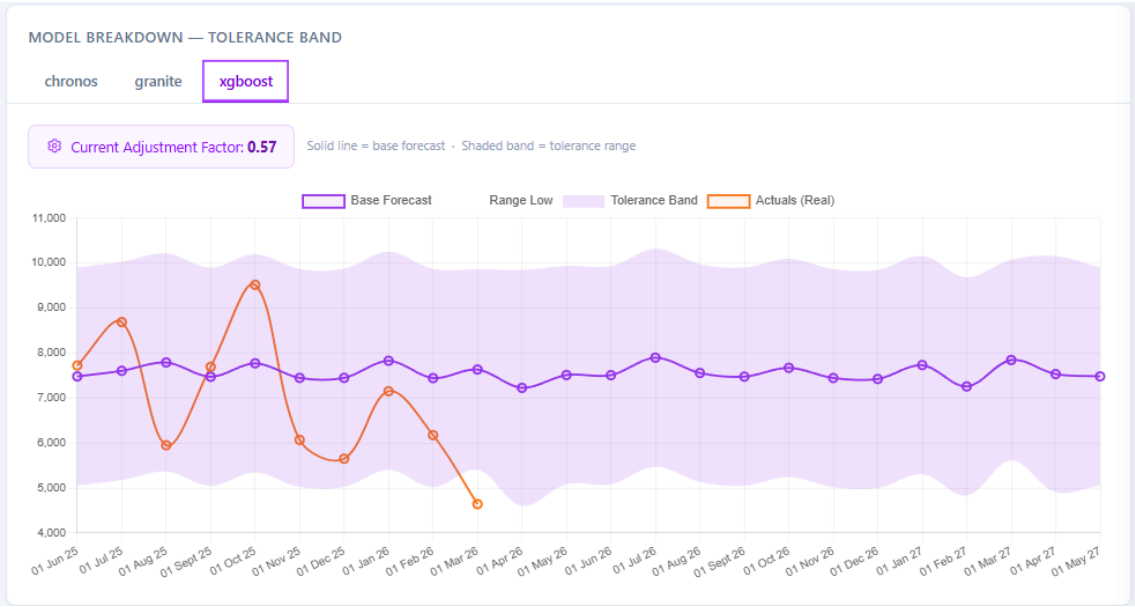


Figure 5.22: Active XGBoost Forecast and Prediction Interval for one year demand after February 2026.

Because the computational core was engineered with high-fidelity decoupling and an Inverse Error-Weighting, this partial algorithmic failure did not halt the platform. When a model fails to execute, the orchestration layer isolates that specific failure to its own processing thread, preventing it from affecting the other five models. The refinement engine then recalculates the mathematical consensus using only the models that completed successfully — in this case, Chronos, Granite, and XGBoost — redistributing the Inverse Error Weighting protocol’s influence across the surviving streams rather than requiring all six to succeed. The central database is updated with this adjusted consensus, allowing the downstream planning grid to continue operating without any manual intervention.

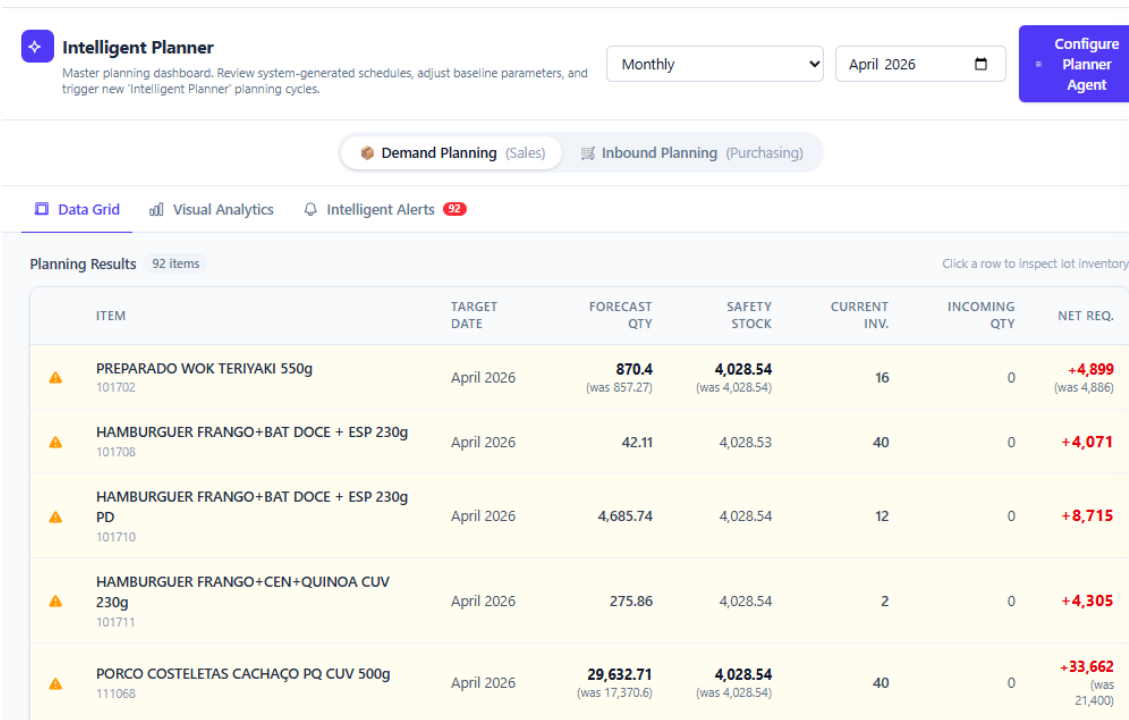


Figure 5.23: Intelligent Planner Interface displaying functional April 2026 Master Planning Results derived from surviving model consensus.

Crisis Stress Testing and Data Constraints

To evaluate the platform’s ability to ingest a sudden event in a live setting, a war scenario simulation was also requested, timed to coincide with the escalating geopolitical tensions of 2026 rather than a purely hypothetical exercise — testing the crisis-simulation feature introduced in Scenario 2 under conditions as close as possible to a genuine, unplanned deployment event. Figures 5.24 and 5.25 display the system’s interactive response to this request.

When the conflict scenario is toggled on (Figure 5.25), the engine immediately injects the corresponding historical anchor modifiers, projecting a distinct, volatile *Machine Learning Scenario* curve overlaying the baseline trend. As in Section 5.3, the dashed line represents this scenario-adjusted projection, layered here onto the same orange actuals and blue baseline forecast used throughout Chapter 5.

5.4. LIVE REAL-TIME VERIFICATION (SCENARIO 3 EXECUTION)

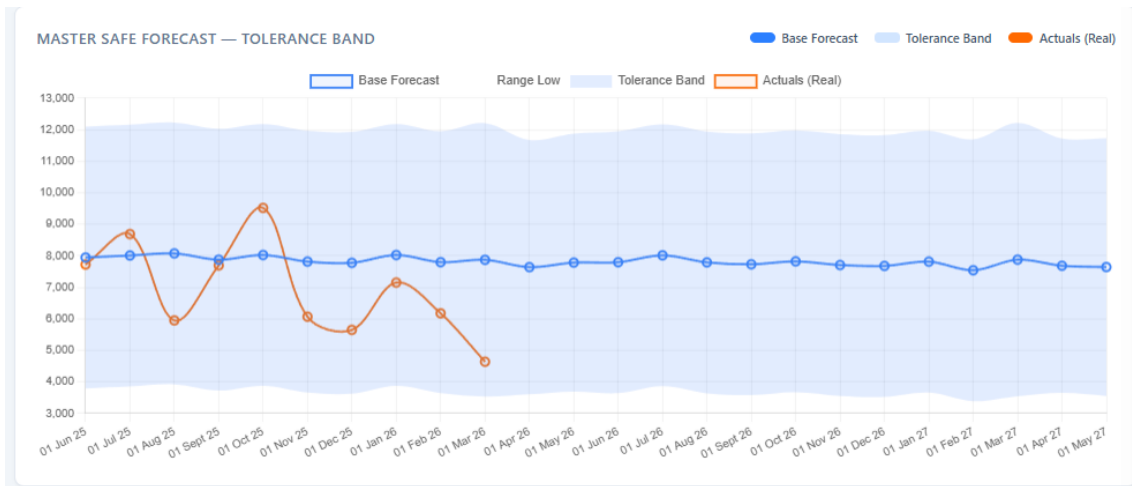


Figure 5.24: Master Forecast vs. Actuals dashboard tracking baseline projections prior to scenario injection.

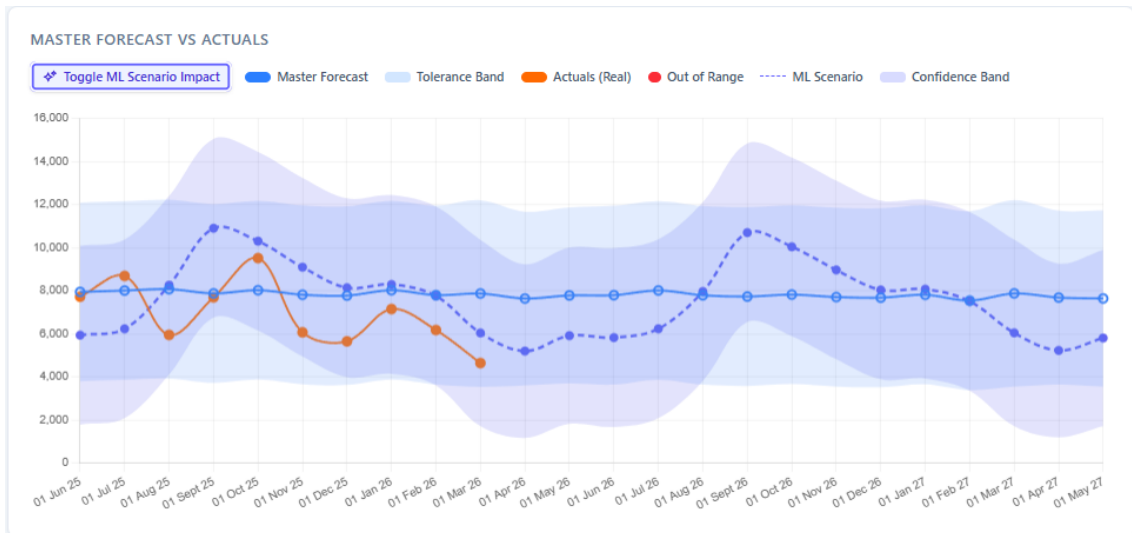


Figure 5.25: Master Forecast vs Scenario Impact Simulation overlays against real-world actual lines.

While the visual transition proved successful, tracking the curves against the orange *Actuals (Real)* line exposes the most significant data constraints and structural limitations encountered during the prototype’s implementation in a live production environment. Within this specific manufacturing sector, maintaining high data density over long horizons is exceptionally difficult due to shifting administrative habits. In practice, factories frequently alter or recycle their internal product identification codes (Item IDs) after some time (months or years), even when the underlying product compositions remain functionally identical. Because predictive algorithms rely entirely on consistent, uninterrupted identifier history to trace consumption behaviors, these frequent ID lifecycles Resets strip the deep-learning models of the long-term data volume required to extract reliable trends and minimize forecasting variance.

To mitigate this severe historical data limitation and improve model precision under live conditions (also verified in the 2025 controlled environment each time refinement process was executed, as demonstrated by the dynamic blue range in Figure 5.8), the system’s architecture implements a dynamic Walk-Forward Validation protocol instead of relying on a static historical baseline. When a user triggers an operational training request, such as an execution window beginning in March 2026, the computational core automatically initializes an iterative simulation that anchors exactly one year in the past. Moving sequentially from this fictional start point of March 2025, the engine executes a continuous monthly rolling loop, training the forecasting models and updating performance evaluation data month-by-month up until February 2026.

By running this progressive sliding simulation right up to the active execution frontier, the system dynamically generates fresh, context-aware error tracking boundaries across the historical transition. The refinement process directly captures these localized, rolling metrics to compute the upper and lower bounds of the prediction interval, ensuring that the resulting forecasting range adapts elastically to recent volatility variations rather than flattening into a rigid, non-adaptive band across the active 2025 scheduling year.

These findings suggest that while the walk-forward layer helps make use of short-horizon datasets, future deployment phases must establish intelligent identifier mapping layers. Future work should focus on building ID bridges that can automatically stitch fragmented product lifecycles back together, ensuring that the foundation models can fully exploit their deep-learning capacities across rolling operational horizons.

These results validate Scenario 3’s central goal: the architecture held under fully unsupervised, raw conditions, isolating three model failures without manual intervention and recalculating a working consensus from the surviving streams — the resilience behavior the decoupled design was built to provide. What the scenario also exposes, and what a controlled test like Scenario 1 could never surface, is the practical cost of raw production data: as shown in Figure 5.24 the Master Forecast fails to track the downward trend in actual demand over the live window, and even the scenario-adjusted projection in Figure 5.25 only partially closes that gap. This divergence is consistent with the identifier-lifecycle fragmentation discussed above: as Item IDs reset, the models lose the continuous history needed to recognize a real trend shift, rather than treating it as noise. Taken with Scenarios 1 and 2, this establishes a clear progression across the three tests: architectural soundness under control, coherent response to a modeled shock, and resilience without accuracy guarantees once identifier continuity breaks down under raw conditions.

5.5 Operational Evaluation and Partner Validation

The validation presented in Scenarios 1 through 3 (Sections 5.2 through 5.4) evaluated the platform against its own design objectives, using historical and live transactional data. To complement this technical assessment with an operational perspective, structured feedback was also gathered from Tecnicamente. This organization facilitated the project on behalf of their client — the food manufacturing organization whose historical data underpinned the prototype and who served as the primary beneficiary of the system. Their responses reflect direct exposure to the system’s outputs across the three validation scenarios and indicate whether the platform can serve as a viable operational tool rather than only as a functional prototype.

To collect this feedback, ten open questions (Appendix A) were prepared in advance and discussed in a meeting with Tecnicamente in August 2026, after the prototype and the three scenarios had been presented. The answers were given by the two managers of Tecnicamente, a department of Tecnopack. Both work directly on software development and were the author’s superiors during the project, and one of them was also the author’s company-side supervisor. Questions Q1 to Q3 asked whether the prototype solves the client’s planning problems and whether its numbers can be trusted, Q4 and Q5 asked about two specific features, Q6 to Q9 about a possible next phase, and Q10 about overall impressions. The answers below are summarized from the author’s notes of the meeting and refer to their question numbers. Since both respondents followed the project closely and are not planners at the client, their answers should be read as the view of an industrial partner, not as an independent evaluation.

Asked whether the system addresses the planning challenges originally described (Q1), seasonal demand, product deterioration rates, and the need to react to sudden market shifts, the partner organization confirmed that, for a first phase, the platform already plans and delivers in line with the proposed objectives. Regarding what this changes relative to the client’s current process (Q2), the feedback centered on three concrete gains: (i) the normalization and automation of purchasing quantities; (ii) tighter control over purchase timing, through explicit windows for when to buy that reduce waste by avoiding premature or delayed acquisition; and (iii) a reduction in the time spent on manual analysis.

On whether the numbers produced by the system are trustworthy enough to act on operationally (Q3), the response was that the presented figures do reflect the quantities actually required to meet market demand for the corresponding items — suggesting the organization views the output as operationally credible rather than merely illustrative.

Feedback on the manual weight-adjustment feature (Q4) — Figure 4.7, Section 4.1.3 — was similarly positive: the partner organization considered it a sensible capability for the client’s planners to use in practice, particularly in cases where the blended forecast output exhibits significant variance from actual demand and specific models are known to perform more accurately for a given item — allowing a planner to adjust model influence and correct the final forecast accordingly. The crisis-simulation feature (Q5, Section 5.3) received a comparable reception: rather than being seen as a demonstration exercise, it was described as a genuinely useful capability for improving the precision and responsiveness of demand planning under abrupt, unplanned disruptions.

Asked to assess the prototype as a first working version (Q6), the partner organization characterized it as a solid foundation for a second development phase, noting that it already provides a strong base to build several further improvements on. When asked what should be prioritized in a subsequent phase (Q7), two specific requests stood out: a feature comparing what was actually spent against what the system had planned, to evaluate and demonstrate the system’s value over time, and a mechanism linking equivalent products when the same product receives a new item code, so that its sales history is not lost. This second point matches the identifier-lifecycle limitation observed during Scenario 3 (Section 5.4), and is addressed further as a priority for future work in Chapter 6.

Regarding what would be needed to deploy the system on live operational data rather than historical data (Q8), the partner organization acknowledged that some errors and bugs would likely surface, but noted that the system would still return usable values and feedback, consistent with what was already demonstrated under live conditions in Scenario 3 (Section 5.4). On ownership within the client organization (Q9), the tool was identified as belonging to the purchasing and planning functions specifically, rather than IT or floor operations.

Asked what most surprised them about the prototype (Q10), the partner organization pointed to the quality of the output values and the tool’s functional value, along with the number of additional features that could realistically be integrated into the existing platform to optimize secondary processes and ease day-to-day operations on the client’s side.

These observations are consistent with the conclusions drawn from Scenarios 1 through 3 (Sections 5.2 through 5.4): the architecture is viable and the outputs are considered actionable by the partner organization, while the identifier-continuity and cost-comparison priorities it raised match the limitations and future-work directions already identified through the platform’s own live testing.

CONCLUSION & FUTURE WORK

Integrating data-driven predictive models into Material Requirements Planning System (MRP system) environments moves food manufacturing planning away from static, rule-based logic toward frameworks that adapt to real demand signals. This thesis developed and evaluated a decoupled, three-tier planning platform designed to address the specific vulnerabilities of food manufacturing operations. By combining automated data ingestion, multi-model ensembles, and time-phased inventory attrition simulations in a single framework, the system uses both historical data and active factory timelines, with the aim of mitigating the bullwhip effect and supporting operational stability.

System Contributions and Empirical Evaluation

The empirical validation executed across the experimental scenarios demonstrated the structural viability of the proposed architecture and provided consistent indications of its resilience, most notably its ability to isolate model failures and continue operating under live, unsupervised conditions in Scenario 3 (Section 5.4). The resilience was assessed qualitatively rather than through a formal quantitative resilience metric, and should be read within that scope. Under the controlled conditions of Scenario 1, the training process established a performance benchmark for the 2025 fiscal year by extracting seasonal trends from historical data dating back to 2014. The refinement process then combined the six statistical and deep-learning forecasts through Inverse Error Weighting into a single forecast with a 90% prediction interval, which the planning engine used as input to the recursive Bill of Materials (BOM) decomposition and the First-Expired-First-Out (FEFO) lot assessment. Scenario 2 demonstrated the prototype's capacity for scenario-based stress testing, showing that anchoring modern operations against localized historical precedents can convert macro-level geopolitical crises into descriptive, actionable operational exceptions.

The validation presented in this thesis is therefore functional: it shows that the prototype's mechanisms work as designed on real historical data, not that they improved the client's operations. The forecasting pipeline was run on real sales data and each model's error was measured through walk-forward validation, but the combined forecast was not evaluated against actual demand across the product catalog. The planning layer, including the recursive BOM decomposition, FEFO allocation, and shortage and waste alerts, operated on reconstructed recipes, simulated lead times, and synthetic inventory lots. These mechanisms can contribute to lower stock levels and less waste, but that contribution was not measured in a real operating context, so this thesis does not claim that stock or waste were actually reduced.

The most significant test of the prototype occurred during the live, uncontrolled testing of Scenario 3 in the active 2026 calendar year. This environment exposed the core engineering vulnerabilities of the data streams while also highlighting the fault-tolerant nature of the architecture. Due to unpredicted data-density limits and environmental constraints, only three of the six models (Chronos, Granite, and XGBoost) completed execution under the live horizon. Because the backend was built with microservice decoupling and fallback routines, this partial failure did not crash the system: the platform isolated the failing pipelines, recomputed the combined forecast and its prediction interval from the remaining models, and generated material schedules. This shows that the architecture tolerates the failure of individual models, although, as Section 5.4 notes, the resulting forecast did not follow the real 2026 demand trend.

Cross-Check Against the Thesis Objectives

Section 1.2 defined four objectives for the integration of Machine Learning (ML) and Artificial Intelligence (AI) into the existing MRP system: improving the prediction of production and raw material needs, optimizing inventory and stock levels, minimizing food waste and production losses, and improving responsiveness and resilience to disruptions. Revisiting these objectives against the validation results presented in this thesis allows their fulfillment to be assessed individually rather than only in aggregate.

The first objective, demand prediction, was addressed through the heterogeneous six-model ensemble and Inverse Error Weighting mechanism (Req-01, Req-03, Section 4.1.2), tested under controlled conditions in Scenario 1 (Section 5.2) and, with reduced model coverage, under live conditions in Scenario 3 (Section 5.4). Each model's error was measured through walk-forward validation, but the accuracy of the combined forecast was assessed visually for a representative item rather than measured across the product catalog. The partner organization (Section 5.5) reported, qualitatively, increased forecasting precision and a reduction in manual analysis time relative to the client's prior process.

The second objective, inventory and stock-level optimization, was addressed through the statistical safety-stock customization and recursive BOM decomposition that drive the planning engine's net-requirement calculations (Req-04, Req-06). The partner organization described the system as normalizing and automating purchasing quantities, but no change in inventory levels or inventory cost was measured. This objective was therefore implemented and demonstrated at prototype level, not quantitatively validated.

The third objective, minimizing food waste and production losses, was addressed through the FEFO stock-attrition logic and lot-expiration tracking built into the planning engine (Req-07, Figures 5.13–5.14). The partner organization cited the explicit purchase-timing windows as a means of reducing waste from premature or delayed acquisition. As with the second objective, the mechanism was implemented and demonstrated at prototype level, on synthetic inventory lots, but no reduction in waste was measured.

The fourth objective, responsiveness and resilience to disruptions, was addressed through the scenario simulation engine (Req-08, Section 5.3) and, independently, through the architecture's demonstrated fault tolerance under live, unsupervised conditions in Scenario 3 (Section 5.4). As noted above, this objective was validated qualitatively: the system responded coherently to an injected shock and continued operating when half of its predictive models failed, but no formal resilience metric was computed across the full SKU catalog. The partner organization also considered the crisis-simulation feature practically useful, describing it as a tool for day-to-day planning rather than a conceptual demonstration.

Taken individually, all four objectives were addressed by working mechanisms and tested through the validation scenarios and industrial partner validation, but none was quantitatively validated in a real operating context. The first objective comes closest, since each model's error was measured; the second and third were demonstrated at prototype level only; and the fourth was demonstrated technically and assessed qualitatively, without a quantified resilience measure. There is an important difference between implementing a mechanism intended to optimize stock or reduce waste and demonstrating empirically that it did so, and closing that gap is the main priority for future work.

Industrial Challenges and Structural Limitations

Despite the structural successes of the prototype, the live deployment exposed fundamental data challenges inherent to the industrial manufacturing sector. The most significant limitation identified centers on identifier lifecycle constraints. In the real-world factory environment, organizations frequently alter, split, or recycle internal product classification codes (Item IDs) after some time (months or years), even when the physical items remain functionally unchanged. Because deep-learning models depend entirely on continuous, long-term historic identifier, these resets reduce the data available to the algorithms, introducing noise and increasing forecasting variance.

The Walk-Forward Validation protocol partly limits the effect of this fragmentation on the prediction interval. By retraining the models month by month over the year before the forecast, the refinement phase can use the error of the fold matching each calendar month rather than a single error for the entire year. It does not, however, restore the lost history, and structural identifier fragmentation remains a key hurdle for long-horizon predictive accuracy in active production environments.

Future Implementation Phases and Enhancements

While the current prototype demonstrates the core calculation, refinement, and material scheduling logic, several practical features have been identified to guide the system's evolution into a production-ready framework.

Improving Predictions for Seasonal and Slow-Moving Items

Future development phases should focus on updating the system to better handle seasonal items or products that experience long periods with zero sales. To make the calculations more precise for these specific items, upcoming versions will test changing the data view from monthly to weekly blocks, which helps smooth out minor weekly variations caused by delivery delays. Furthermore, the system should include specific logic built for irregular sales patterns. This includes adding standard methods like Croston's Method to predict both when a sale might happen and how much will be sold, as well as testing a two-stage approach that uses a simple checker to assess whether a sale is likely before running the main calculation engine to estimate the final volume.

Product Code Cleanups and Quality Report Tracking

To permanently address the issues caused by factories changing their item codes every year or two, future work should build a code history layer in the database. This component would act as a link table that automatically maps new or modified product codes back to their original historical codes, so that the forecasting models always have access to continuous historical data. Concurrently, the data collection should expand to read Credit Note reports and factory quality control logs. By tracking historical product return rates, the system can display a return risk warning on screen, letting planners know which specific item batches face a higher risk of being rejected or wasted.

Real-Time UI Progress Bars and Manufacturing Constraints

From a system communication perspective, the backend layout can be upgraded by building a direct data connection line inside the FastAPI layer. Moving away from the current method, where the database is constantly polled for updates, a real-time messaging pipeline would allow external services, such as the Quarkus gateway or the frontend screens, to track exactly how far along a calculation is and stream progress bars directly to the user interfaces.

On the manufacturing-constraints side, the main planning screen can expand to include real-world factory limits. While the current prototype checks material requirements against product expiration dates, a production-ready system must also factor in active assembly-line capacities, machine maintenance schedules, and changing supplier delivery delays. By combining these operational bottlenecks with basic cost tracking, the platform would evolve from a simple demand calculator into an intelligent scheduling system that helps minimize the total cost of maintaining the factory's inventory.

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PARTNER FEEDBACK QUESTIONNAIRE

The following questions were prepared in advance and discussed in a meeting with Tecnicamente in August 2026.

- Q1. Does the prototype address the planning challenges you originally described to us (seasonal demand, perishability, reacting to sudden shifts)? Which of these does it handle well, and which still feel unresolved?
- Q2. Compared to how planning is done today, what does this actually change — is it something that would save time, catch things you currently miss, or mostly automate what you already do manually?
- Q3. Would your team actually trust the numbers this system produces enough to act on them — for example, ordering raw materials based on its forecast — or does it currently feel more like a proof of concept than something to rely on operationally?
- Q4. The system lets a manager manually adjust how much weight each forecasting model gets. Is that something your planners would realistically use, or is it too technical for day-to-day use?
- Q5. The war/crisis simulation feature lets you test "what if a disruption happened right now." Is that something you'd find useful in a real situation, or does it feel more like a demonstration than a practical tool?
- Q6. Given this is a first working version (with some features still limited or simplified), does it feel like a credible starting point worth continuing, or too early to judge?
- Q7. If this moved to a second phase, what would you want prioritized first? (For reference, things not yet built include: real-time progress indicators, per-role dashboards for floor workers, and better handling of product codes that change over time.)
- Q8. What would need to happen for your team to actually pilot this on live data, rather than historical data, even on a limited scale?

- Q9. Who inside your organization would this tool belong to — operations, IT, planning — or is that not yet obvious?
- Q10. What surprised you about the prototype, in either direction — anything you expected to see that wasn't there, or anything better than you expected?



2026

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João Alves

